PAMS

Pakistan Army Military Symbology

Backbone for rendering Military Symbology

Project Overview

Future Proof & Backward Compatible

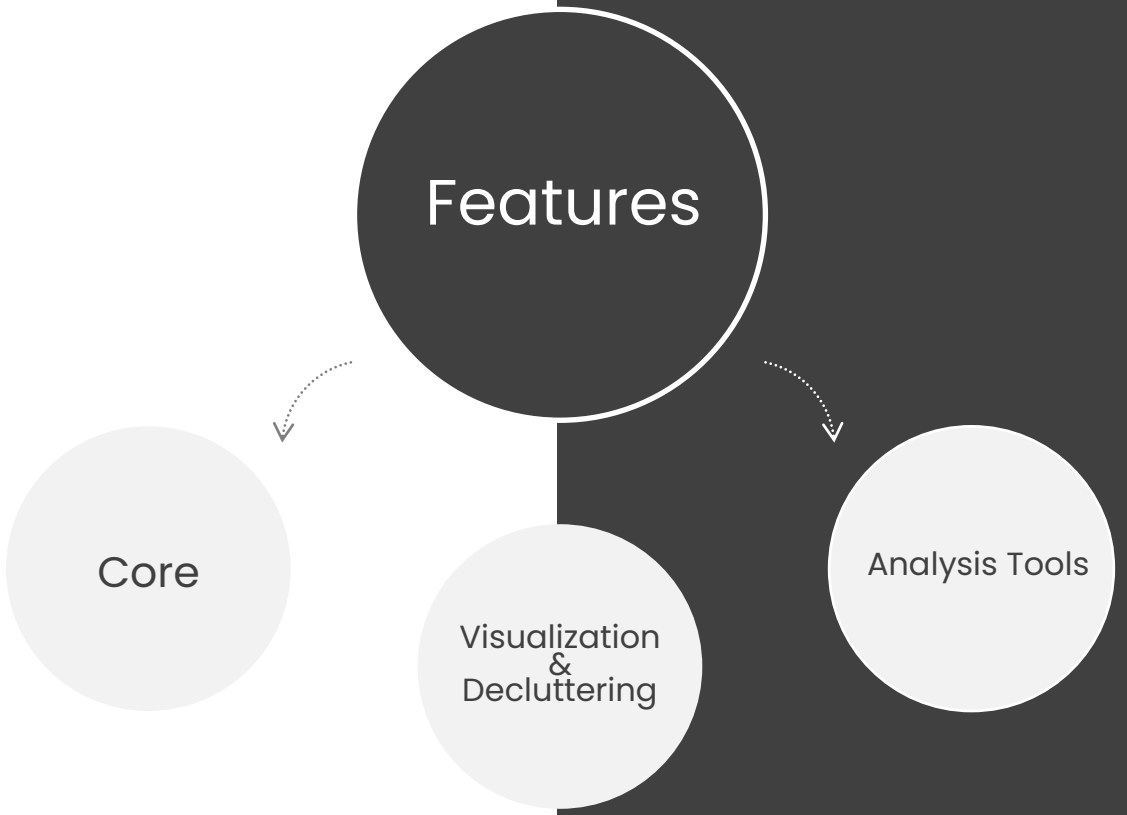
- Acts as the standard for drawing Military Symbols for all the applications
- 2K+ Unique Symbols
- Based on **latest ArcGIS API for JavaScript 5**
- **Fully compatible** with previous version of Symbology
- WebGL based Rendering Engine
- All feature fully compliant with **2D and 3D views**
- Fully **Export/ Import** Support: Plans and GeoJSON

Enhanced User Drawing & Analysis Experience

- Proximity Engine keeps user **aware** of the surroundings
- Deployment Manager for **kick starting Plan Marking**
- Key Terrain Identifier extracts **keys avenues**
- Where to place an OP and is it defensible?
- UAV Flight Planning
- Decluttering Tools

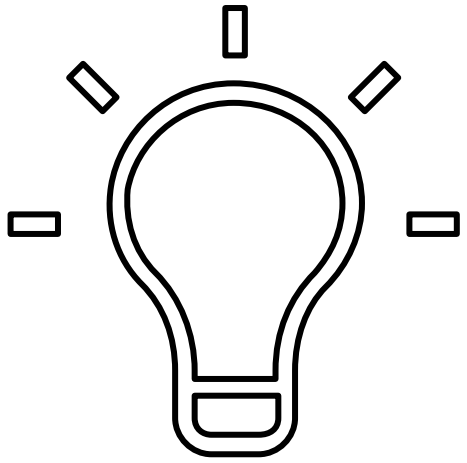
Features

PAMS is packed with a comprehensive suite of advanced features and intelligent tools designed to help users **plan**, **analyze**, and **execute tactical operations** with greater speed, precision, and situational awareness.



Core

Batch Edit Operations
Selection
&
Alignment
Deployment Manager
Control Point Editing



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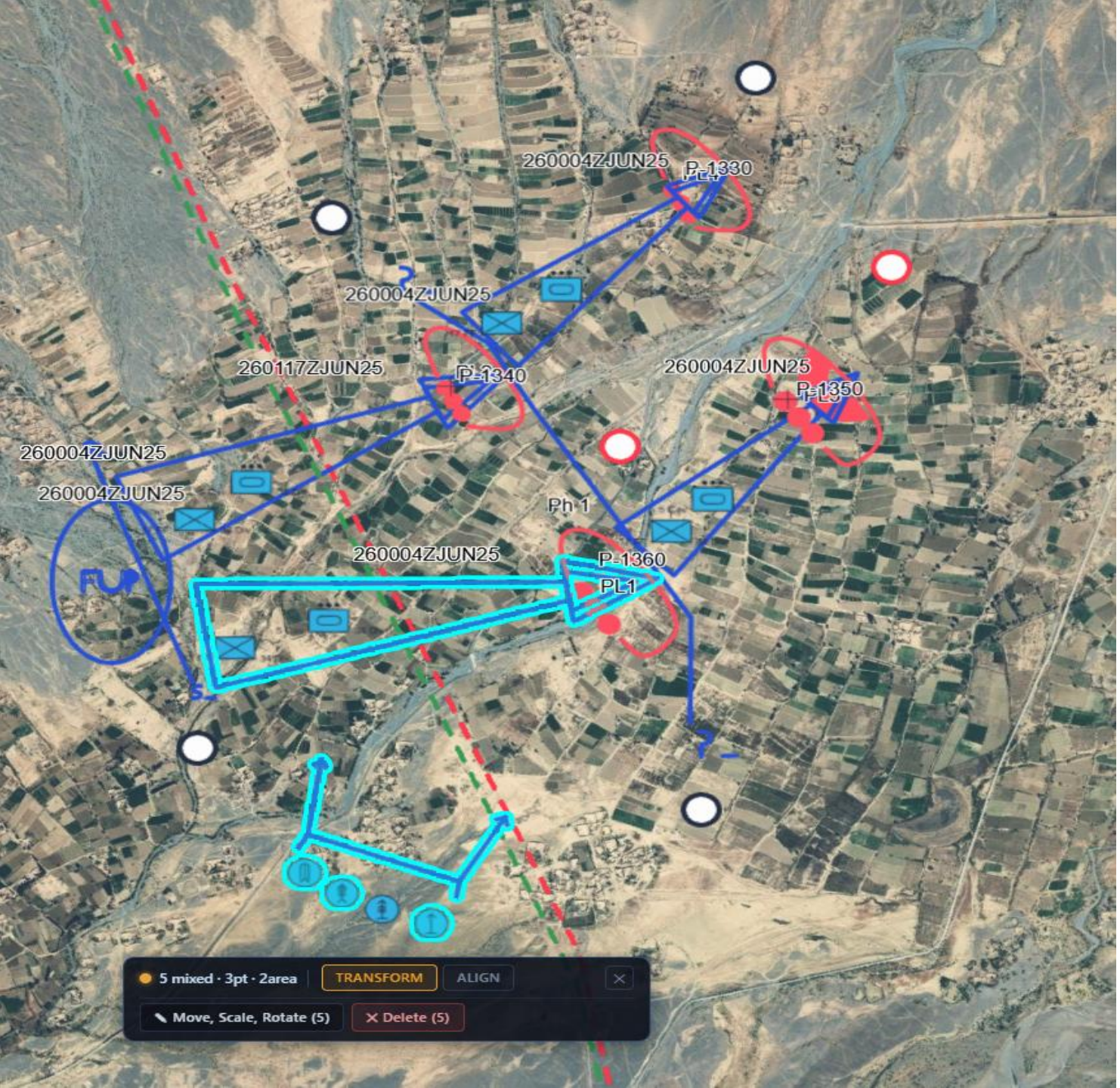

Batch Edit Operations

Edit and Move

Copy and Paste with Offset

Save and Load

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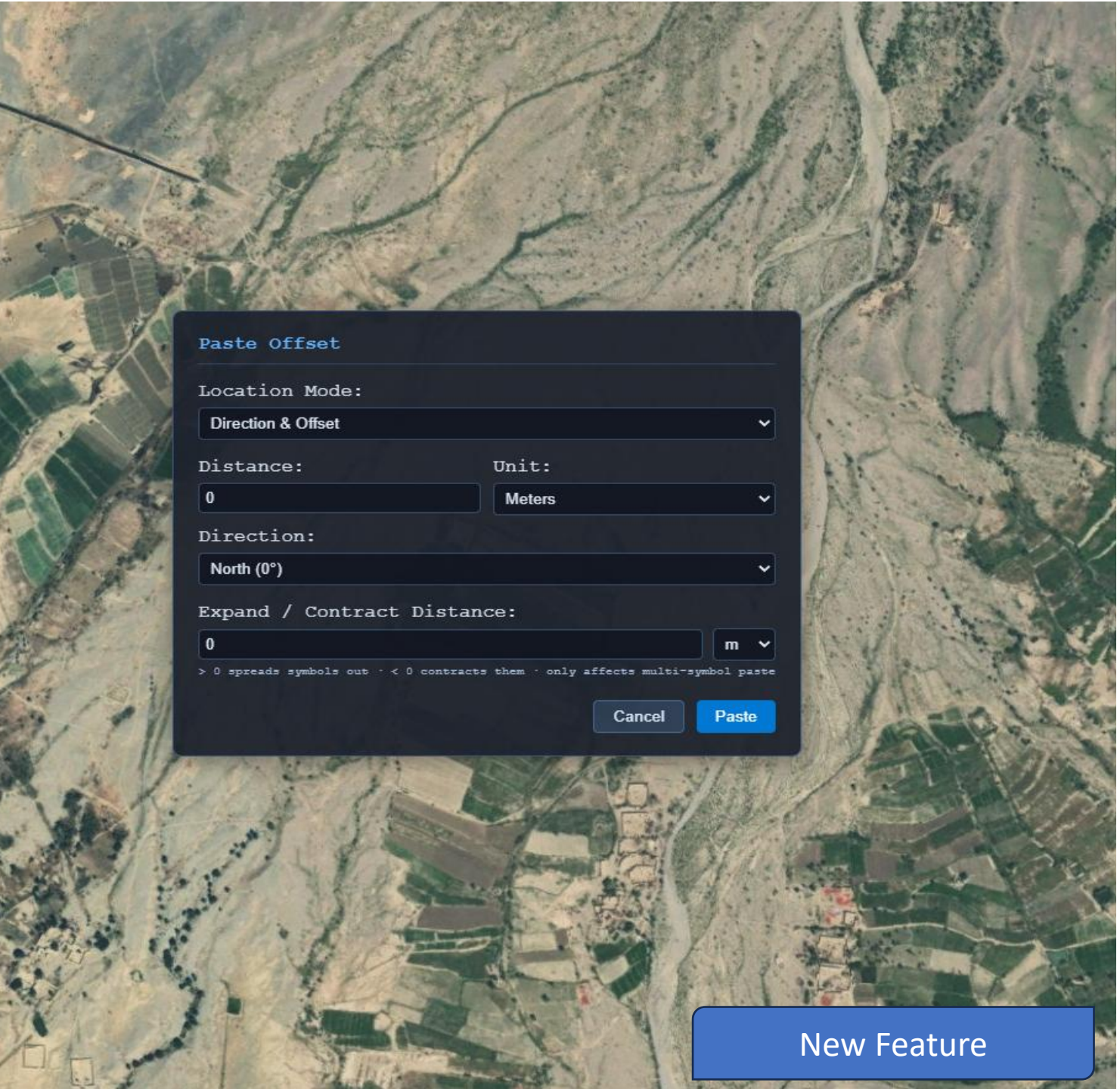

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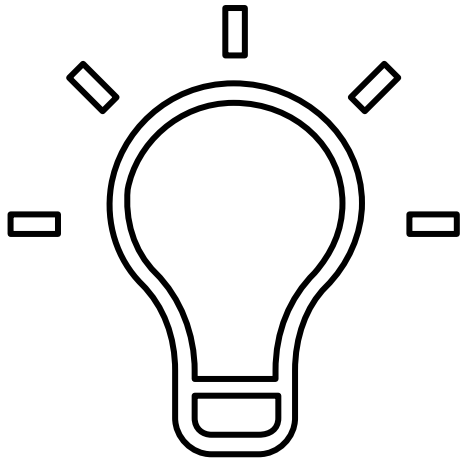

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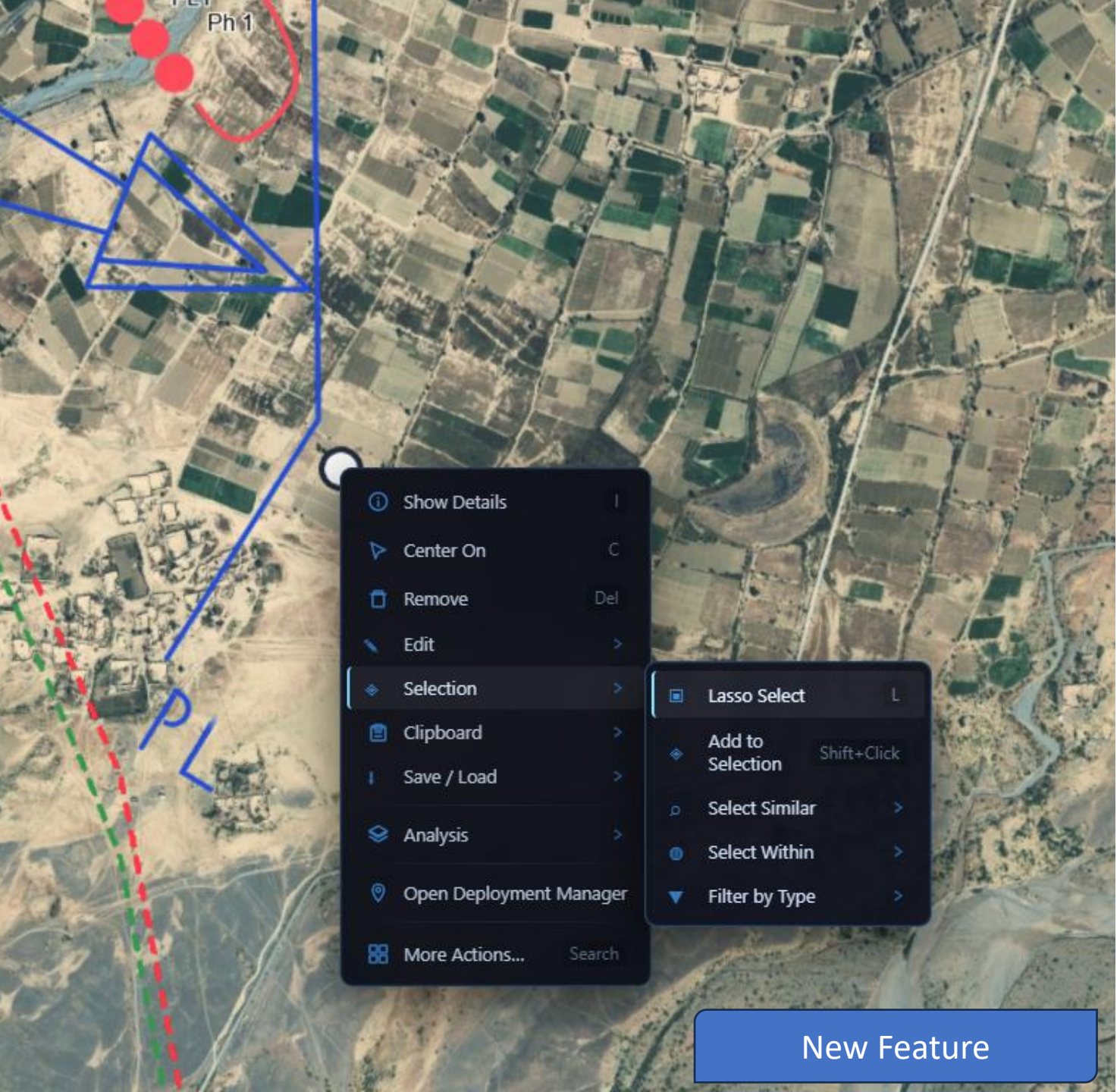
Selection & Alignment



Selection & Alignment

Selection

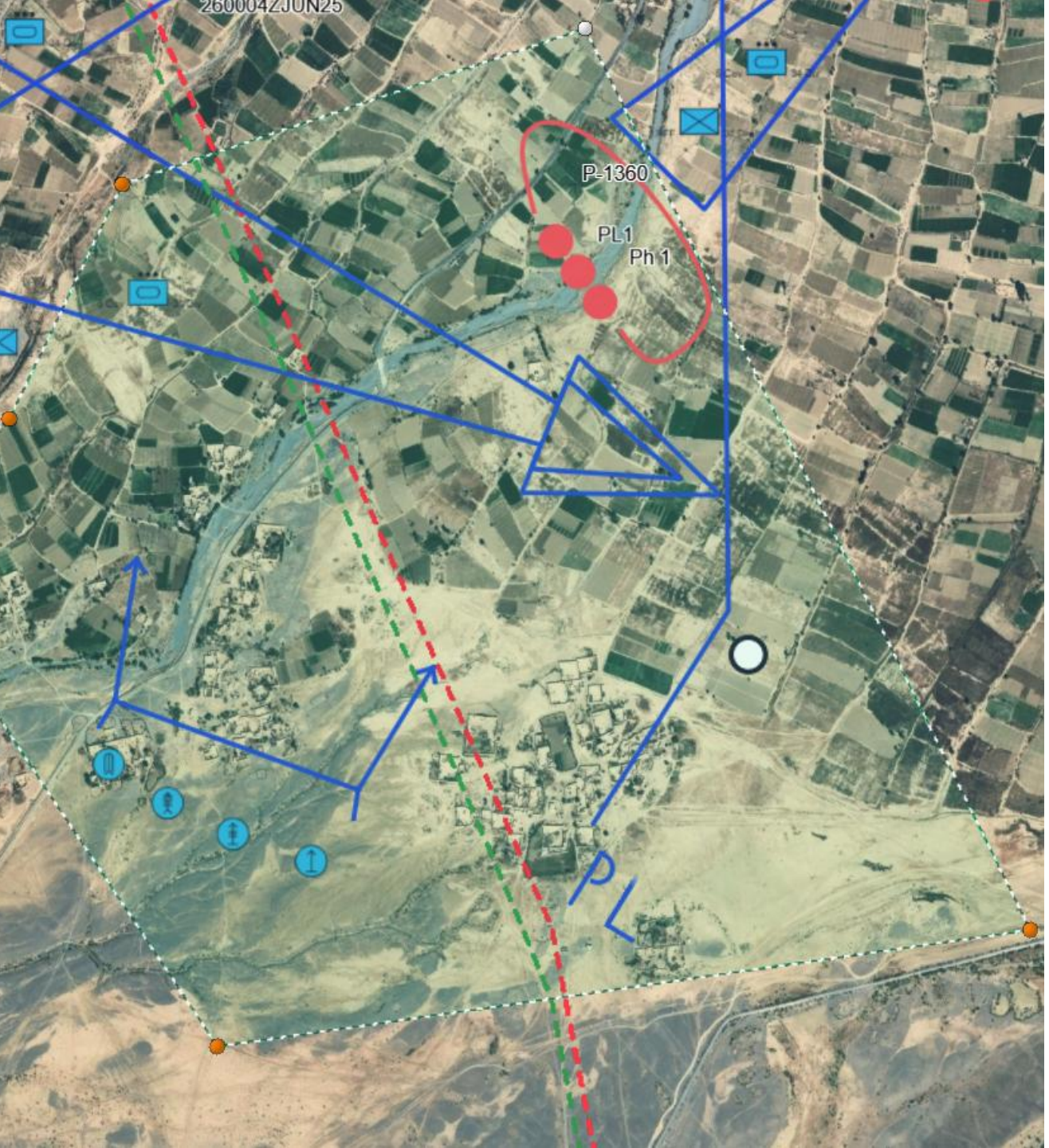
- Lasso: draw polygon to capture multiple symbols
- Select by SIDC code: Same Symbols
- Select by Echelon: (all platoon/company/battalion etc.)
- Select Own / Enemy / Neutral in one operation
- Select by geometry type: Points, Lines or Areas
- Select Within: grab every symbol inside a drawn polygon

Alignment & Distribution

- Batch Align Left / Right / Top / Bottom
- Batch Center on X-axis or Y-axis
- Distribute and Arrange across a selection group

New Feature

Core



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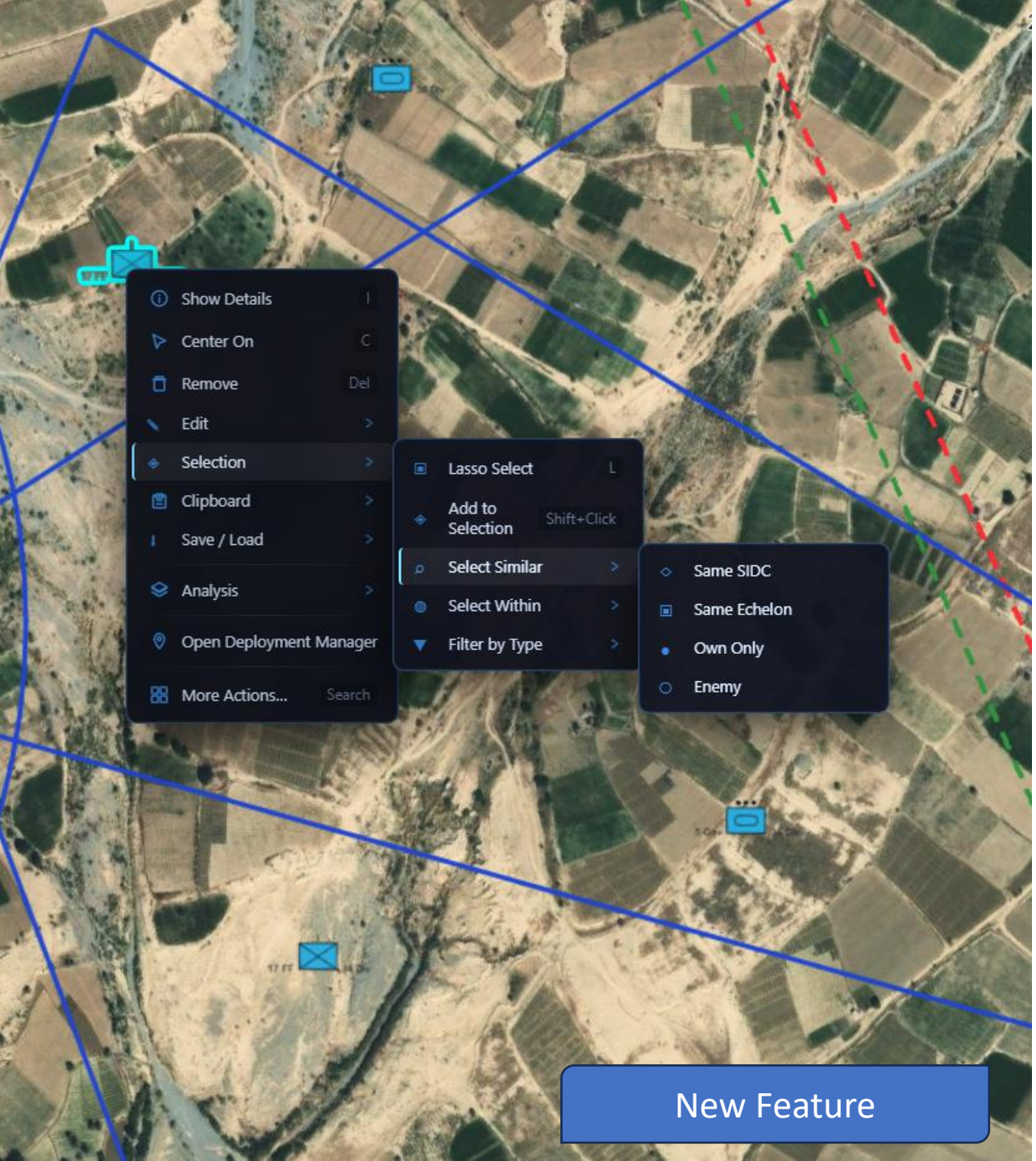

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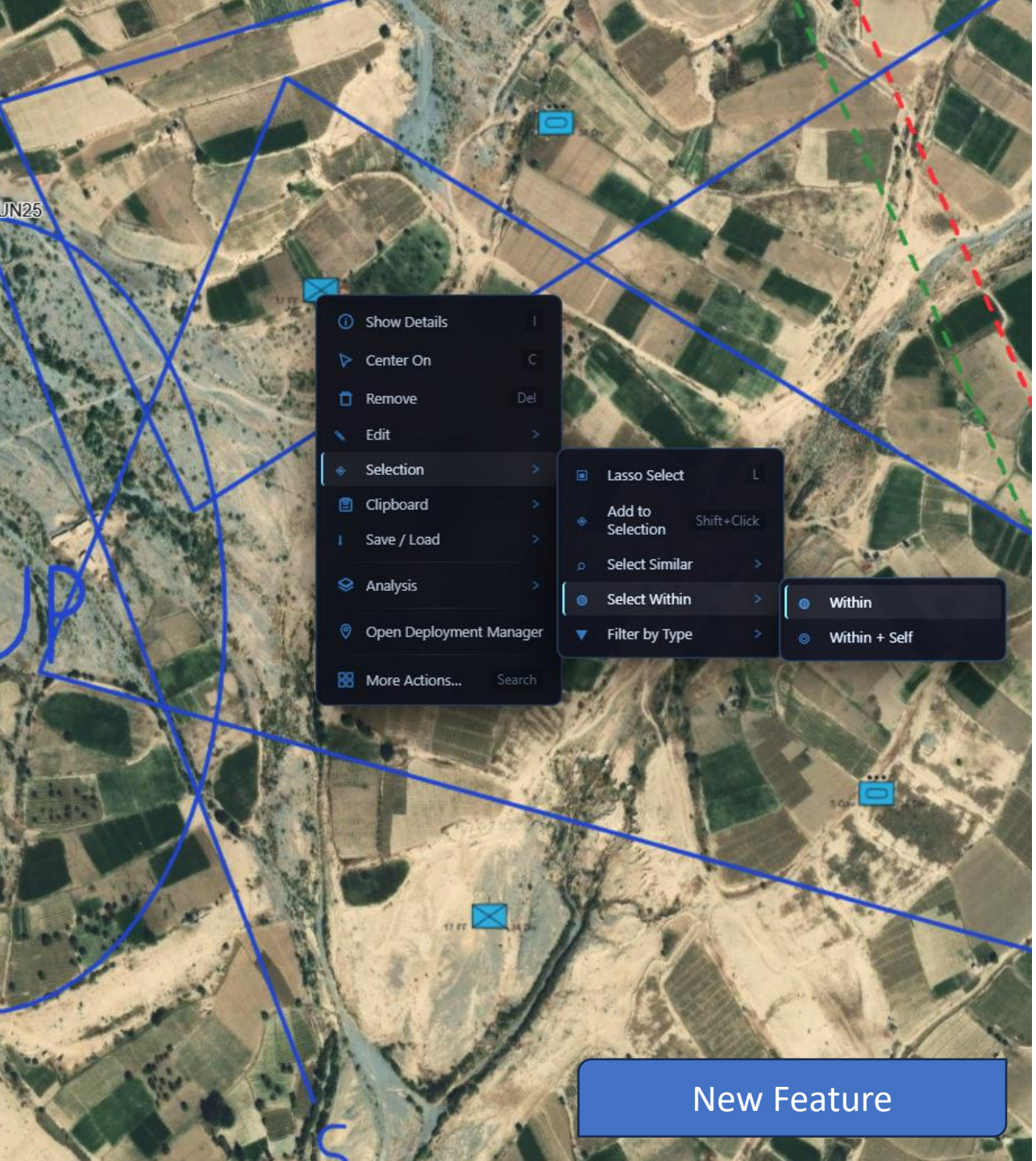

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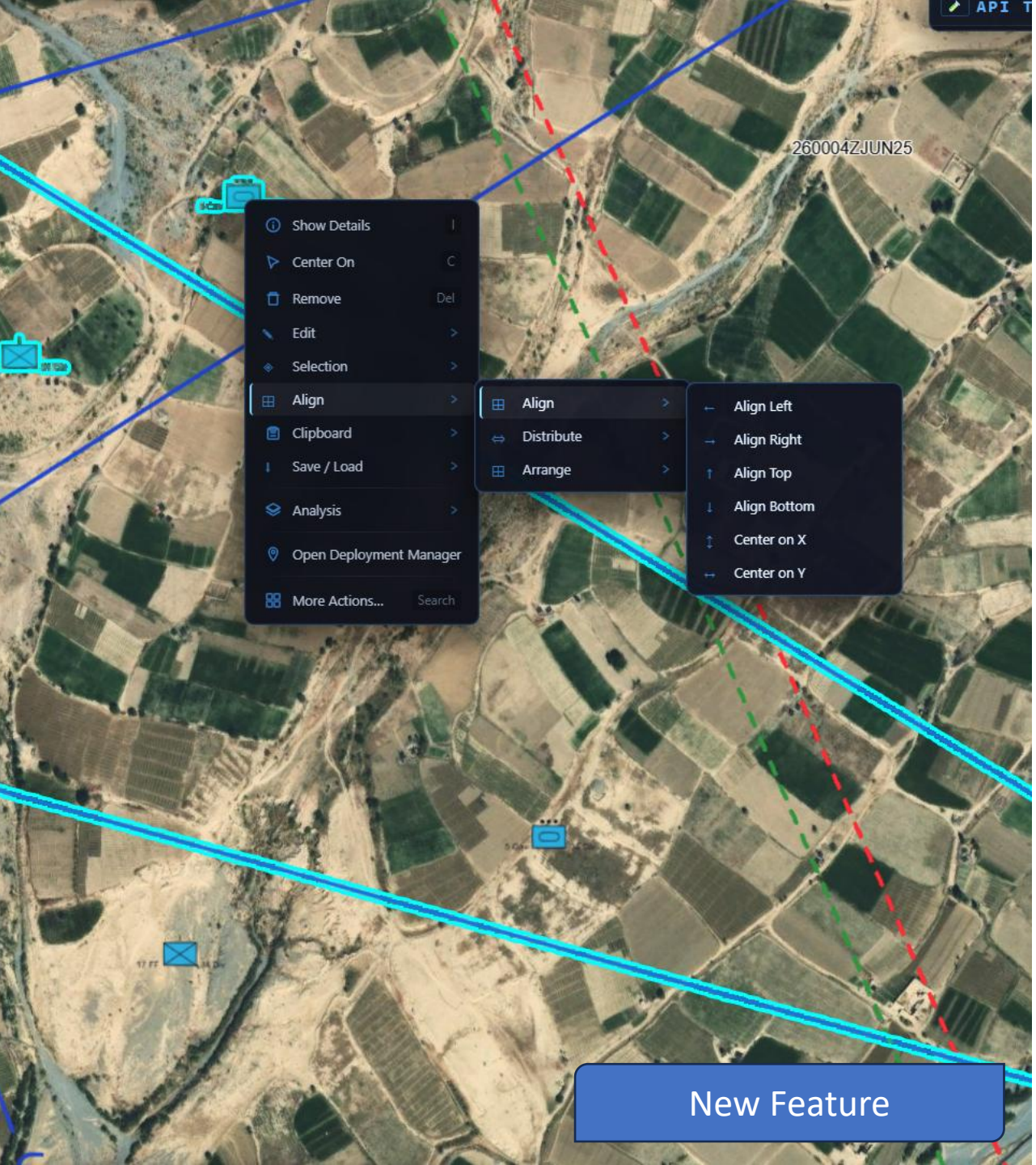

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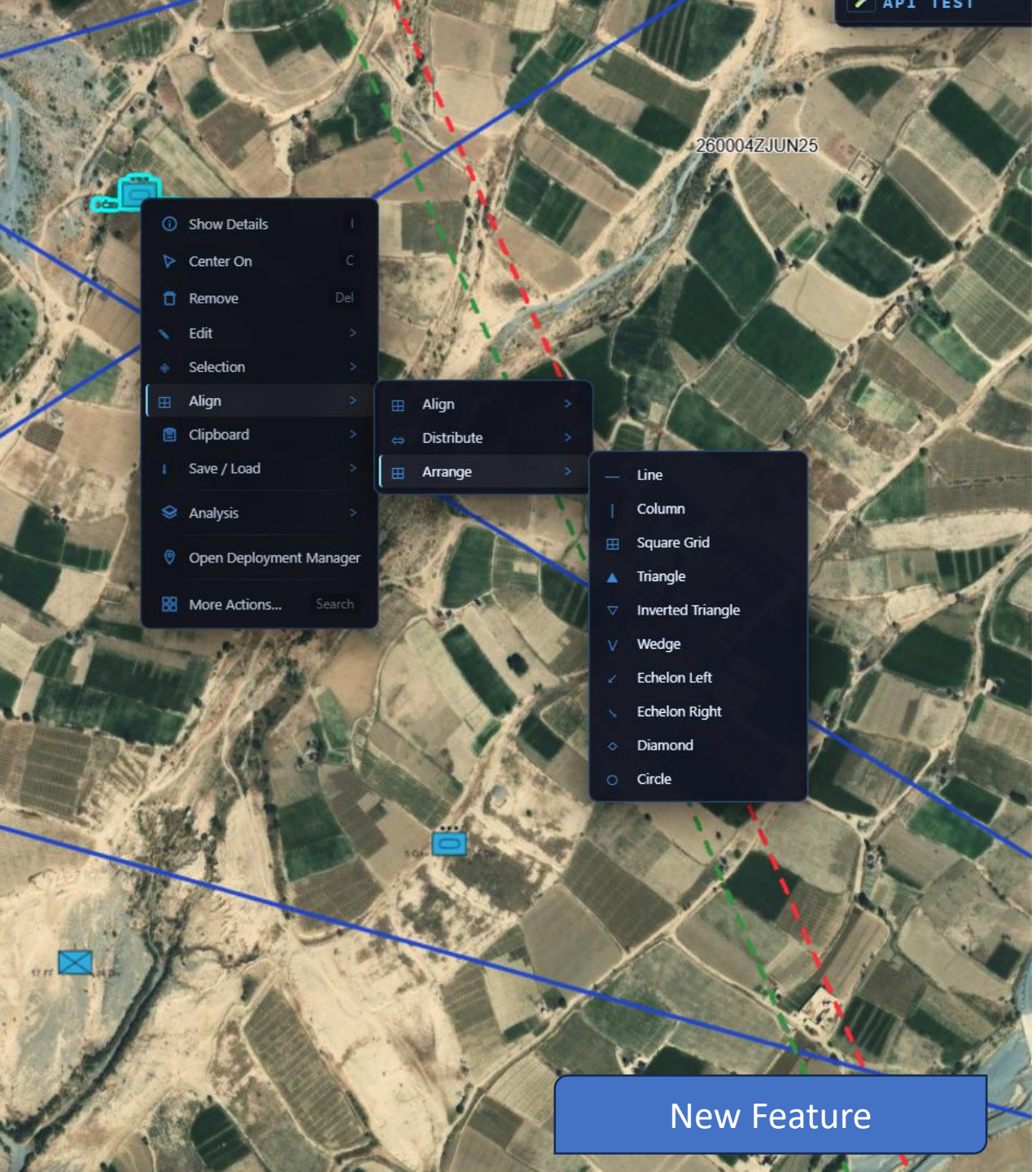

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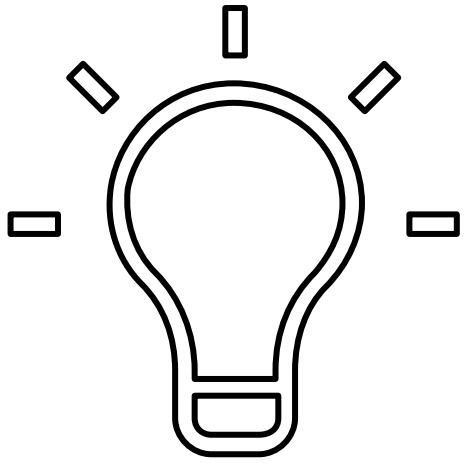

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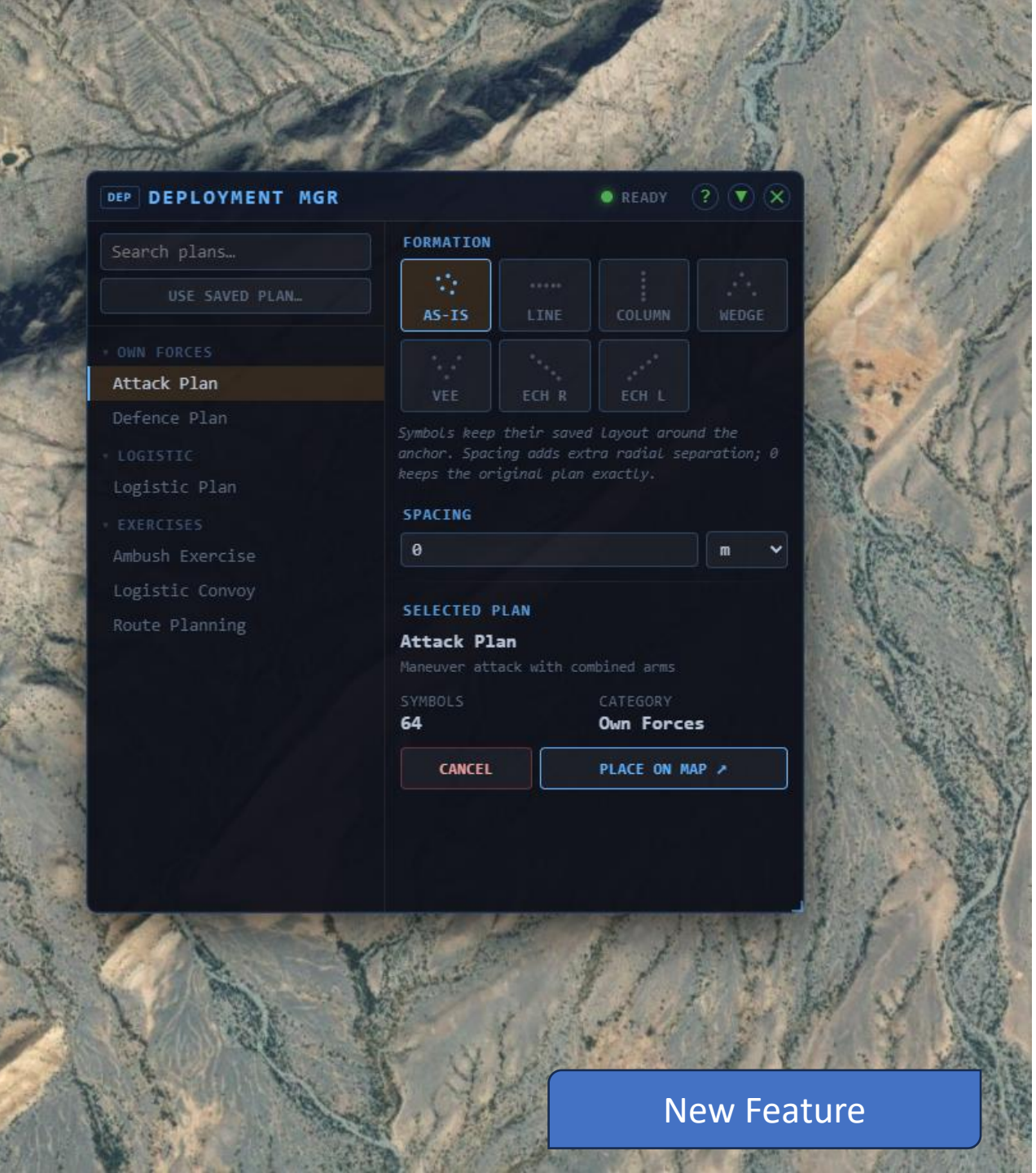
Deployment Manager



New Feature

Deployment Manager

- Singleton engine that deploys pre-built tactical plans onto the map at a user-chosen anchor point and bearing
- Works with both 2D MapView and 3D SceneView; auto-resets on view switch

Formation System

- 7 formation types: As-Is, Line, Column, Wedge, Vee, Echelon-Right, Echelon-Left
- Each formation defined by [lateral, forward] slot offsets relative to the anchor
- Spacing in m / km / mi applied uniformly across all slots
- Ghost Preview

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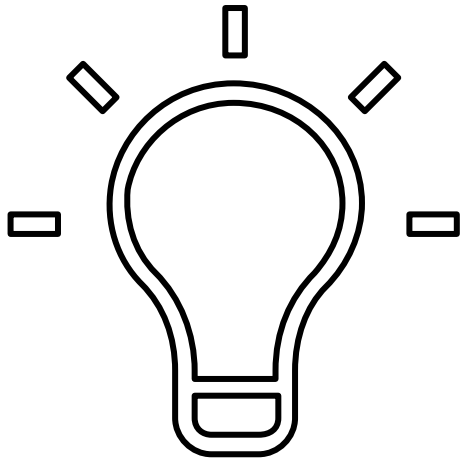

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Control Point Editing

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- Drag individual control points to reshape lines, areas, and tactical graphics in real-time
- Move, Scale, and Rotate handles available as a separate edit mode
- Full undo/redo support — every reshape is a reversible stack entry
- Switches cleanly between 2D and 3D views without losing edit state

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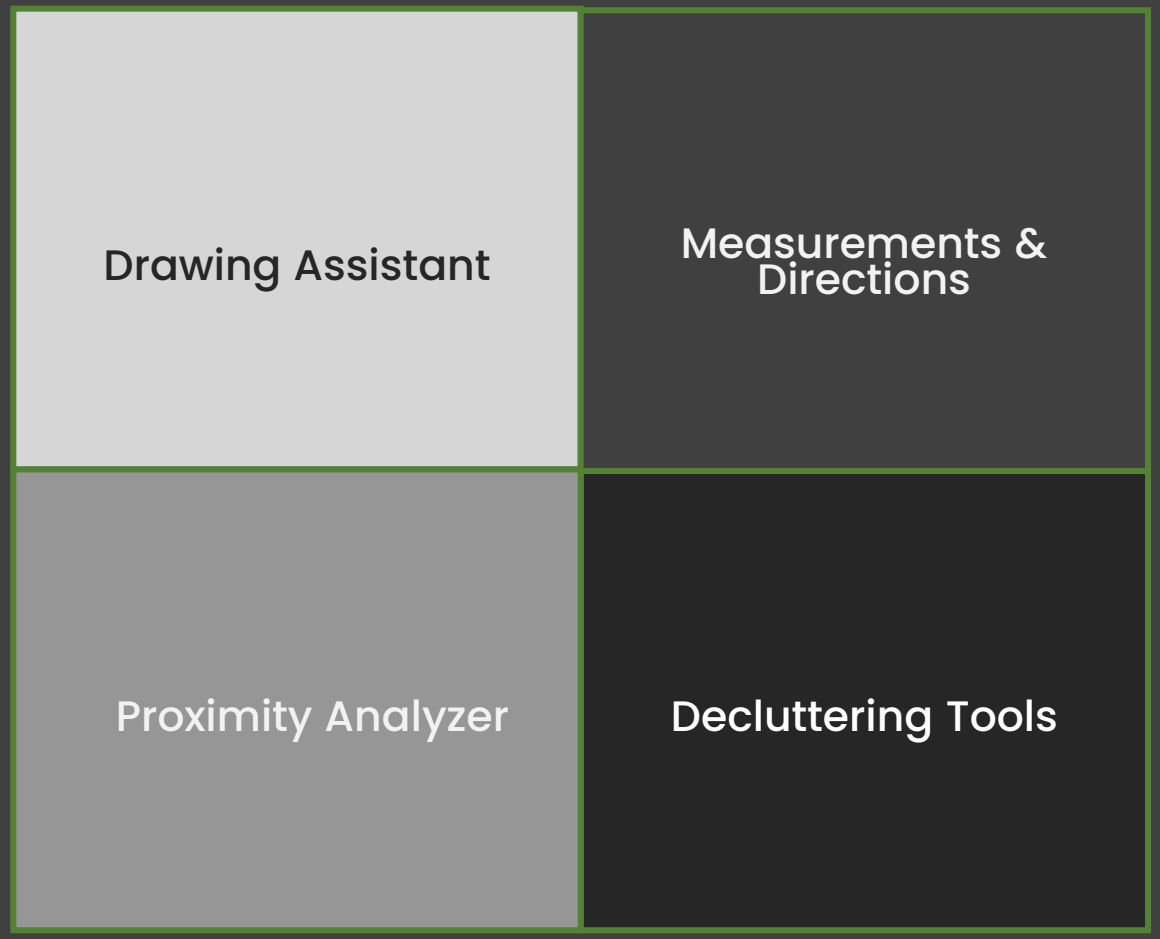

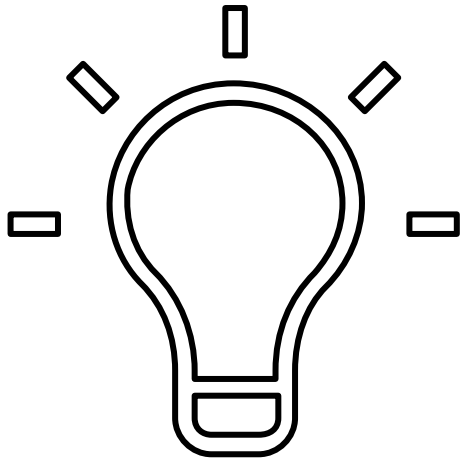

Visualization & Decluttering

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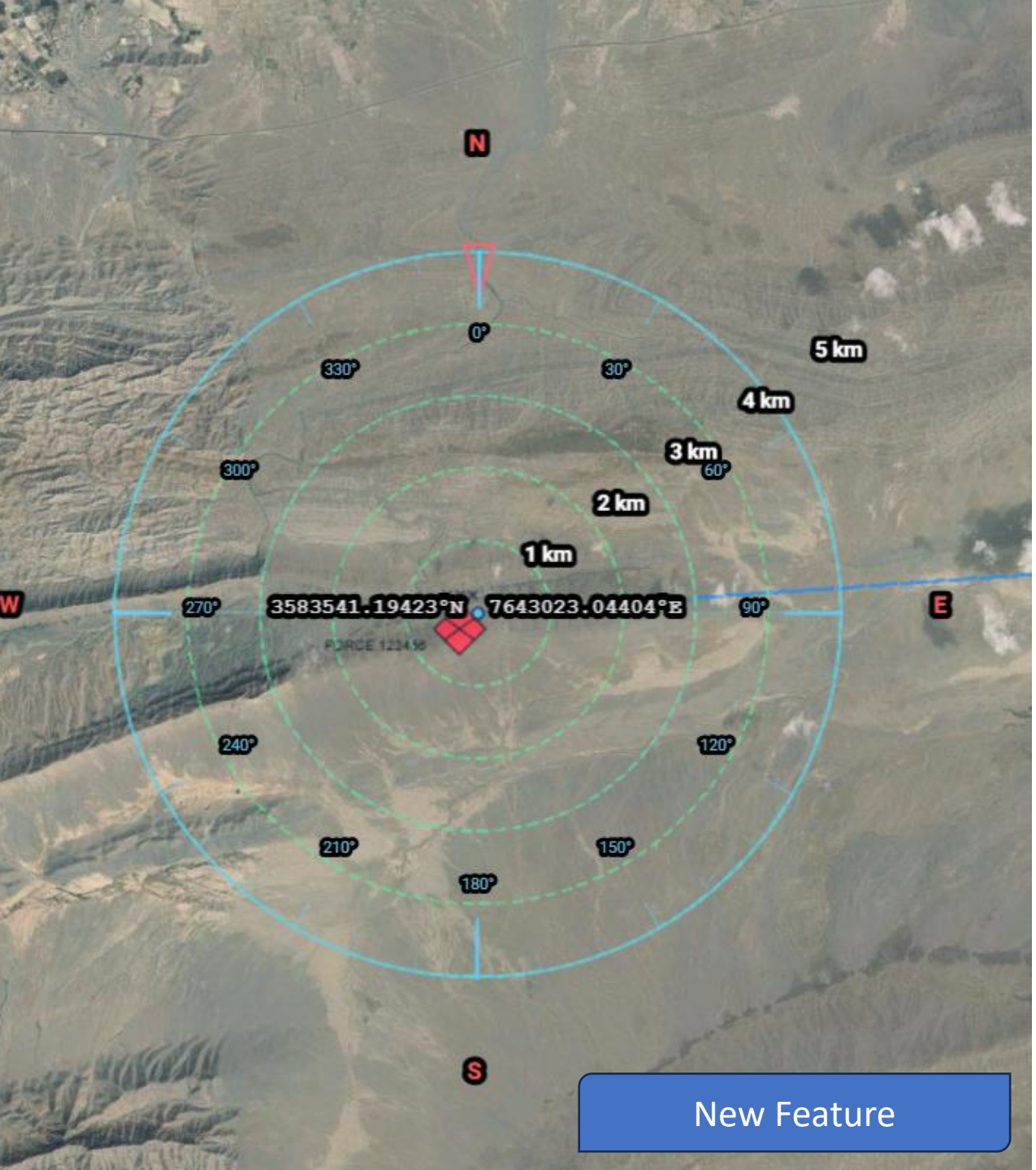


Drawing Assistant

Distance Rings

- Adaptive Incremental Rings

Nearby Symbol

- Color-codes by distance: Red (near) → Amber (mid) → Green (far)

Magnetic Compass

- Placeable interactive compass overlays on the map

Bearing Arc

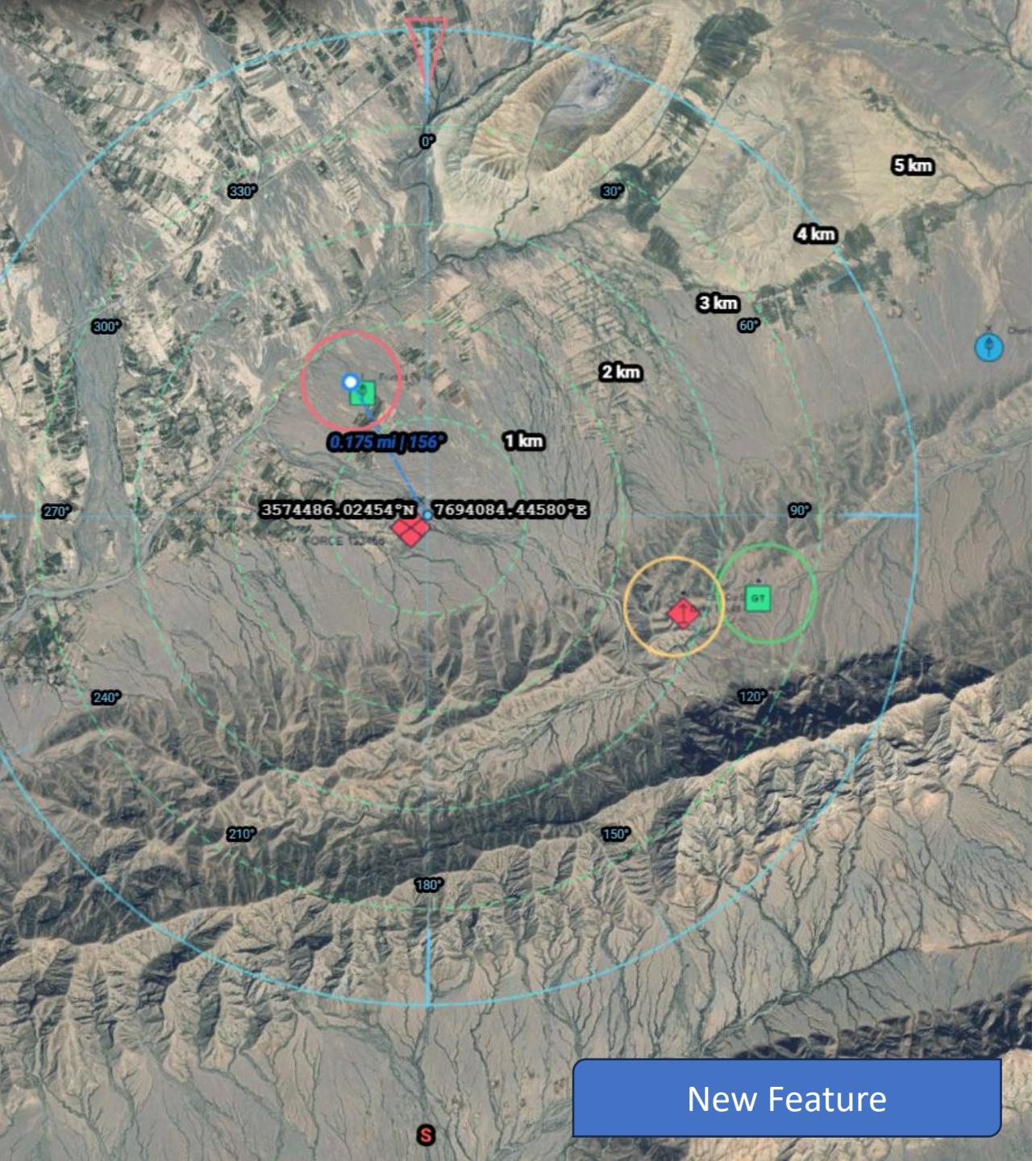
- Full circular protractor centered on last control point

Coordinate Display

- Floating lat/lon label that follows the cursor

Rubber Band

- Live dashed line from last control point to cursor



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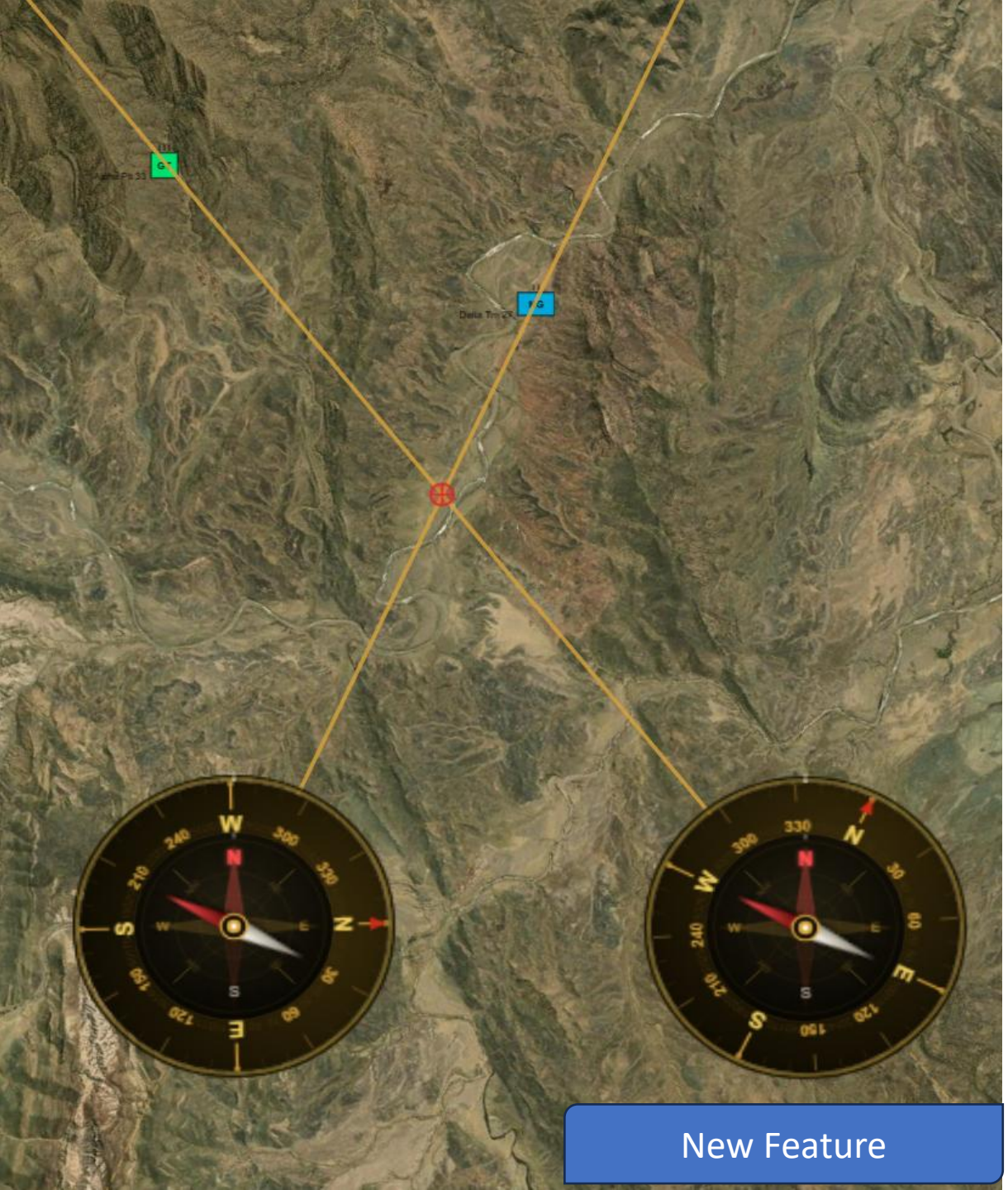

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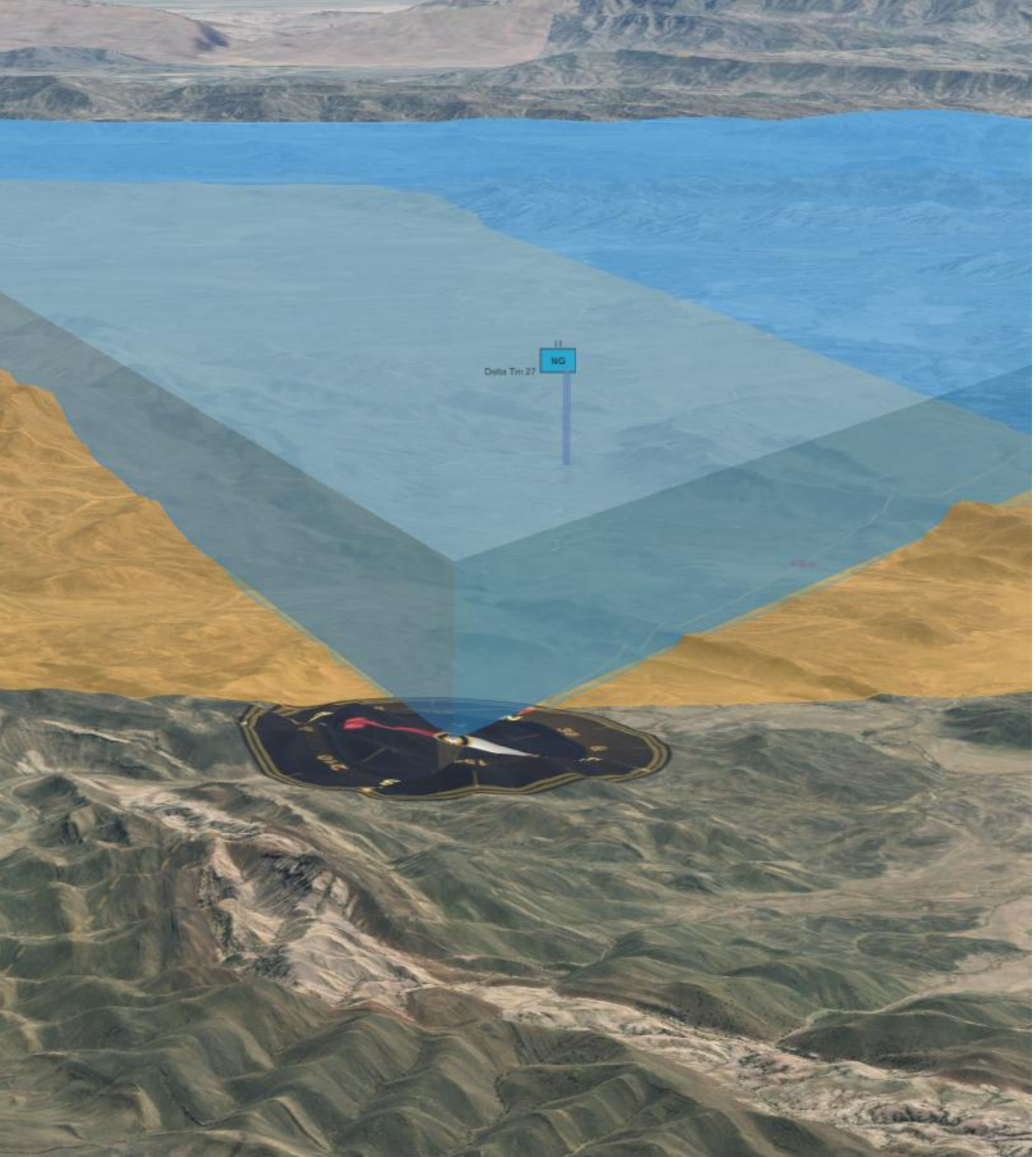

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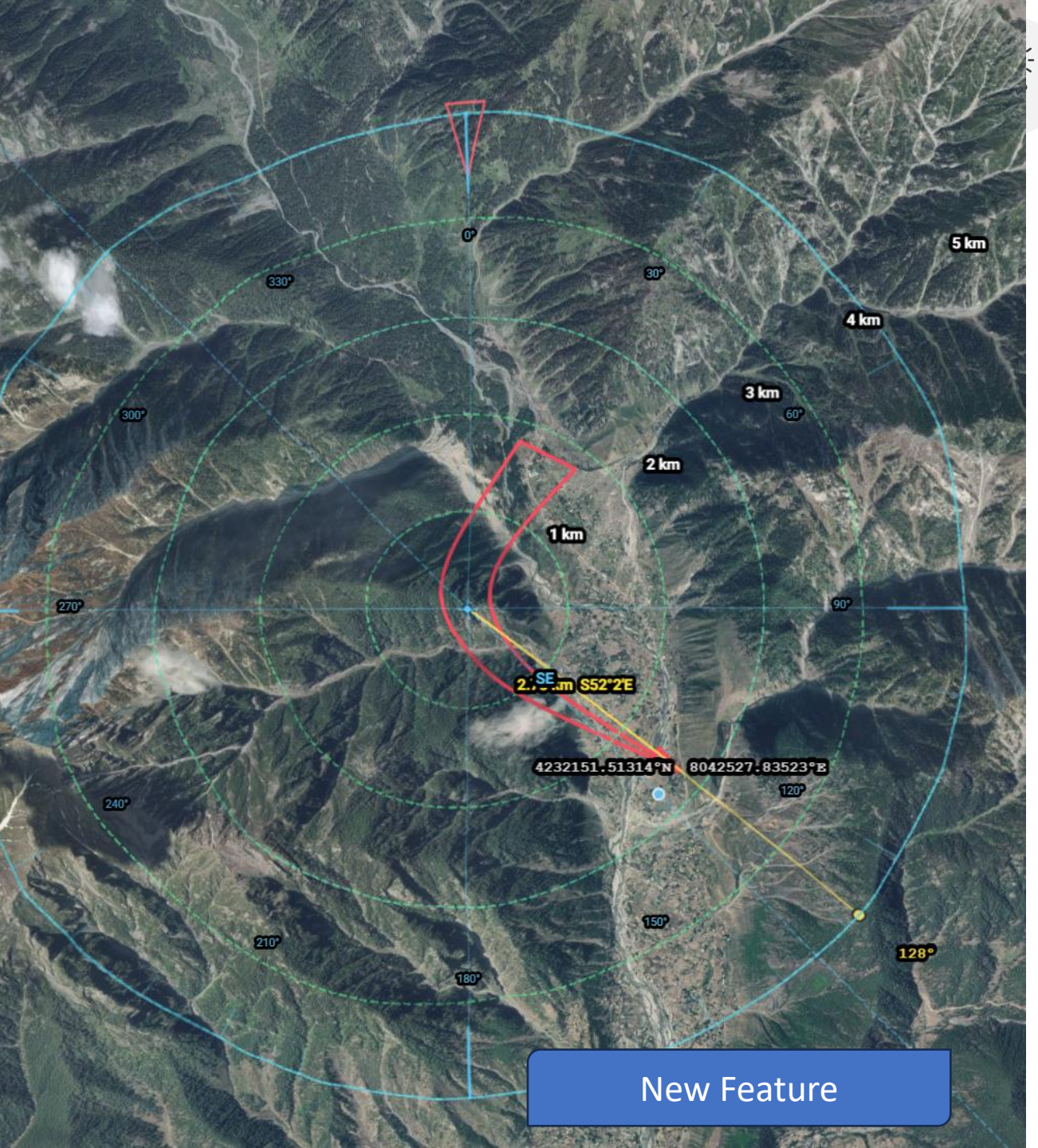

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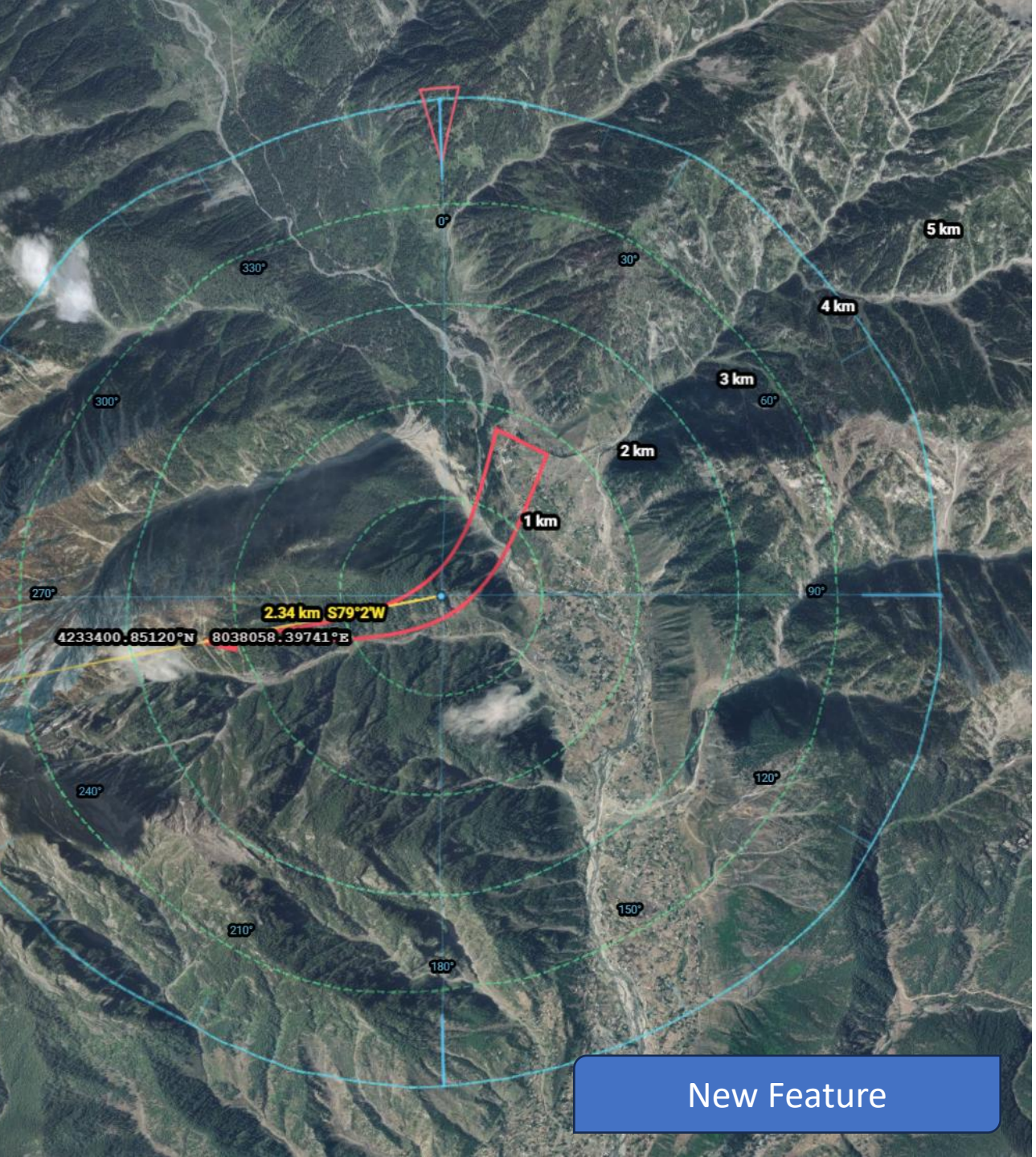

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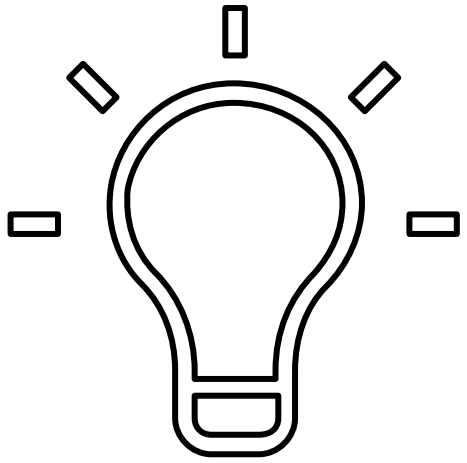

Visualization & Decluttering

Measurements & Directions

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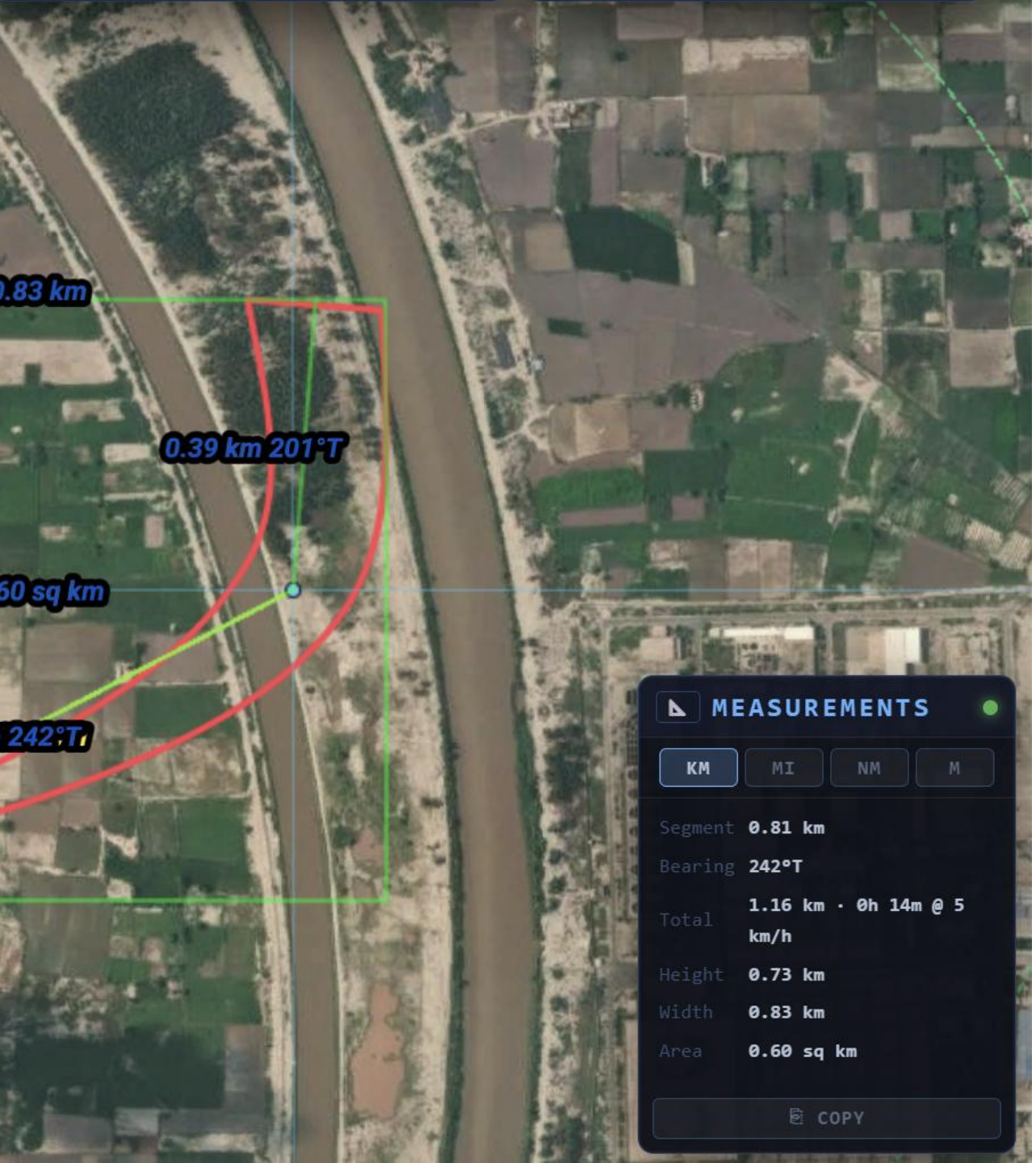
Measurement Tools while drawing



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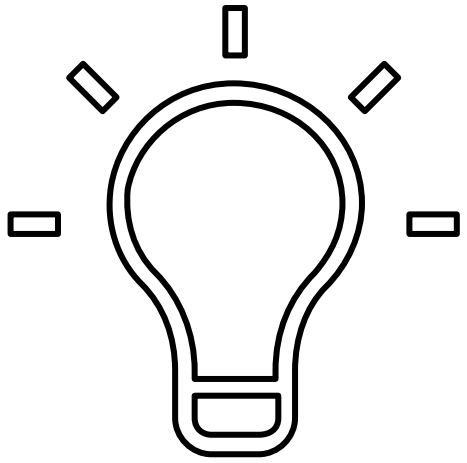

Visualization & Decluttering

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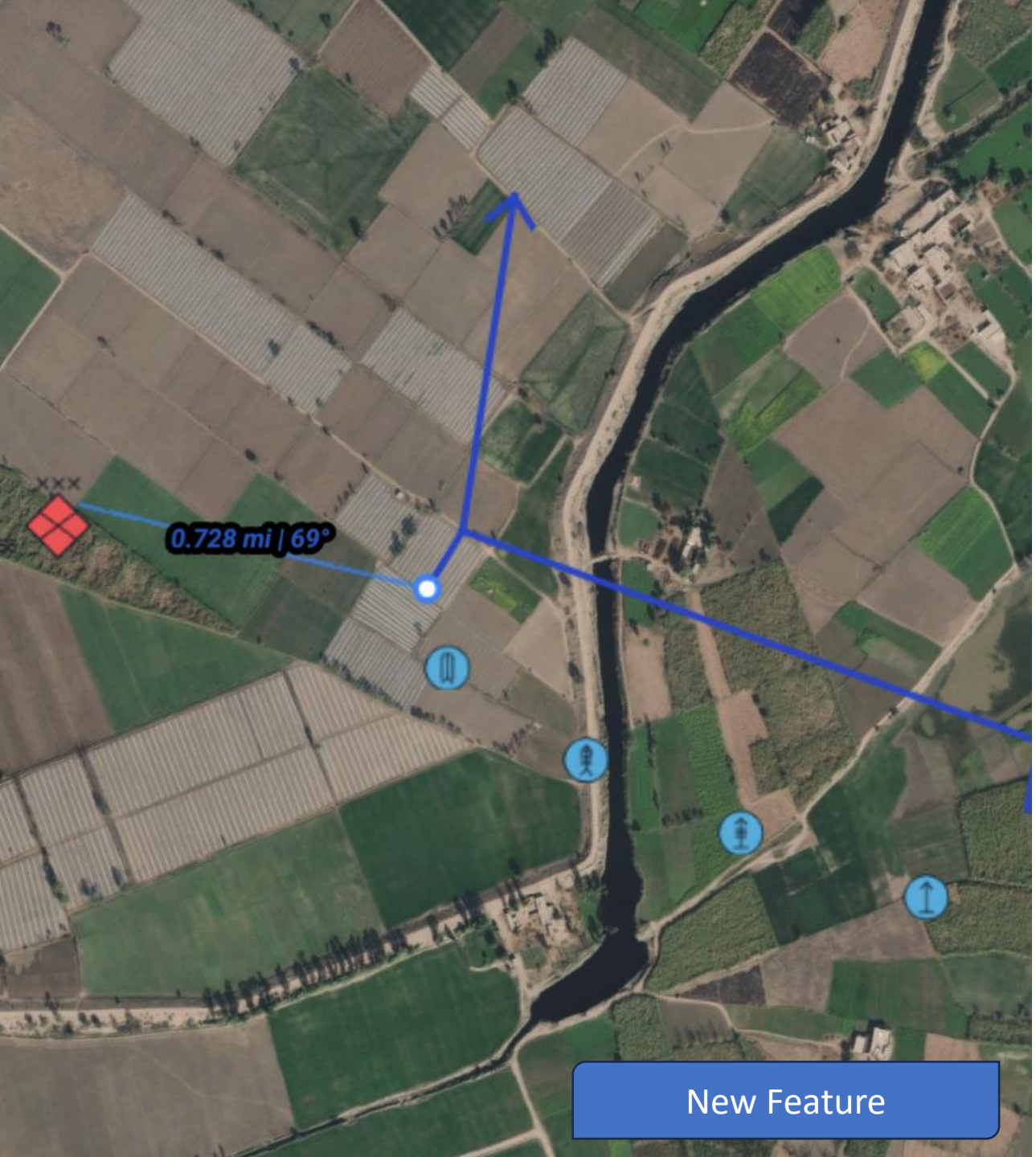


Proximity Analyzer

Precision Placement

- Snaps to nearby symbols automatically while drawing.

Live Feedback

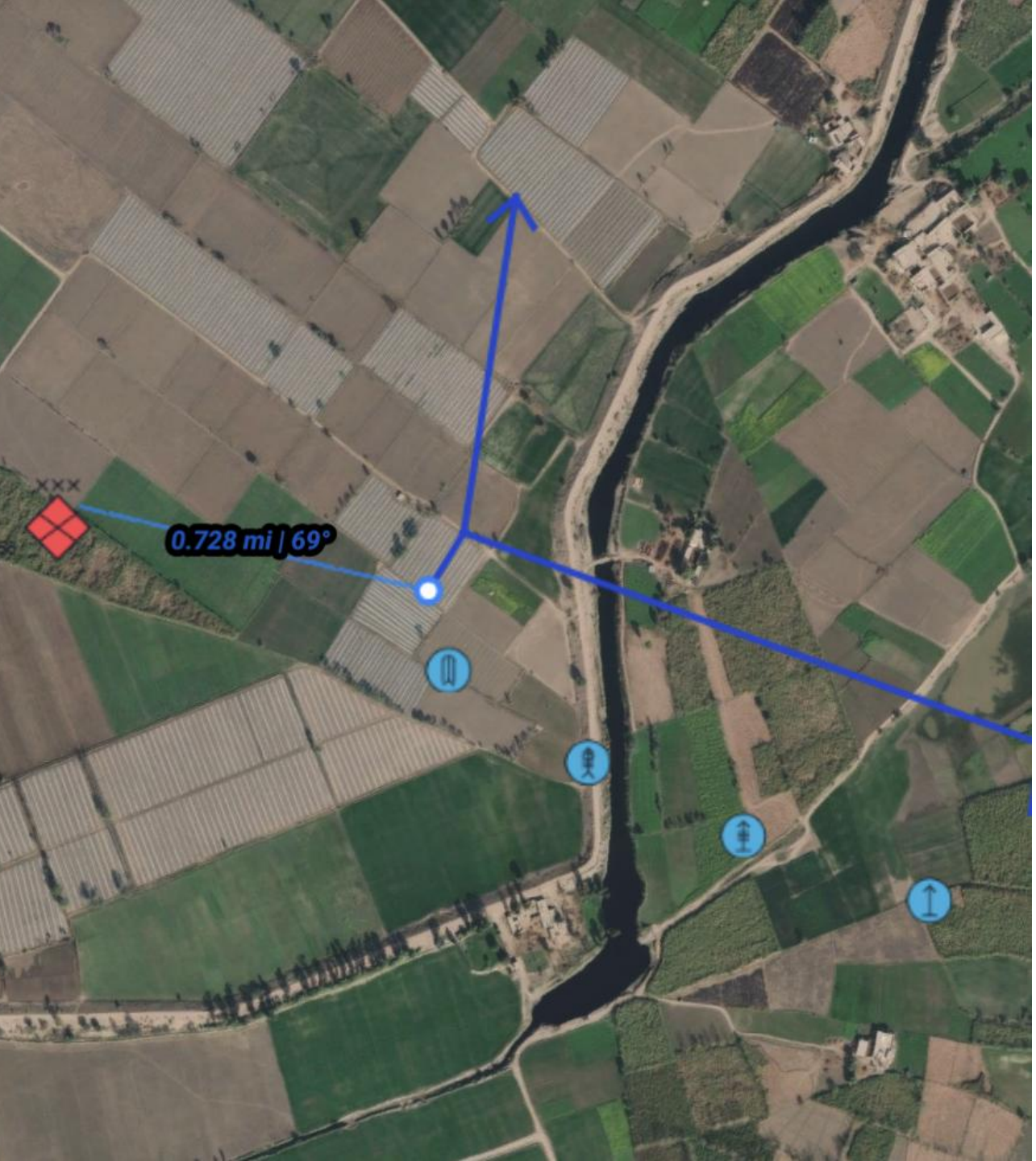
- Shows distance to the nearest symbol in real time.

Non-Disruptive

- Runs silently in the background, toggle on/off anytime.

Operator Control

- Hold Alt to bypass snap instantly.



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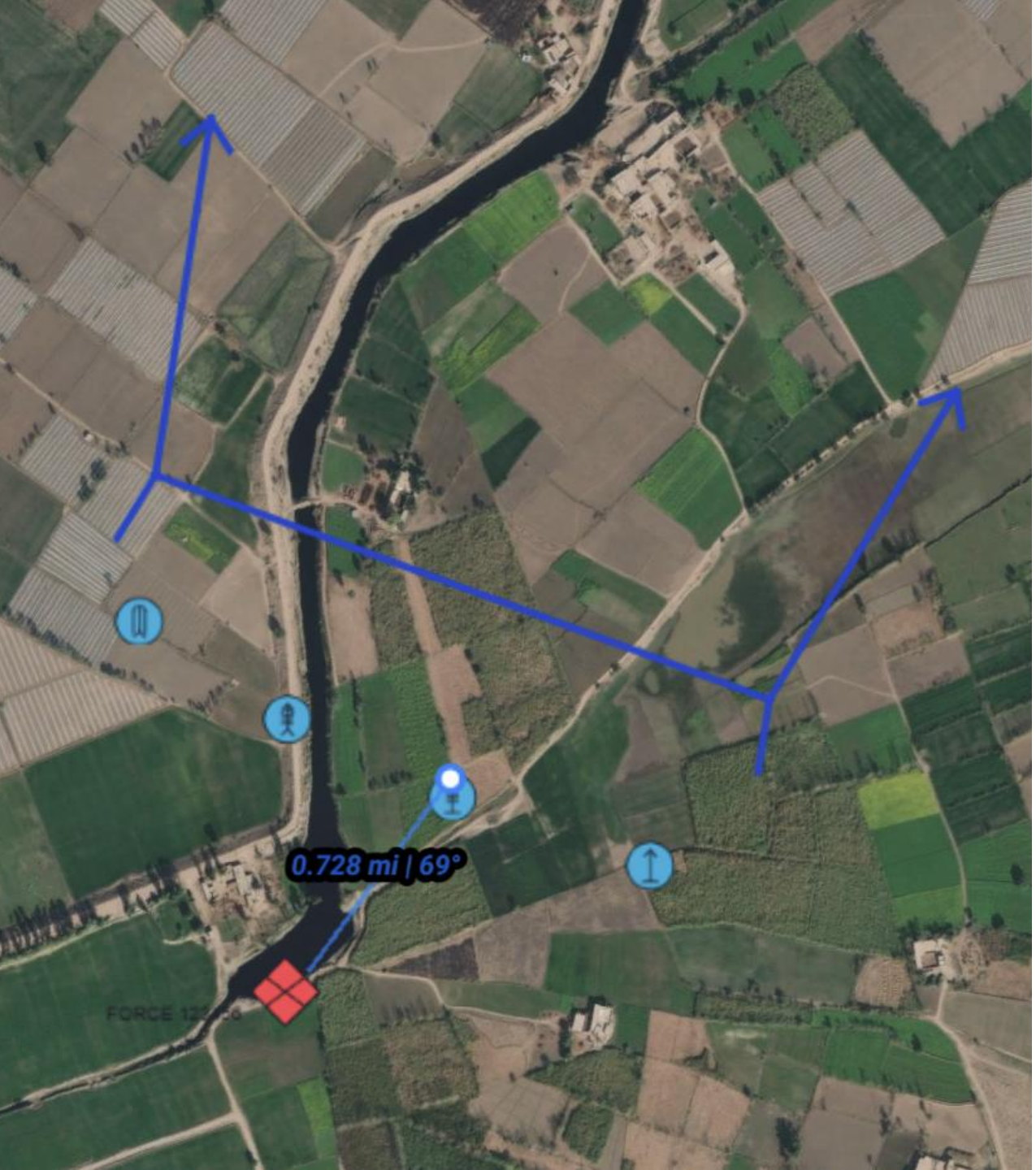

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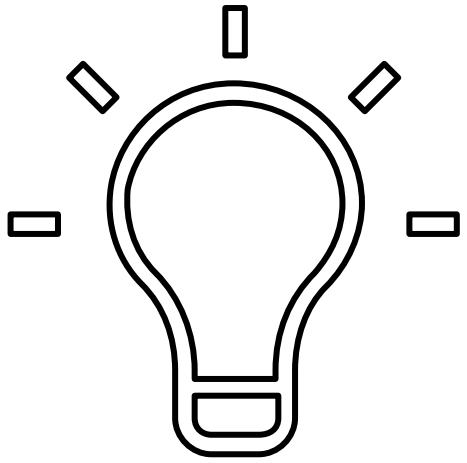

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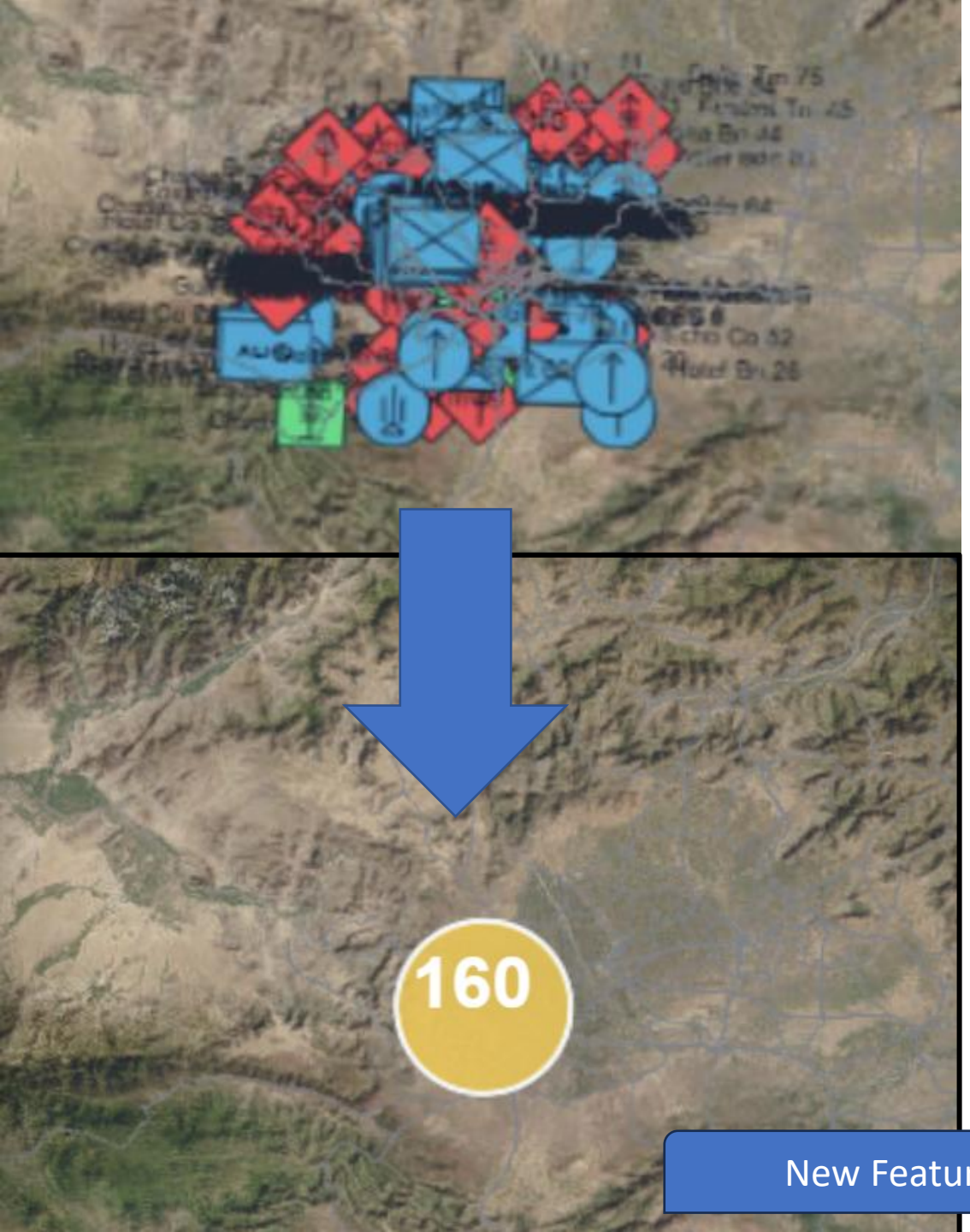
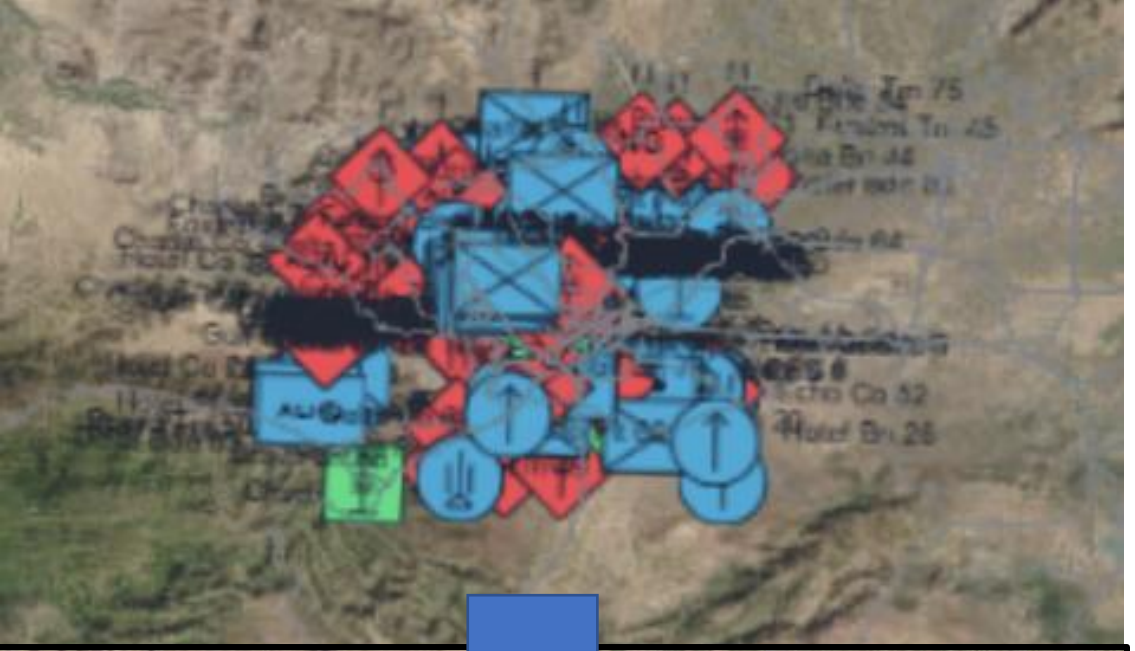
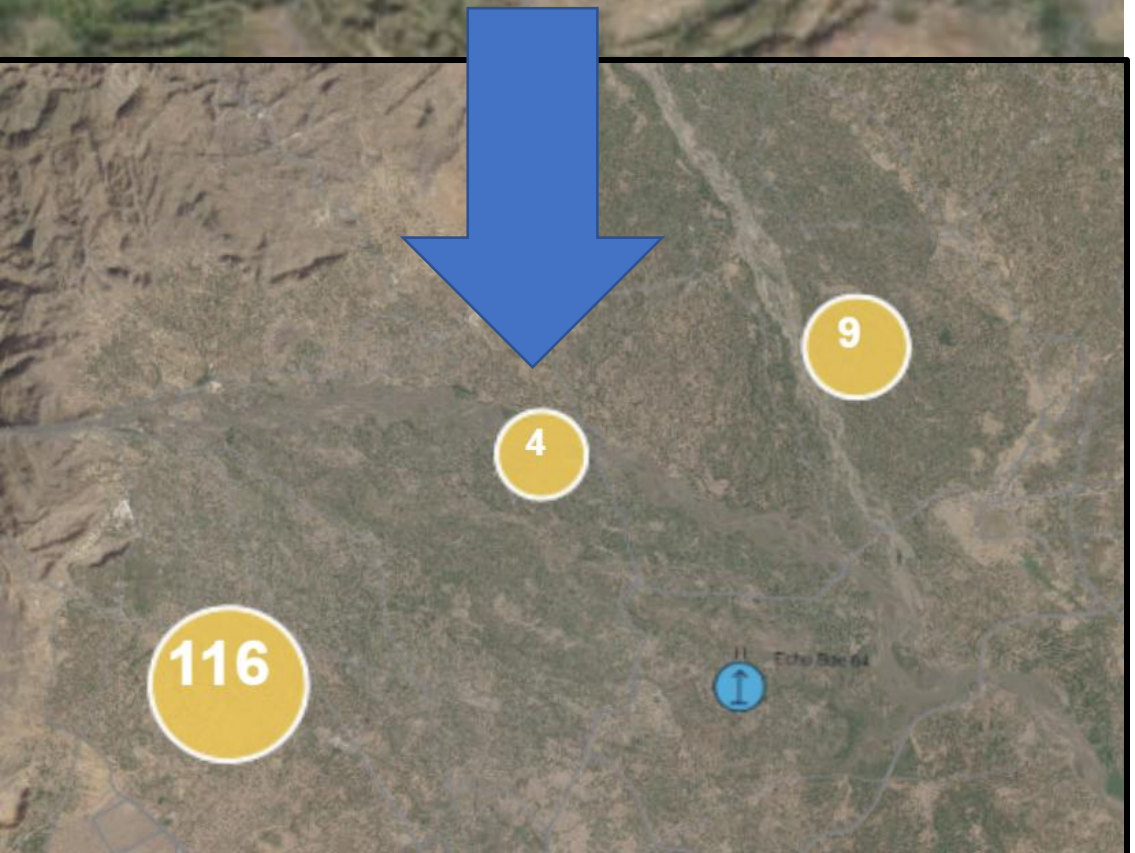


Decluttering Tools

- **Cluster Badges**
  Groups nearby symbols into a single count badge at low zoom to reduces visual noise in dense scenes.
- **Disperse Markers**
  Fans out symbols stacked at the same coordinate into a radial spread so each one is individually visible.
- **Ladder (Flag Halyard)**
  Gathers same-identity symbols into a vertical stack connected by a spine line, sorted by echelon like flags on a halyard.
- **Echelon-Based Filtering**
  Shows or hides symbols based on echelon and zoom level
- **Label Placement**
  Intelligently positions labels around symbols using an 8-position algorithm.





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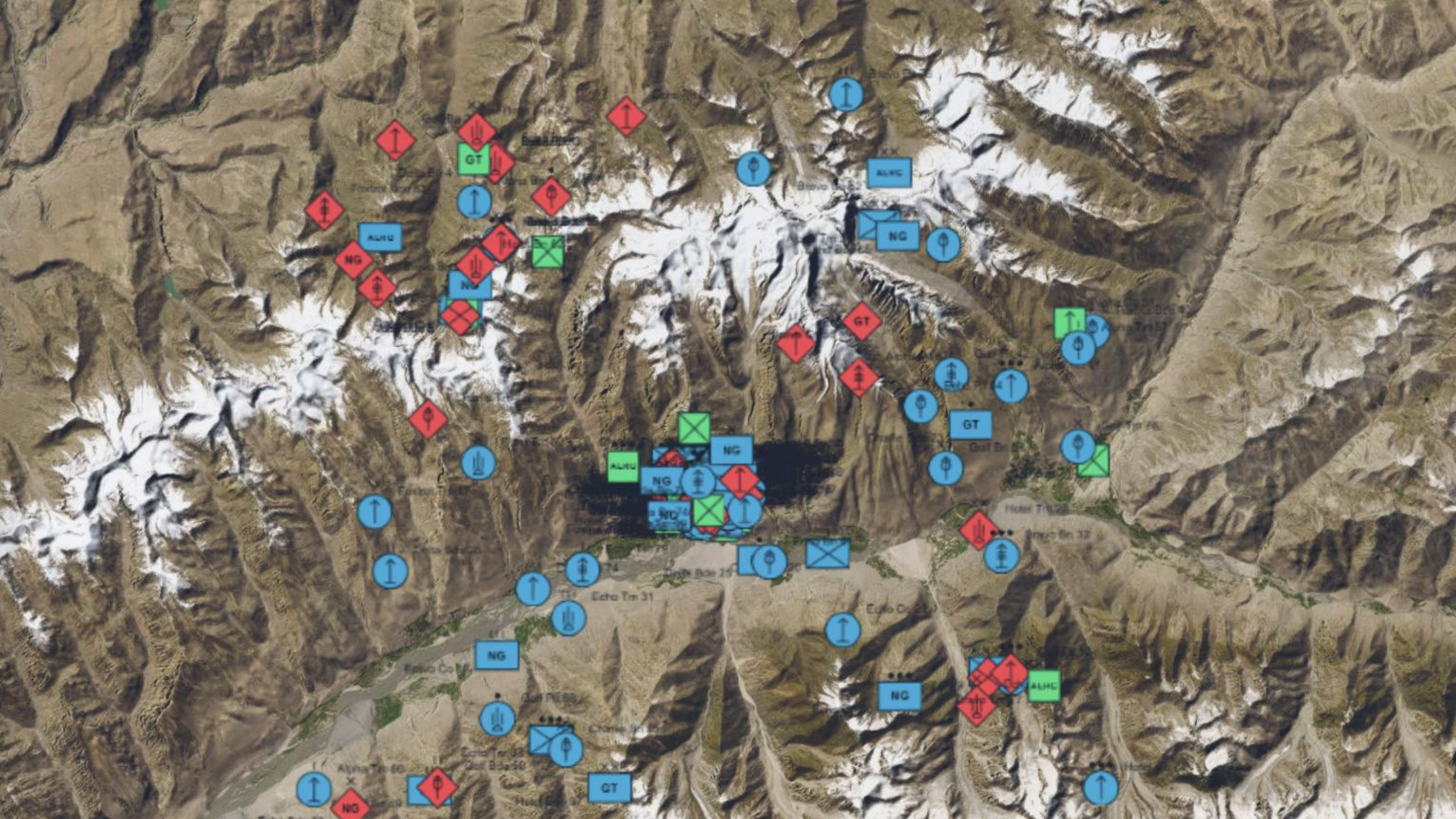
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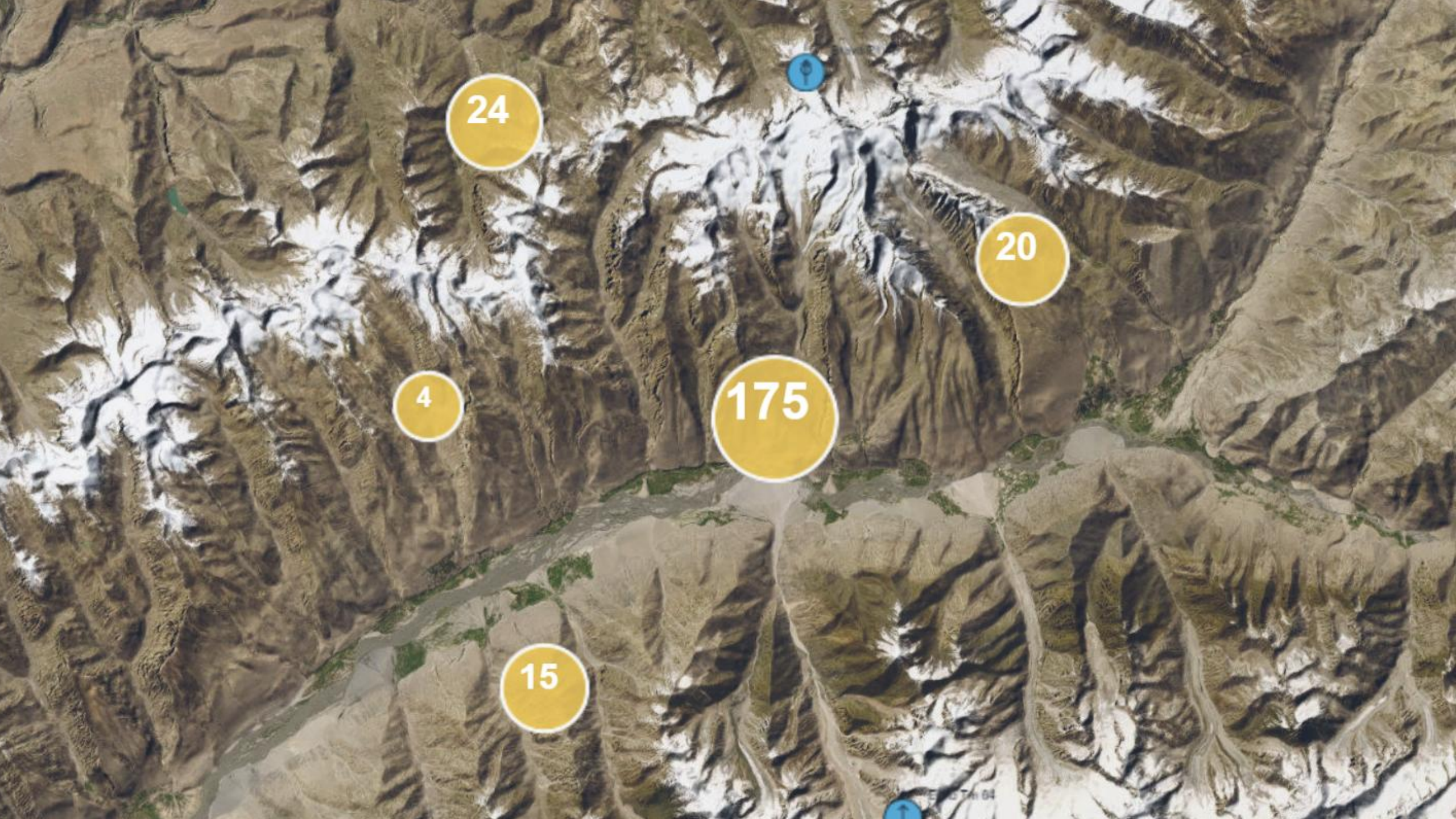
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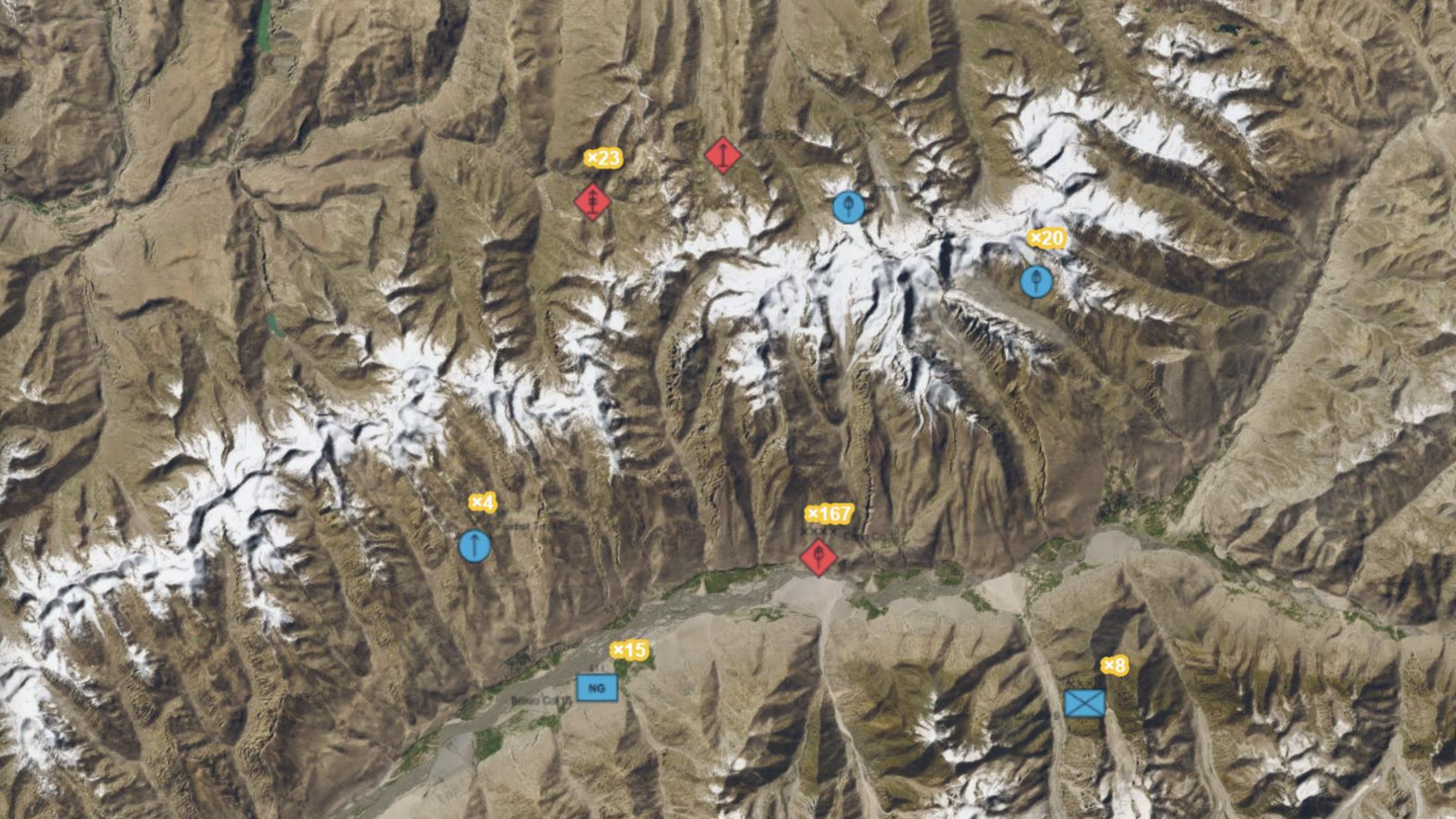
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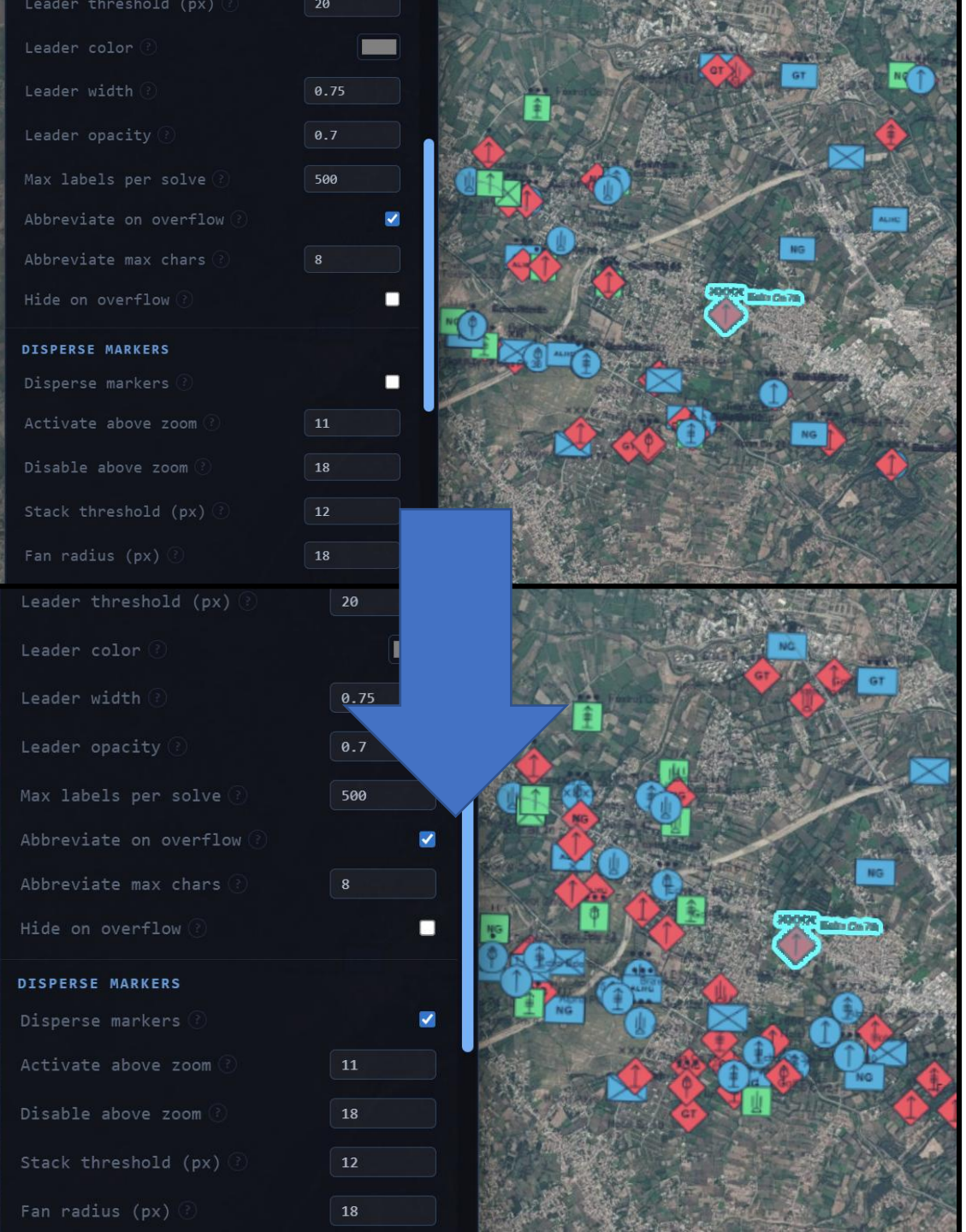

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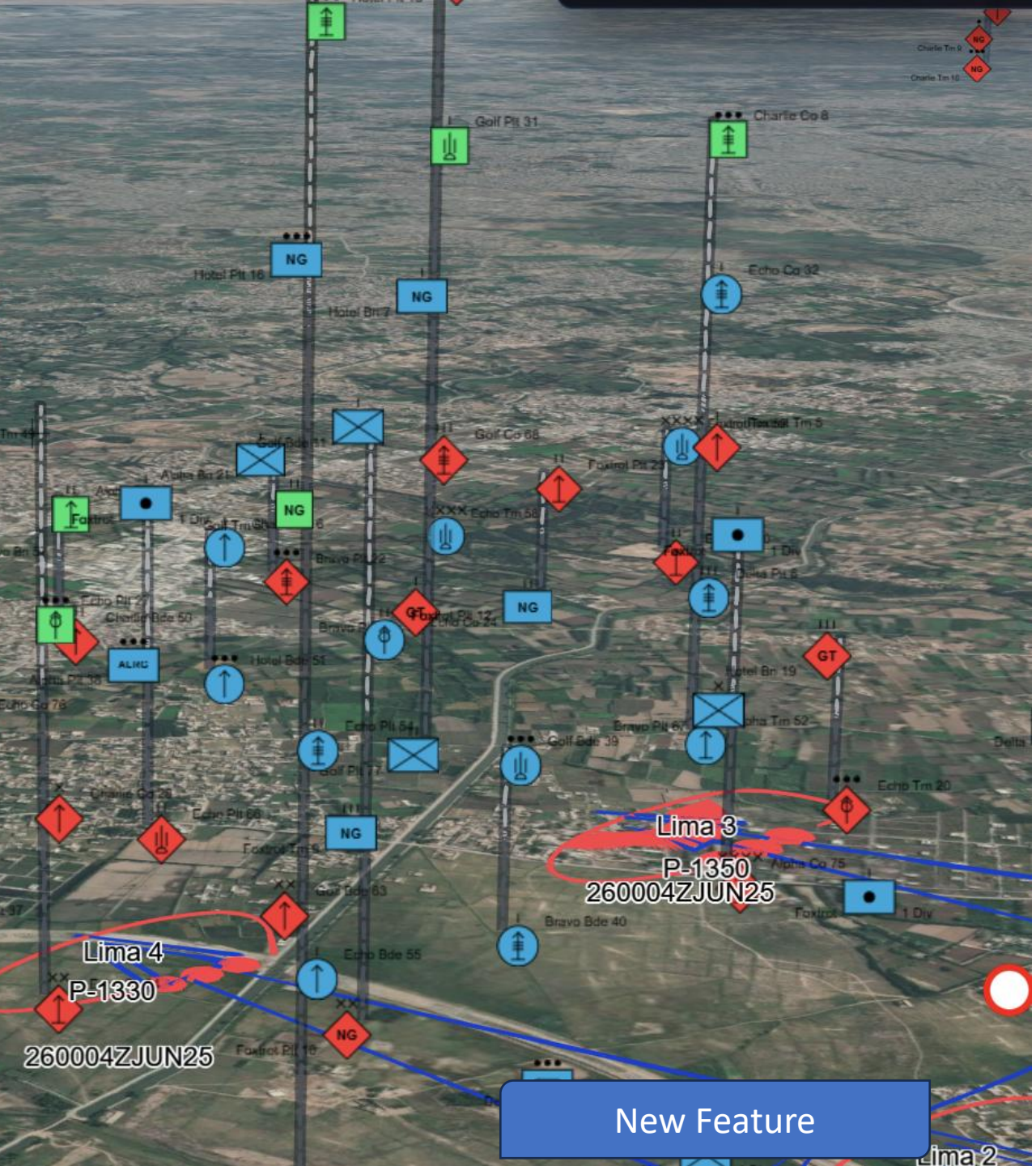
  Fans out symbols stacked at the same coordinate into a radial spread so each one is individually visible.
- **Ladder (Flag Halyard)**
  Gathers same-identity symbols into a vertical stack connected by a spine line, sorted by echelon like flags on a halyard.
- **Echelon-Based Filtering**
  Shows or hides symbols based on echelon and zoom level
- **Label Placement**
  Intelligently positions labels around symbols using an 8-position algorithm.



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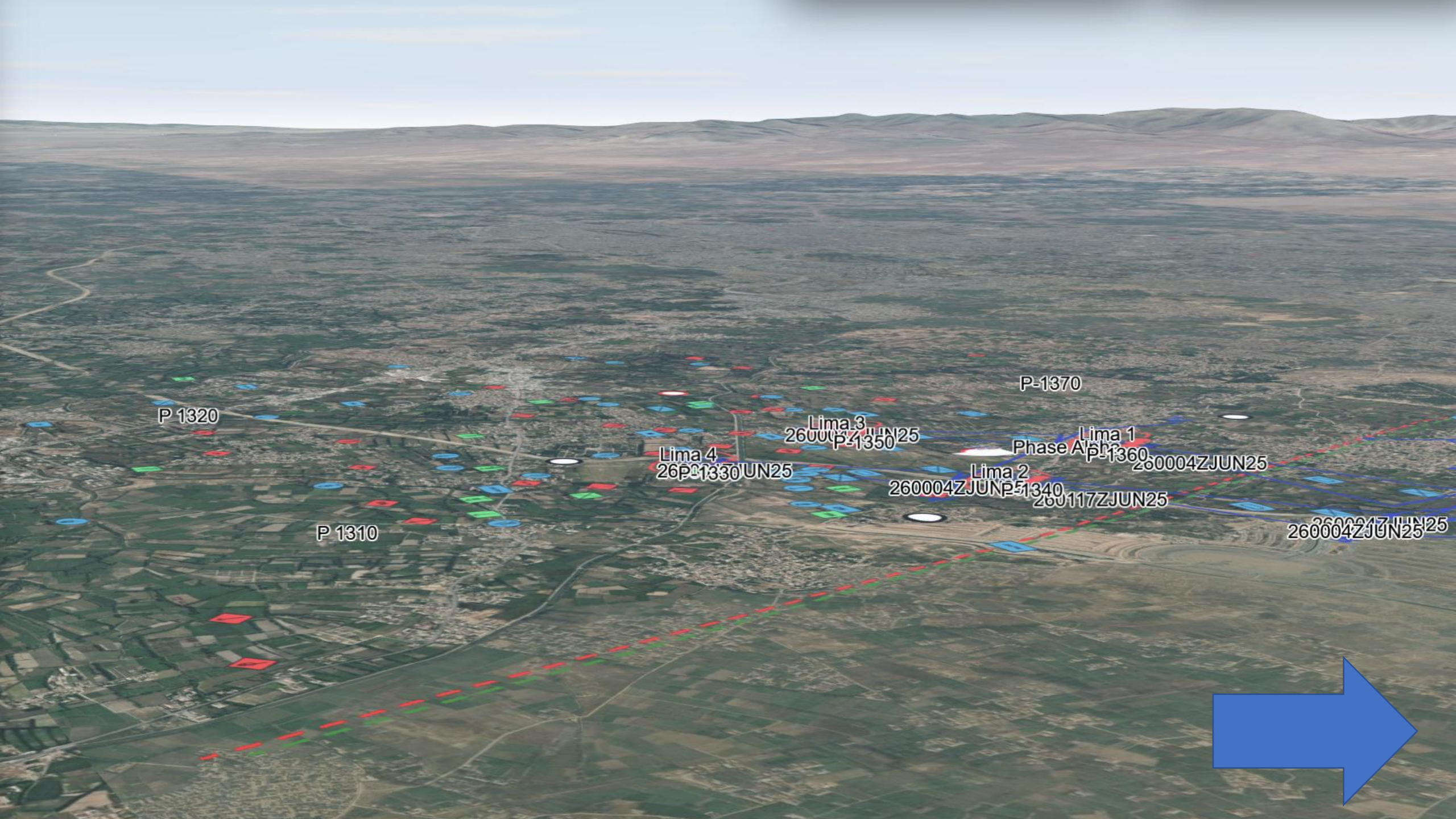
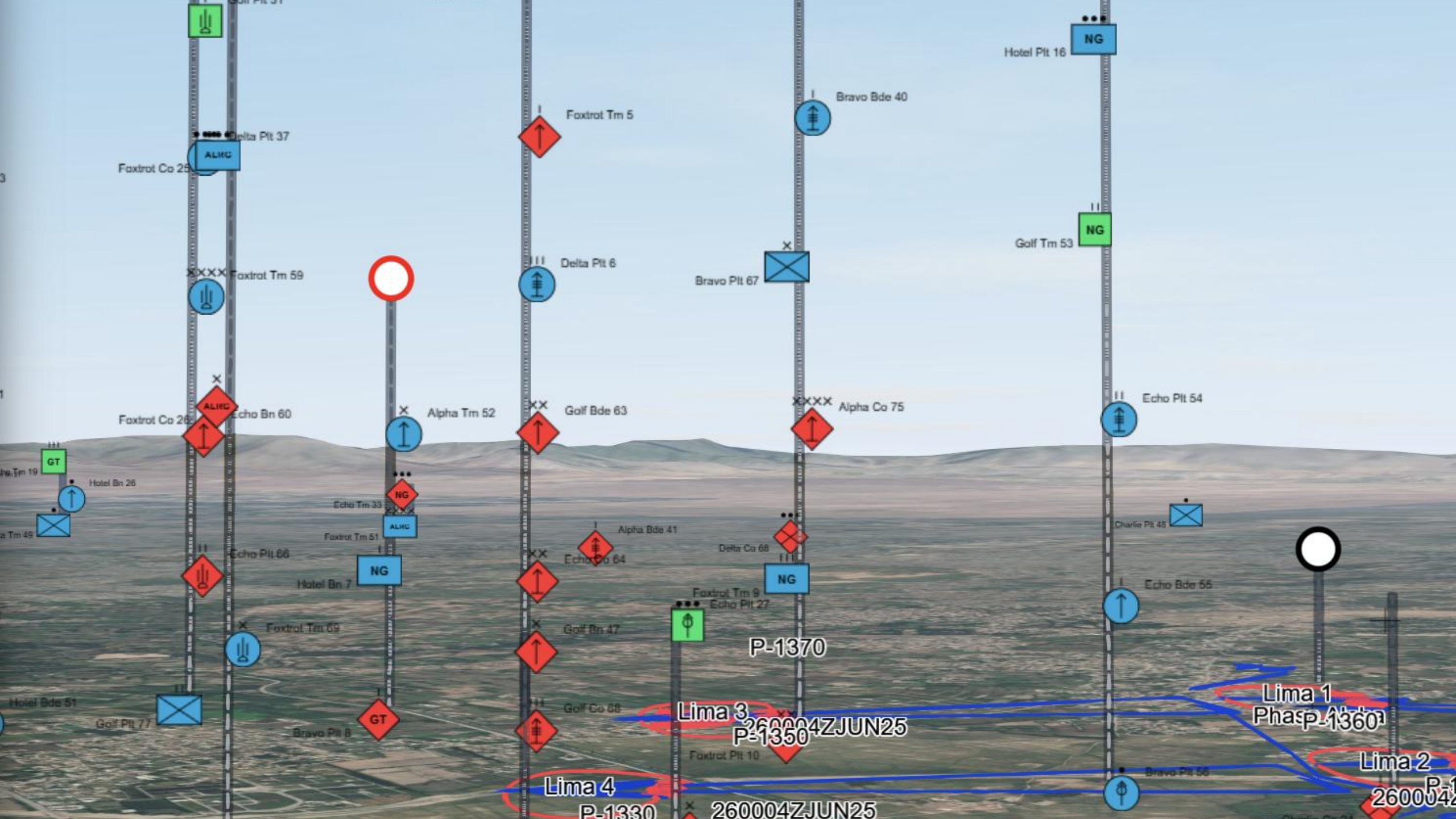
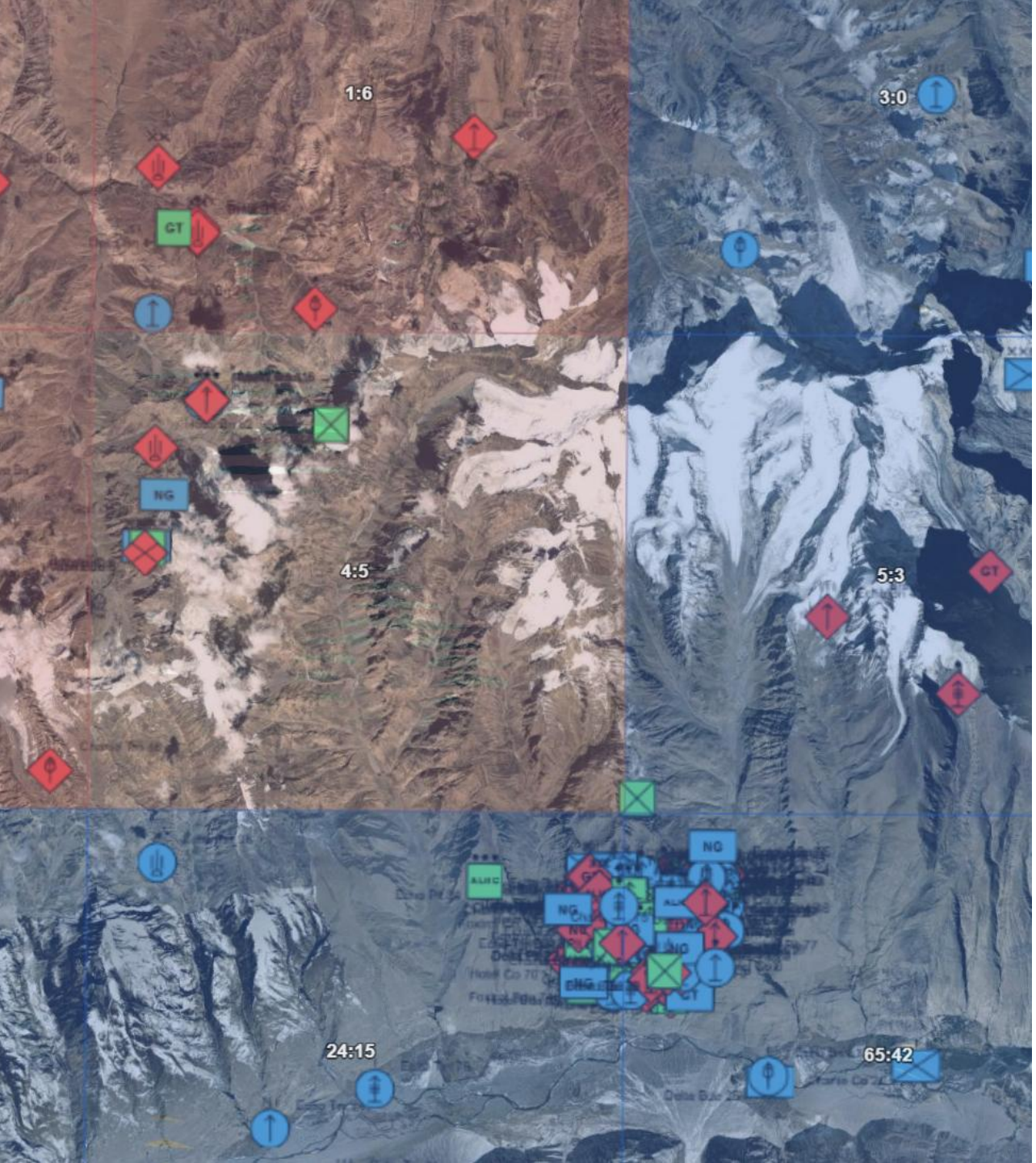

P-1370
P 1320
Lima 3
260004ZJUN25
P-1350
Lima 1
Phase Alpha
P-1360
Lima 4
260004ZJUN25
Lima 2
260004ZJUN25
P-1340
260117ZJUN25
P 1310
260004ZJUN25


Hotel Plt 16
NG
Bravo Bde 40
Foxtrot Tm 5
Delta Plt 37
ALHC
Foxtrot Co 25
Golf Tm 53
NG
Delta Plt 6
Bravo Plt 67
Foxtrot Tm 59
Alpha Tm 52
Golf Bde 63
Alpha Co 75
Echo Plt 54
Foxtrot Co 26
Echo Bn 60
ALHC
GT
Hotel Bn 26
Echo Tm 33
NG
ALHC
Foxtrot Tm 51
Alpha Bde 41
Charlie Plt 48
Delta Co 68
NG
Echo Co 64
Echo Plt 66
Hotel Bn 7
NG
Foxtrot Tm 9
Echo Plt 27
Echo Bde 55
Foxtrot Tm 69
Golf Bn 47
P-1370
Hotel Bde 51
Golf Plt 77
GT
Bravo Plt 8
Golf Co 68
Lima 3
260004ZJUN25
P-1350
Lima 1
P-1360
Foxtrot Plt 10
Lima 2
Lima 4
P-1330
260004ZJUN25



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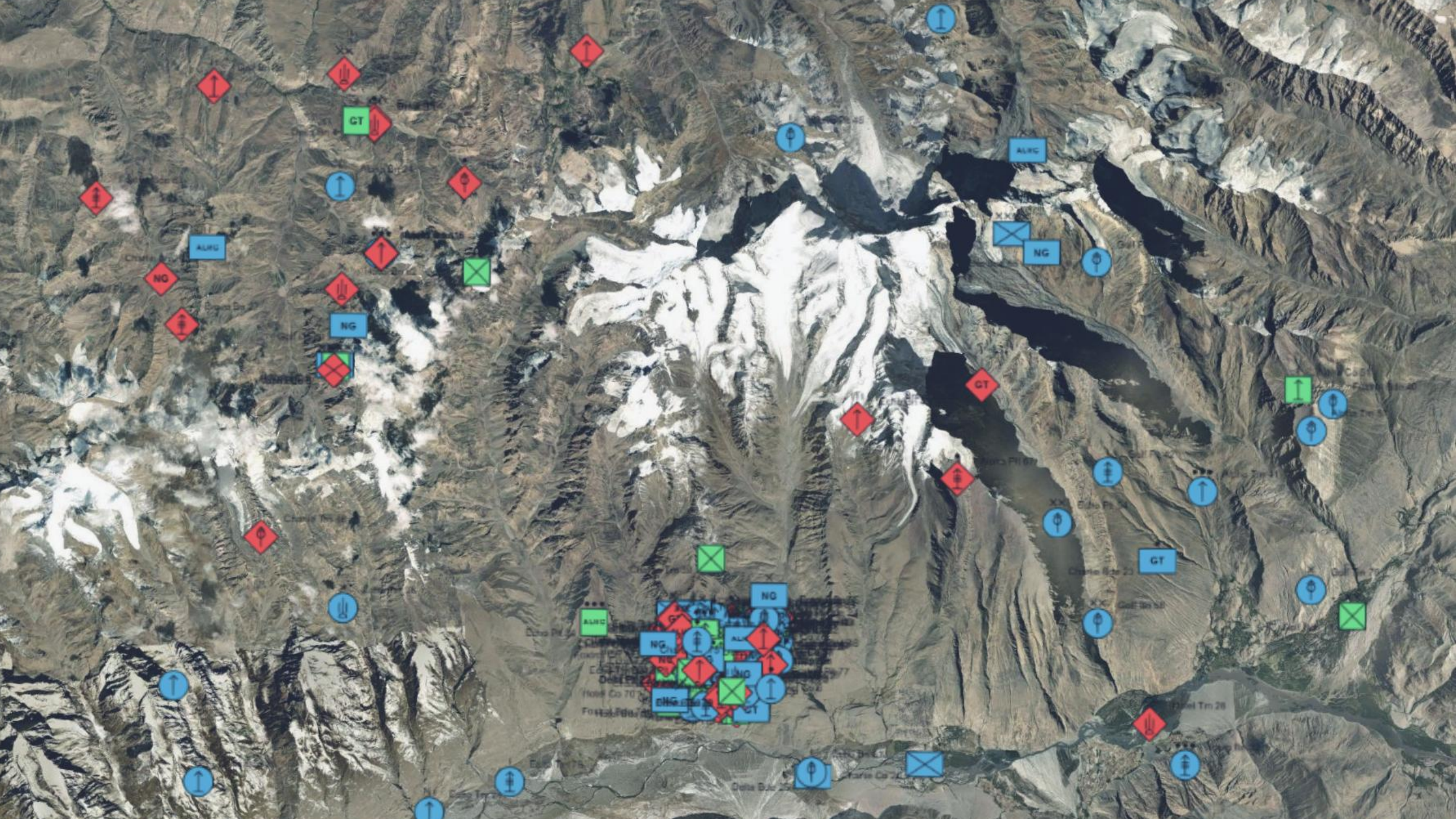
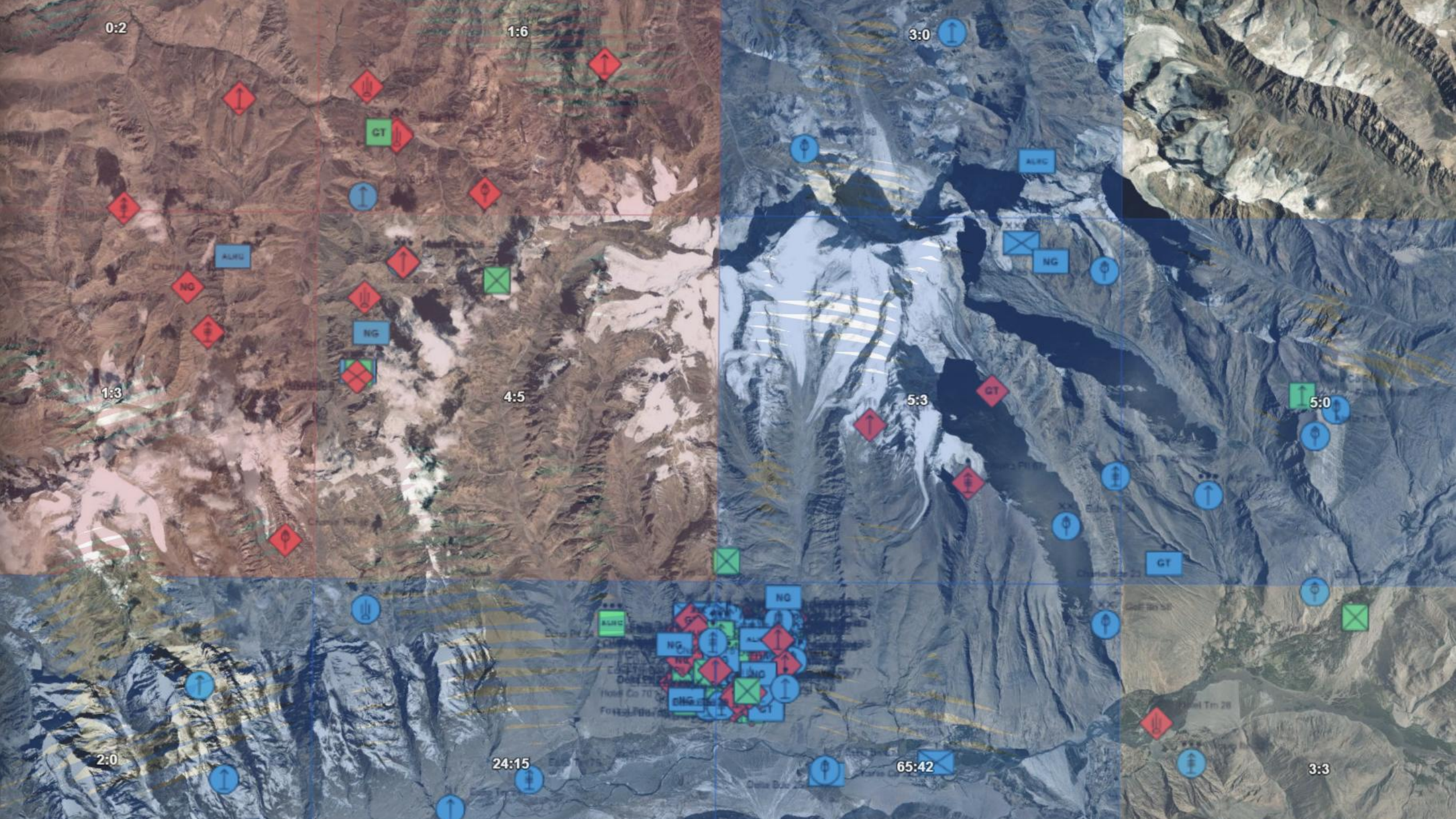
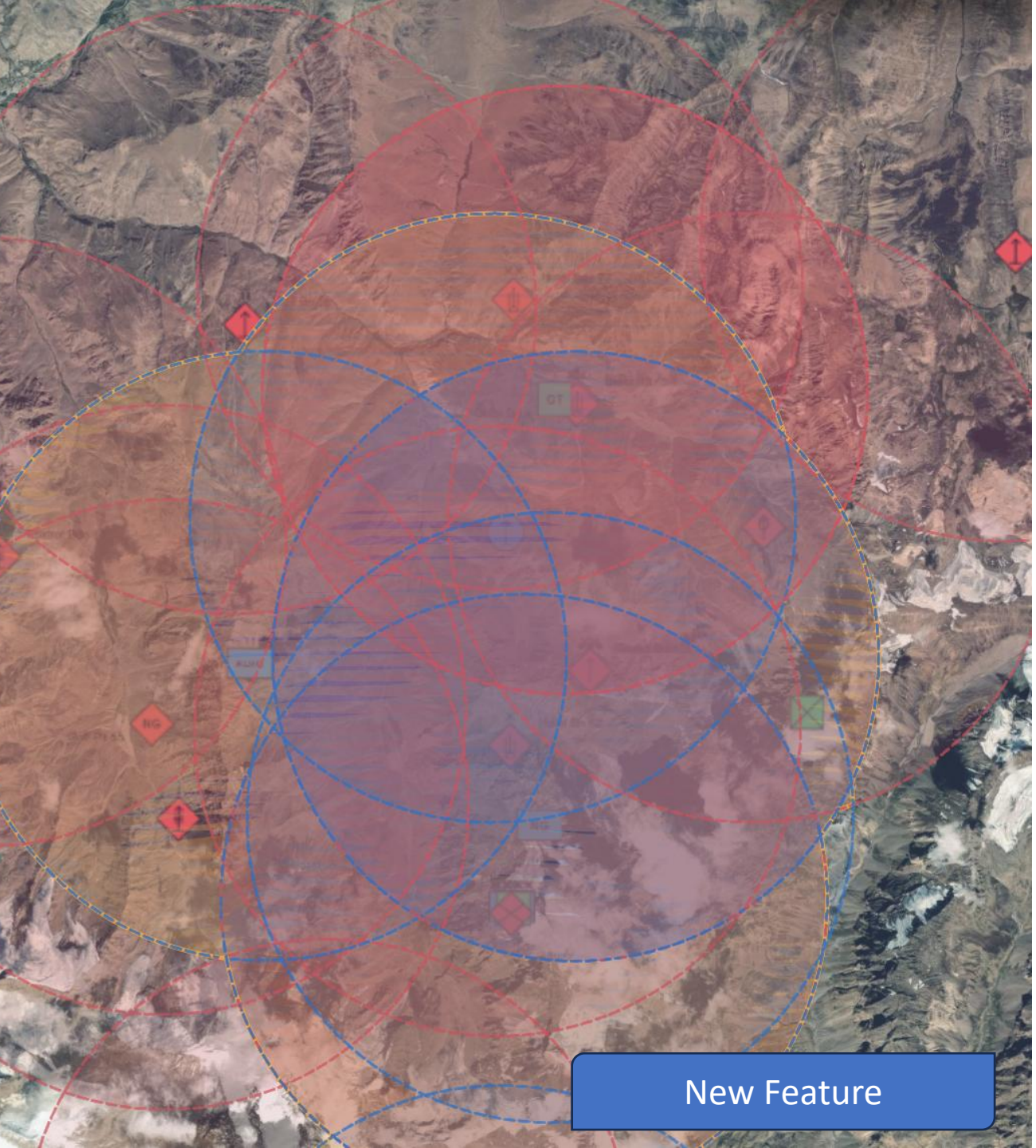

GT
ALRG
NG
NG
ALRG
NG
GT
GT
NG
ALRG
NG


0:2
1:6
3:0
1:3
4:5
5:3
5:0
2:0
24:15
65:42
3:3
GT
NG
ALHG



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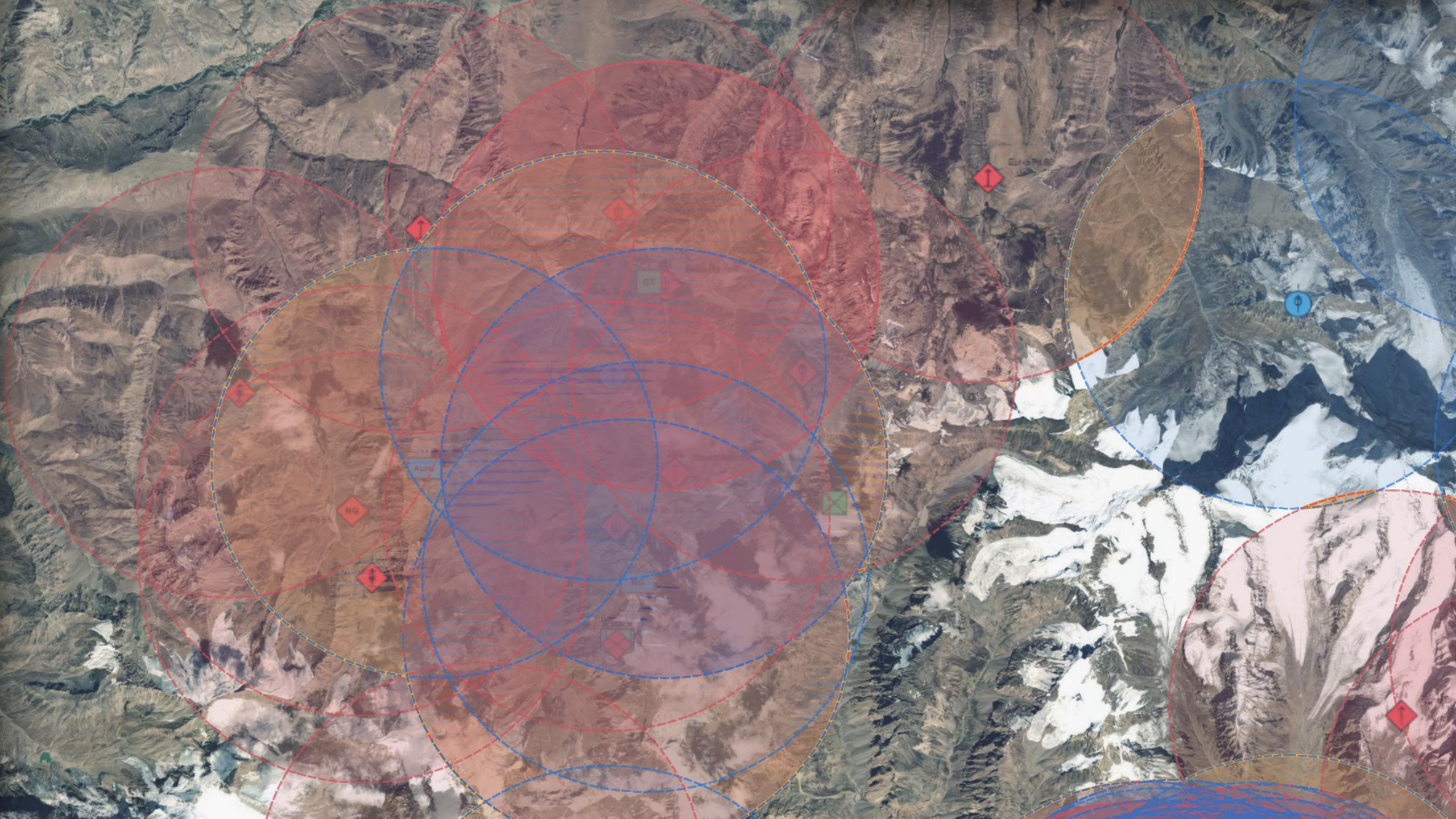
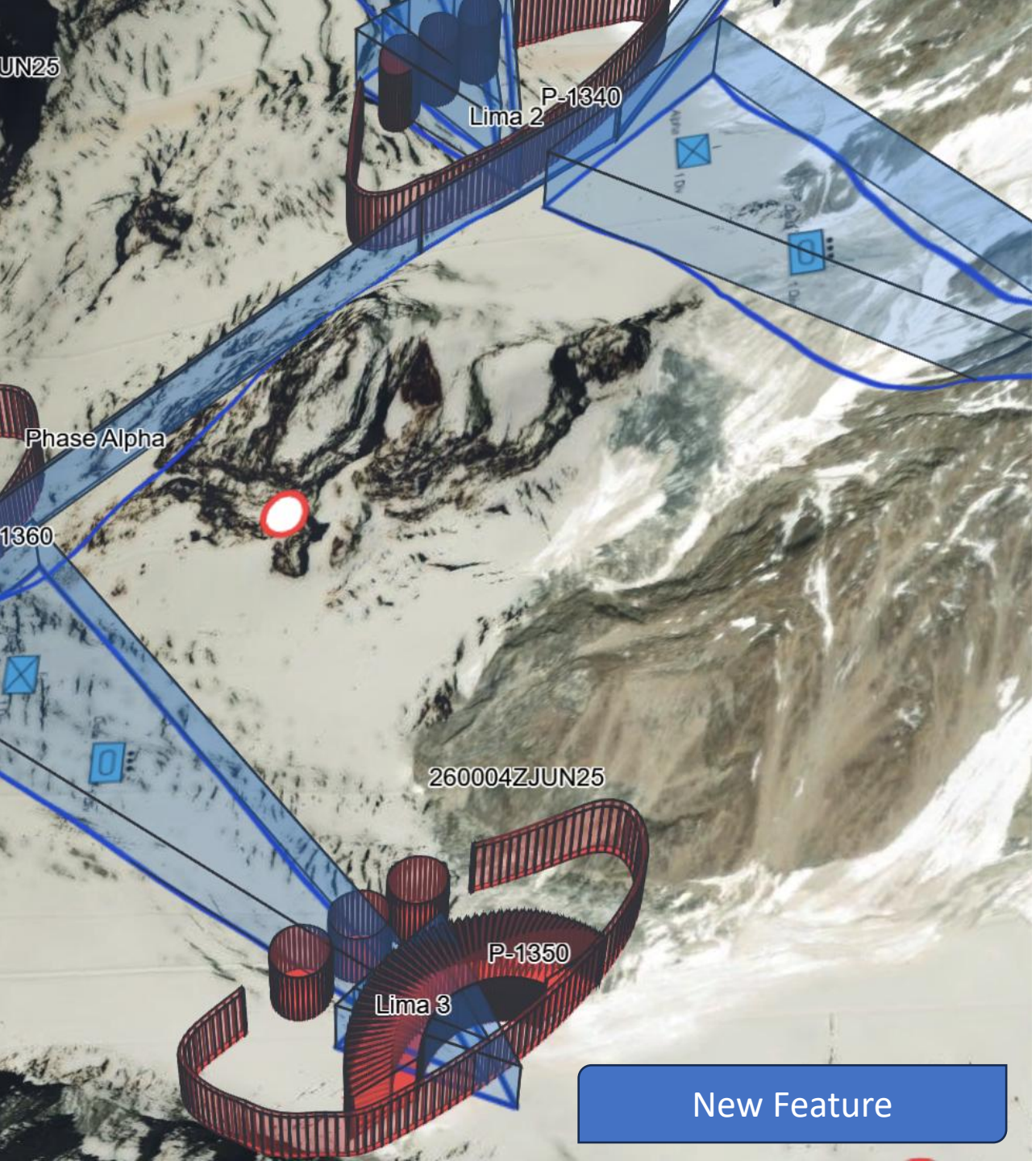
GT
NG
NG





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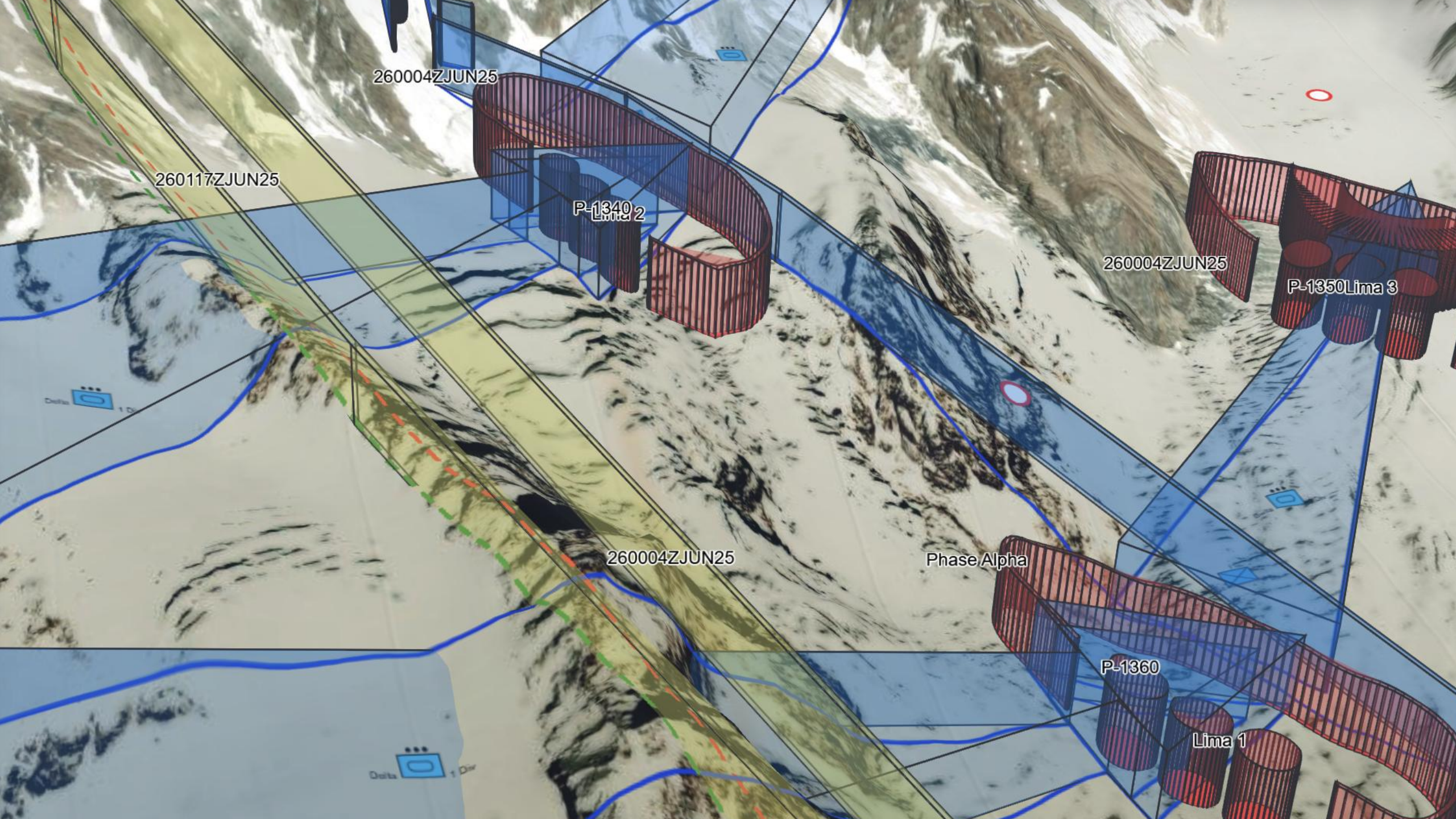
260117ZJUN25
P-1340
Lima 2
260004ZJUN25
P-1350Lima 3
260004ZJUN25
Phase Alpha
P-1360
Lima 1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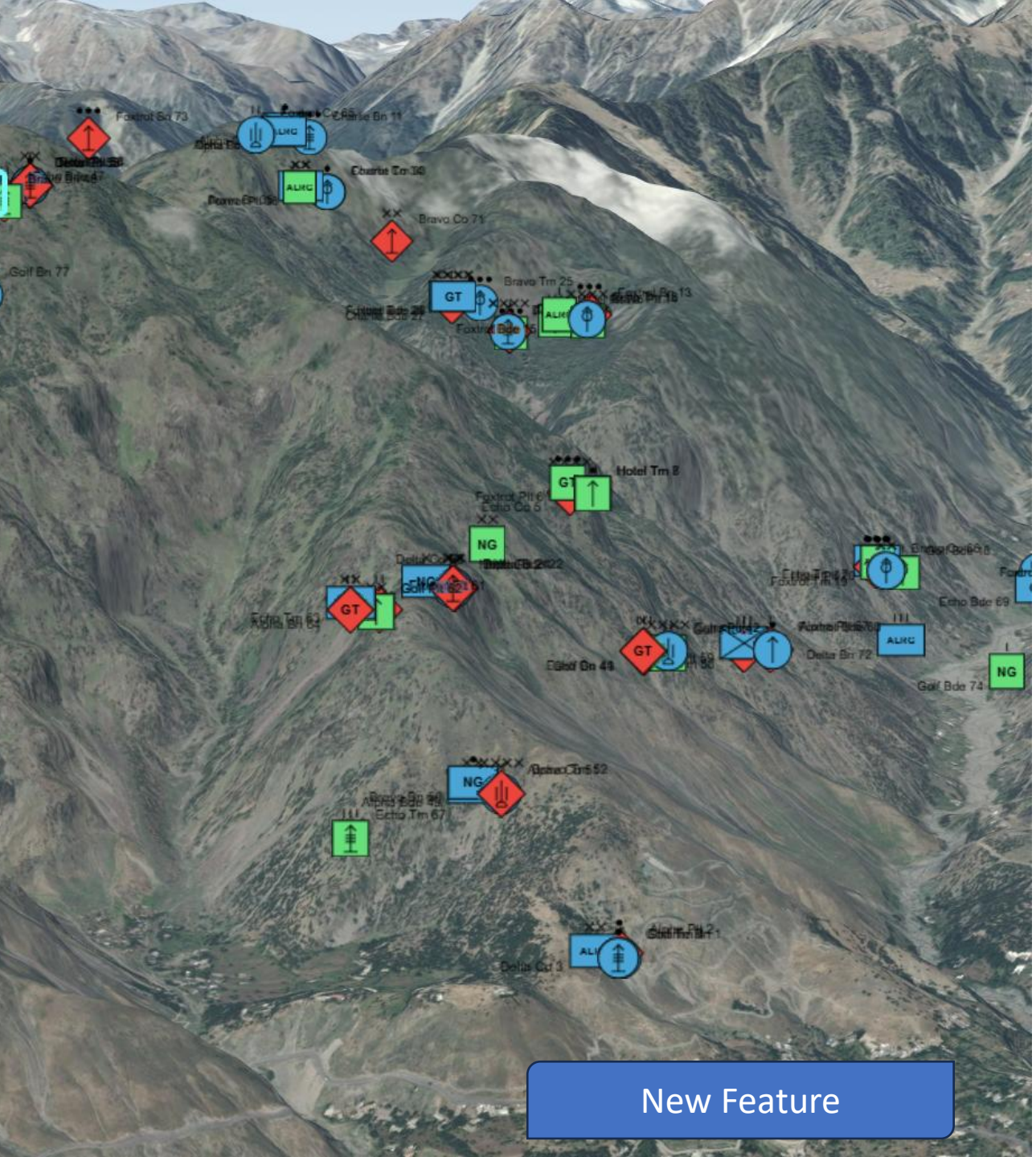

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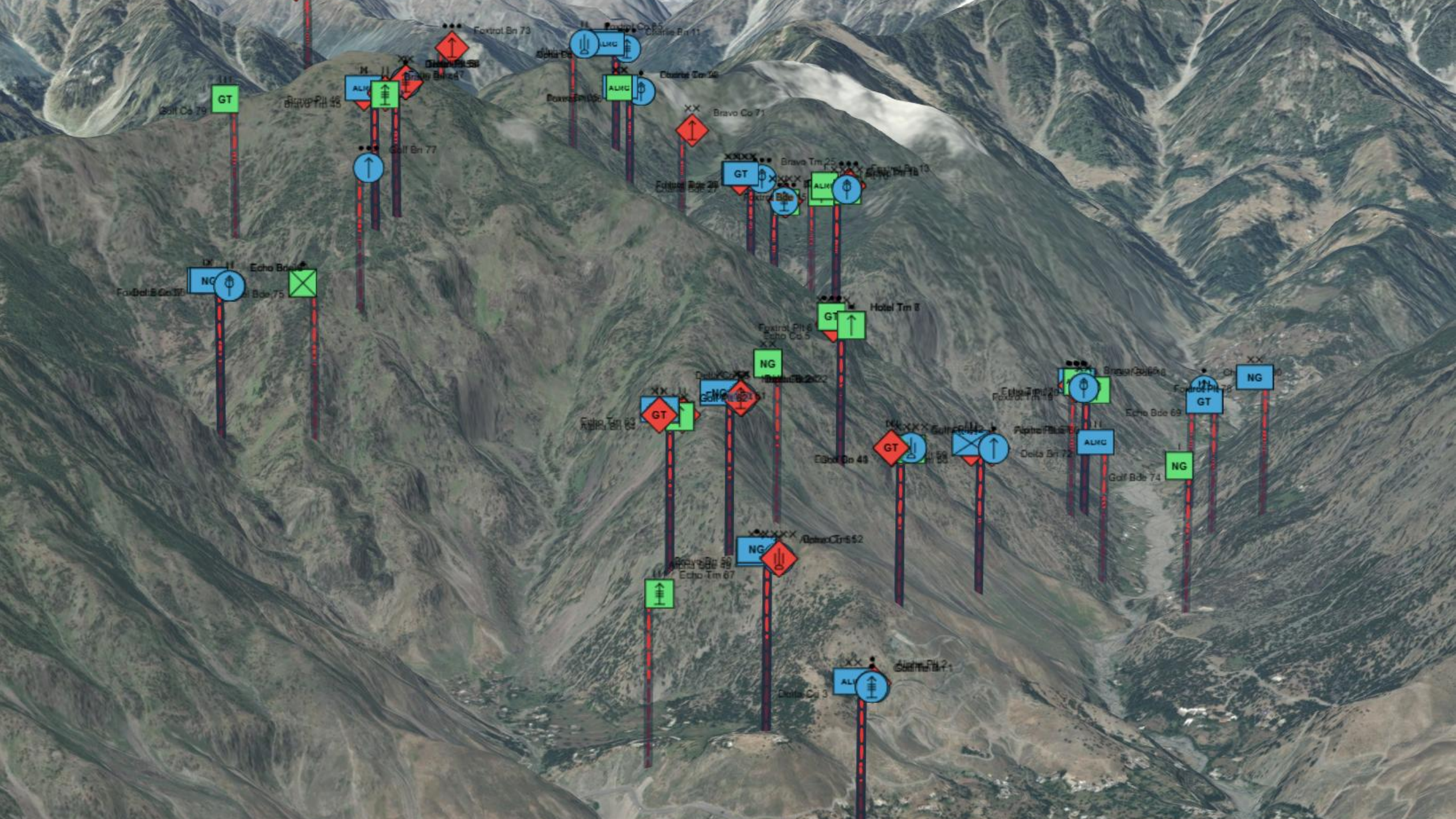
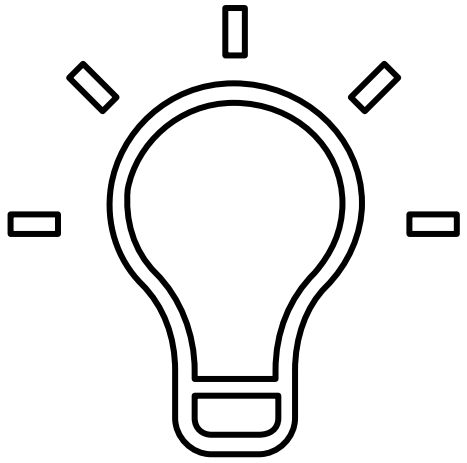

Foxtrot Bn 73
Charlie Bn 11
Charlie Tm 14
Bravo Co 71
Golf Bn 77
Bravo Tm 25
Hotel Tm 8
Foxtrot Plt 6
Echo Co 5
Golf Bde 75
Golf Bn 43
Delta Bn 72
Golf Bde 74
Echo Bde 69
Echo Tm 67
Delta Co 3
GT
NG
ALHC

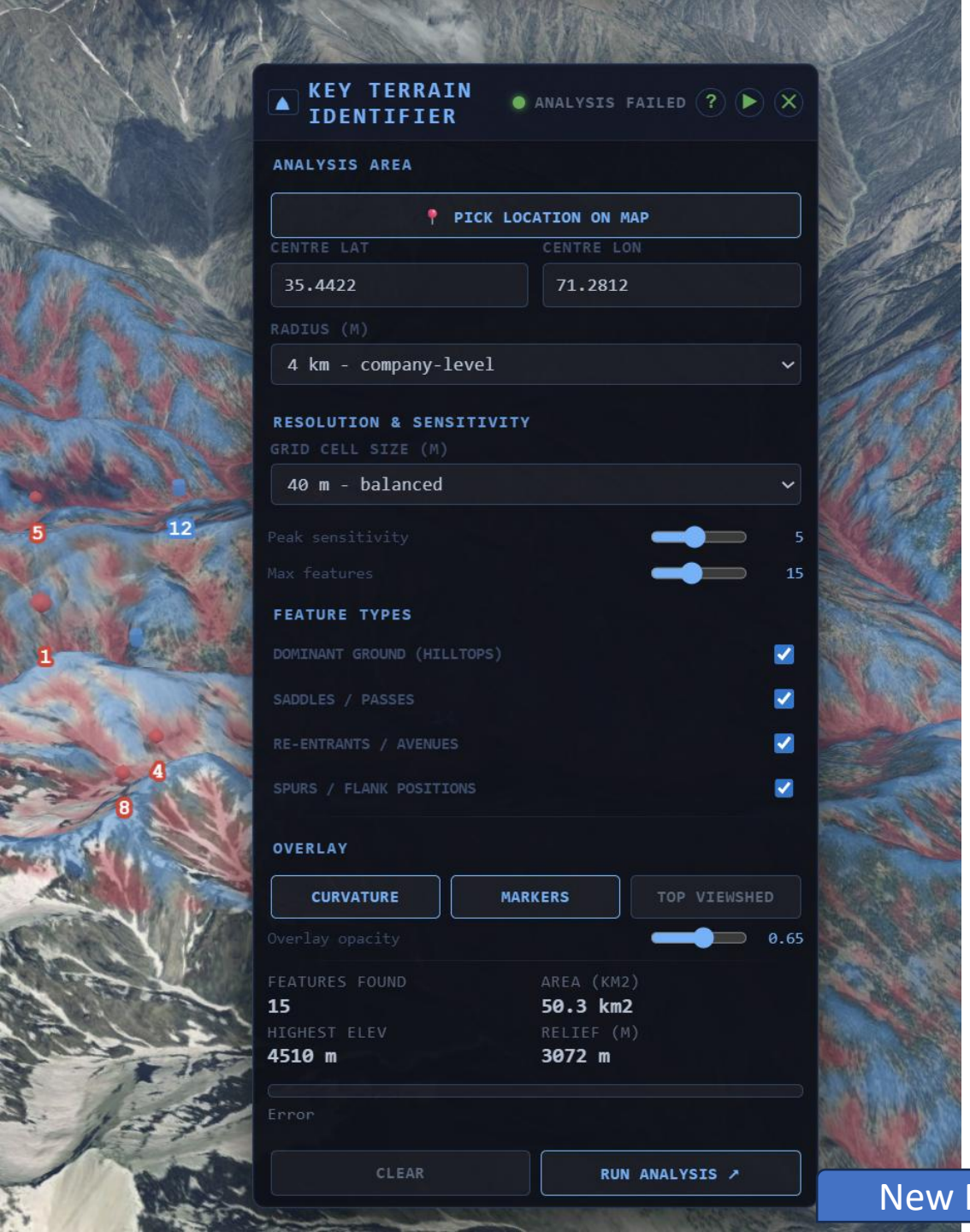
Foxtrot Bd 73
Charlie Bn 11
Charlie Co 10
Bravo Co 71
Golf Co 79
Golf Bn 77
Bravo Tm 25
Hotel Tm 8
Echo Co 5
Delta Co 3
Echo Tm 67
Echo Bde 69
Delta Bn 72
Golf Bde 74
GT
NG
ALRG

Analysis Tools



Analysis Tools

Terrain Intelligence



New Feature

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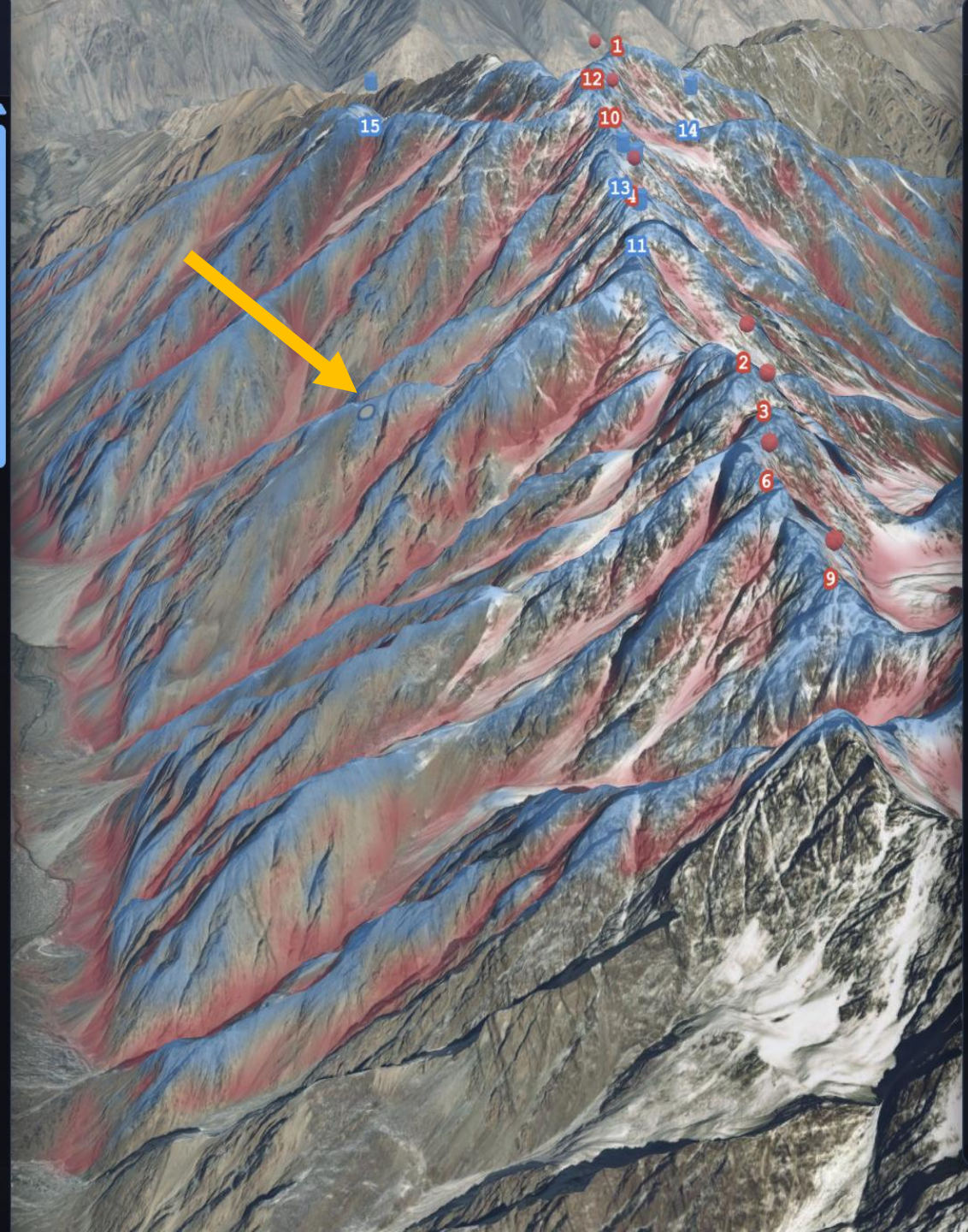
Analysis Tools



KEY TERRAIN — 15 FEATURES RANKED BY TACTICAL VALUE

RANKED

1 ▲ Dominant ground — 99
4984 M MSL · 1% VISIBLE

Prominence		619m
Viewshed		1%
Elevation		4984m

Controls 1% of AO. Seizure gives observation and fires over all lower ground. Must be held or denied.

2 ▲ Dominant ground — 97
5074 M MSL · 1% VISIBLE

Prominence		576m
Viewshed		1%
Elevation		5074m

Controls 1% of AO. Seizure gives observation and fires over all lower ground. Must be held or denied.

3 ▲ Dominant ground — 93
5105 M MSL · 1% VISIBLE

Prominence		508m
Viewshed		1%
Elevation		5105m

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4 ▲ Dominant ground — 93
5030 M MSL · 1% VISIBLE

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Viewshed		1%
Elevation		5030m

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5 ▲ Dominant ground — 92
4755 M MSL · 5% VISIBLE

Prominence		526m
Viewshed		5%

KEY TERRAIN IDENTIFIER

DONE

ANALYSIS AREA

PICK LOCATION ON MAP

CENTRE LAT: 35.8982
CENTRE LON: 70.5807

RADIUS (M): 4 km - company-level

RESOLUTION & SENSITIVITY

GRID CELL SIZE (M): 40 m - balanced

Peak sensitivity: 5
Max features: 15

FEATURE TYPES

- [x] DOMINANT GROUND (HILLTOPS)
- [x] SADDLES / PASSES
- [x] RE-ENTRANTS / AVENUES
- [x] SPURS / FLANK POSITIONS

OVERLAY

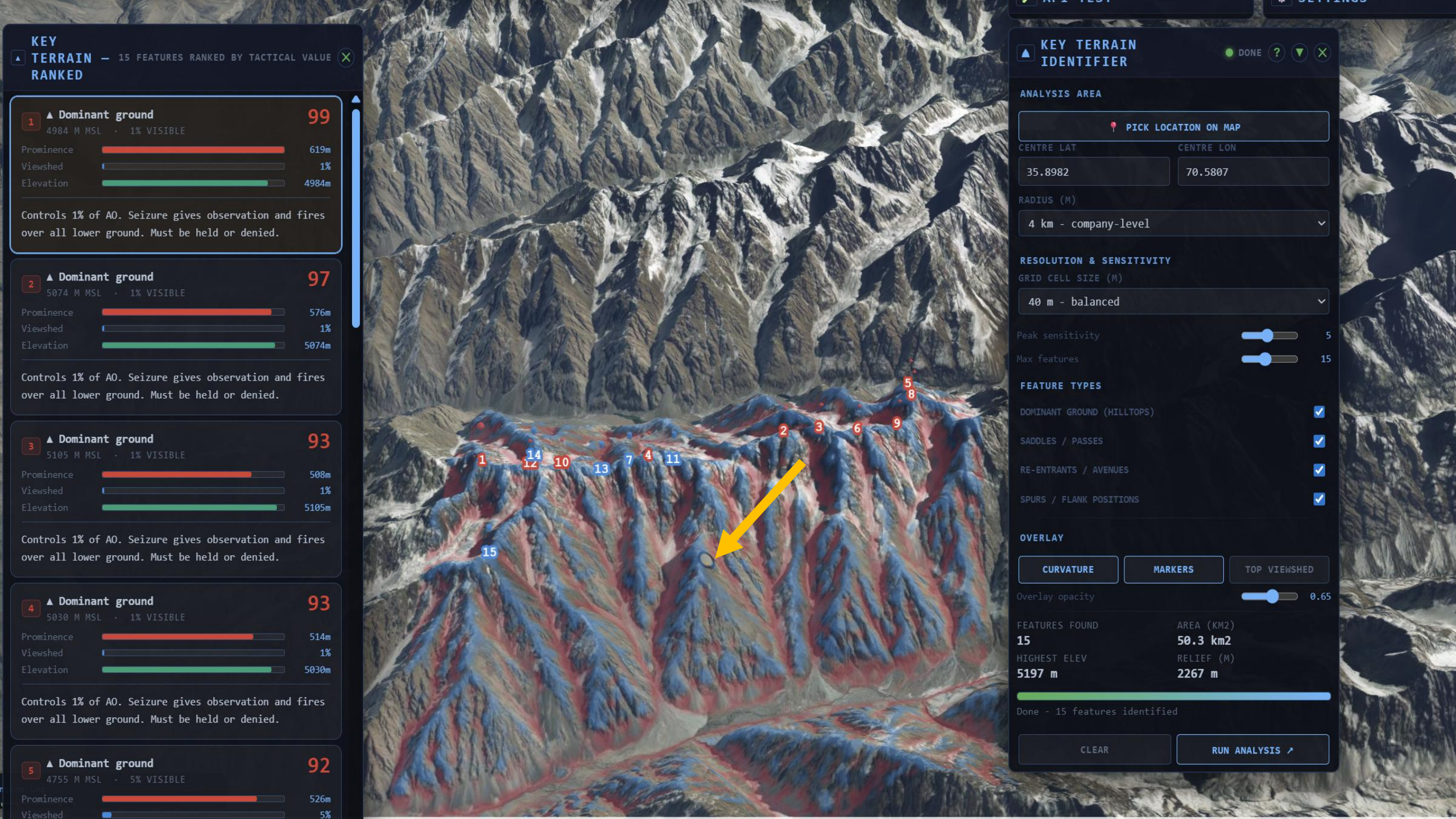
CURVATURE | MARKERS | TOP VIEWSHED

Overlay opacity: 0.65

FEATURES FOUND	AREA (KM2)
15	50.3 km2
HIGHEST ELEV	RELIEF (M)
5197 m	2267 m

Done - 15 features identified

CLEAR | RUN ANALYSIS ↗



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GRID CELL SIZE (M)
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Max features 15
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OVERLAY
CURVATURE
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CLEAR
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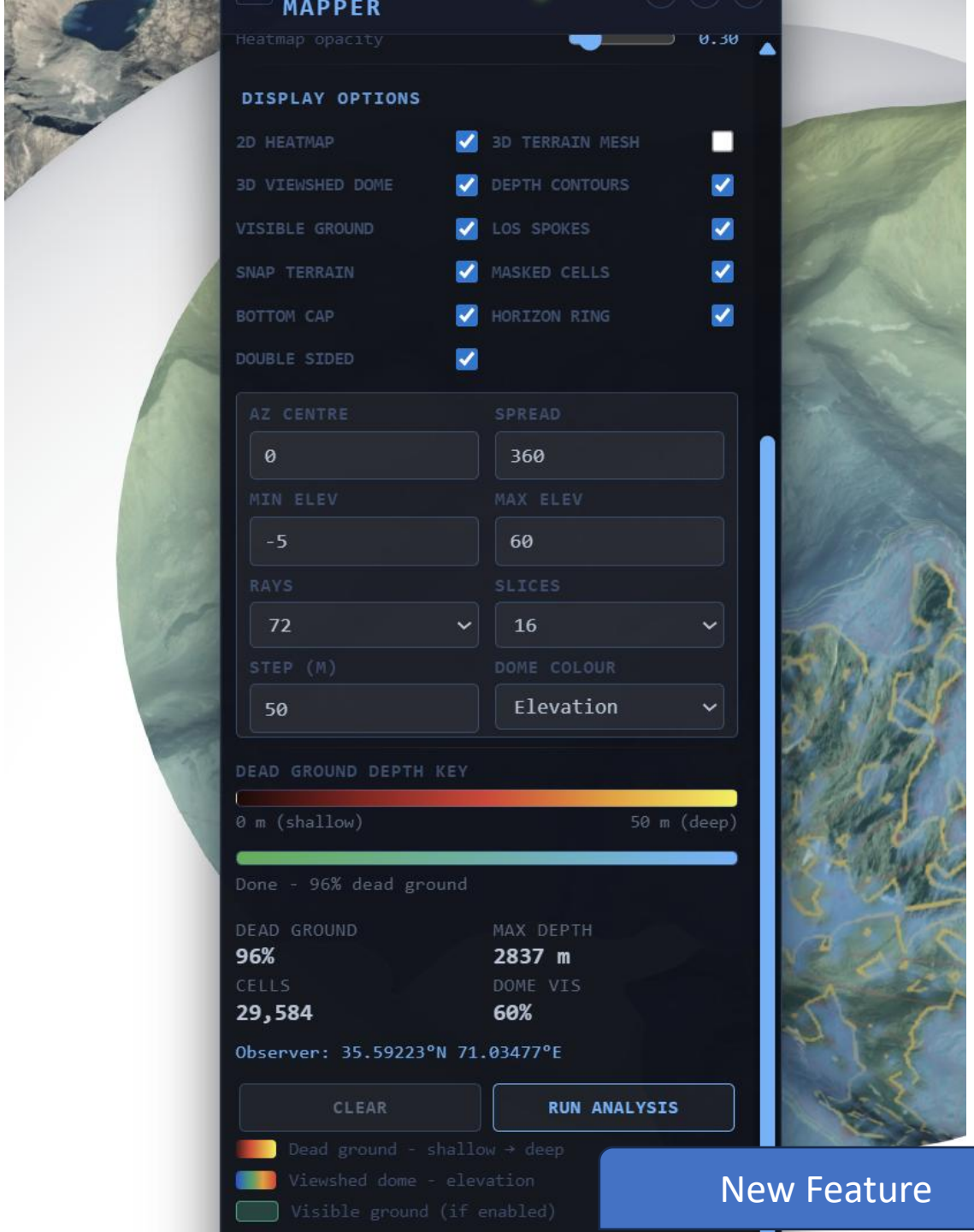

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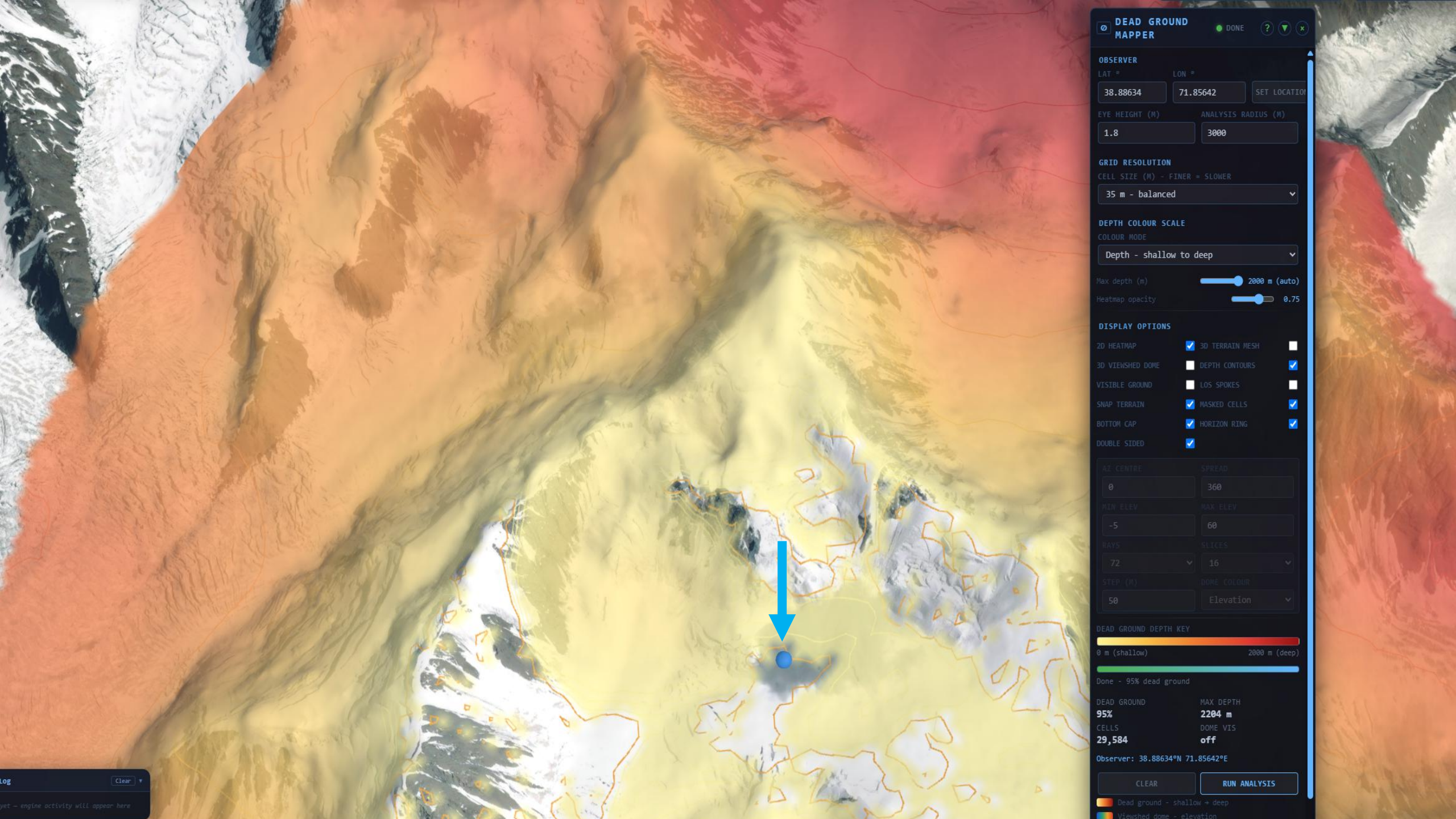
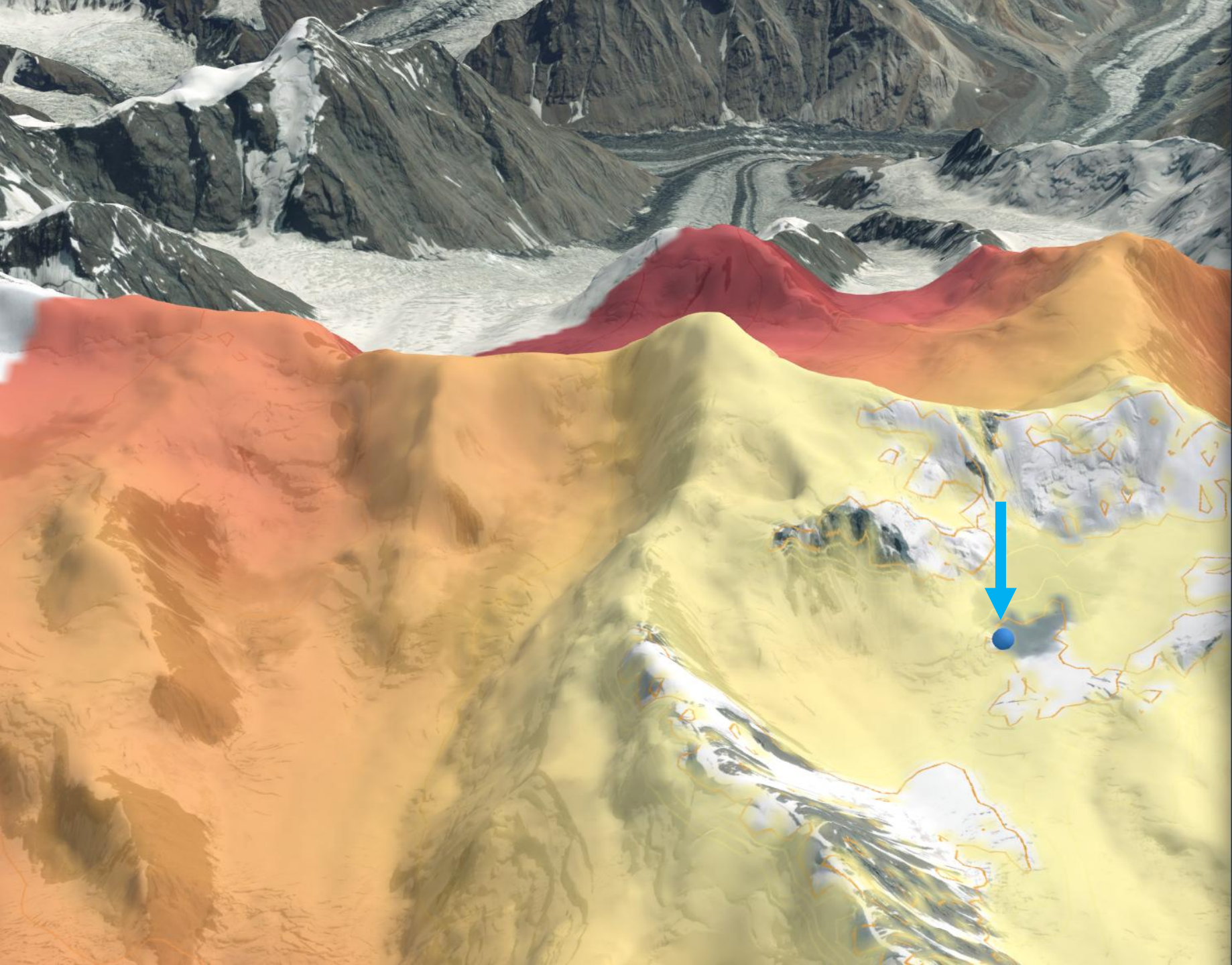

DEAD GROUND MAPPER
DONE
OBSERVER
LAT °
LON °
38.88634
71.85642
SET LOCATION
EYE HEIGHT (M)
ANALYSIS RADIUS (M)
1.8
3000
GRID RESOLUTION
CELL SIZE (M) - FINER = SLOWER
35 m - balanced
DEPTH COLOUR SCALE
COLOUR MODE
Depth - shallow to deep
Max depth (m)
2000 m (auto)
Heatmap opacity
0.75
DISPLAY OPTIONS
2D HEATMAP
3D TERRAIN MESH
3D VIEWSHED DOME
DEPTH CONTOURS
VISIBLE GROUND
LOS SPOKES
SNAP TERRAIN
MASKED CELLS
BOTTOM CAP
HORIZON RING
DOUBLE SIDED
AZ CENTRE
SPREAD
0
360
MIN ELEV
MAX ELEV
-5
60
RAYS
SLICES
72
16
STEP (M)
DOME COLOUR
50
Elevation
DEAD GROUND DEPTH KEY
0 m (shallow)
2000 m (deep)
Done - 95% dead ground
DEAD GROUND
95%
MAX DEPTH
2204 m
CELLS
29,584
DOME VIS
off
Observer: 38.88634°N 71.85642°E
CLEAR
RUN ANALYSIS
Dead ground - shallow → deep
Viewshed dome - elevation
Log
Clear
yet — engine activity will appear here



DEAD GROUND MAPPER

OBSERVER PLACED - CLICK RUN

OBSERVER

LAT °: 38.88702
LON °: 71.85530
SET LOCATION

EYE HEIGHT (M): 1.8
ANALYSIS RADIUS (M): 3000

GRID RESOLUTION

CELL SIZE (M) - FINER = SLOWER

35 m - balanced

DEPTH COLOUR SCALE

COLOUR MODE

Depth - shallow to deep

Max depth (m): 2000 m (auto)
Heatmap opacity: 0.75

DISPLAY OPTIONS

2D HEATMAP	☑	3D TERRAIN MESH	☐
3D VIEWSHED DOME	☐	DEPTH CONTOURS	☑
VISIBLE GROUND	☐	LOS SPOKES	☐
SNAP TERRAIN	☑	MASKED CELLS	☑
BOTTOM CAP	☑	HORIZON RING	☑
DOUBLE SIDED	☑		

AZ CENTRE: 0
SPREAD: 360
MIN ELEV: -5
MAX ELEV: 60
RAYS: 72
SLICES: 16
STEP (M): 50
DOME COLOUR: Elevation

DEAD GROUND DEPTH KEY

0 m (shallow) — 2000 m (deep)

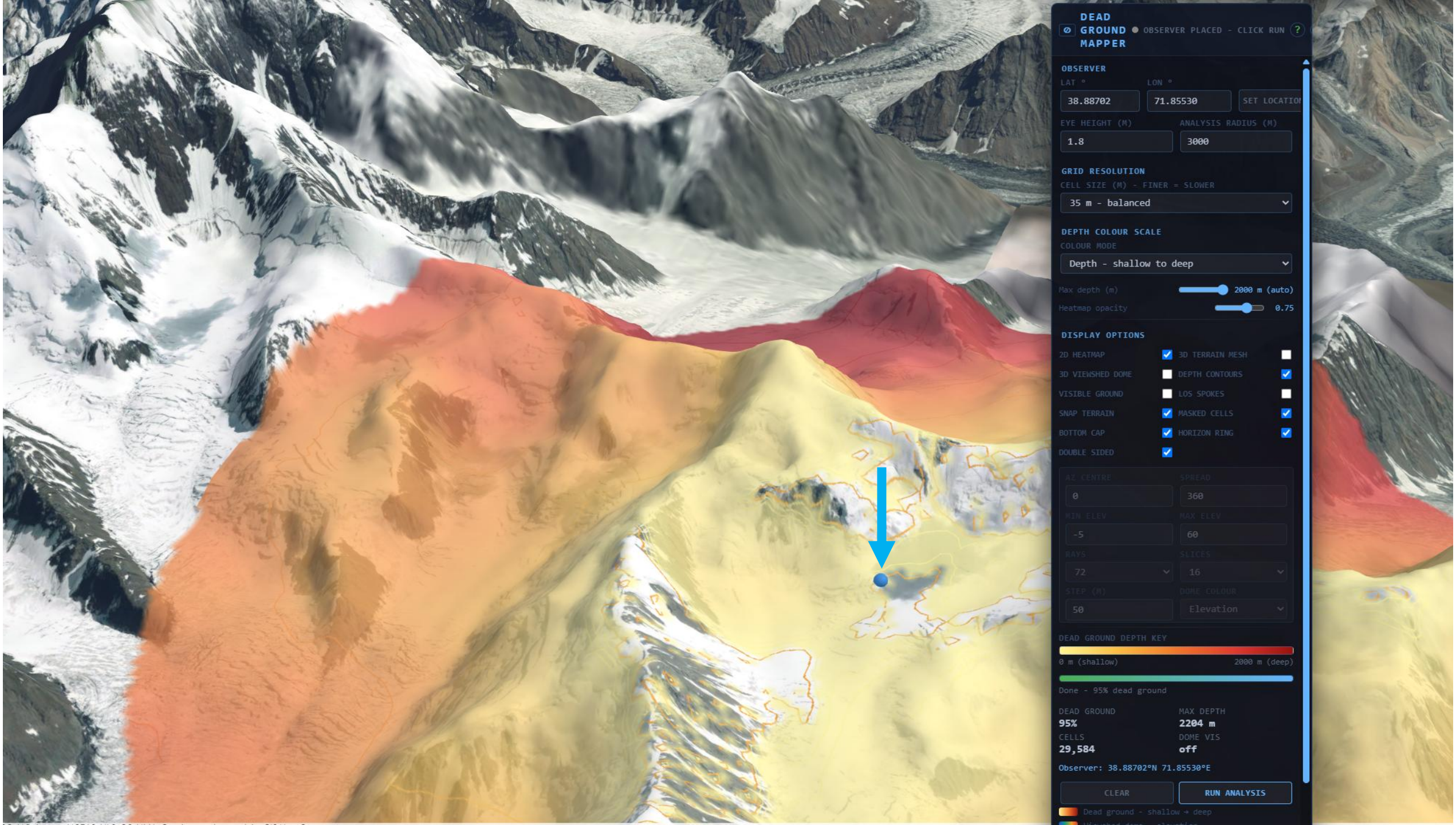
Done - 95% dead ground

DEAD GROUND	MAX DEPTH
95%	2204 m
CELLS	DOME VIS
29,584	off

Observer: 38.88702°N 71.85530°E

CLEAR | RUN ANALYSIS

Dead ground - shallow → deep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LON °
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VISIBLE GROUND
LOS SPOKES
SNAP TERRAIN
MASKED CELLS
BOTTOM CAP
HORIZON RING
DOUBLE SIDED
AZ CENTRE
SPREAD
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360
MIN ELEV
MAX ELEV
-5
60
RAYS
SLICES
72
16
STEP (M)
DOME COLOUR
50
Elevation
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DEAD GROUND
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MAX DEPTH
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DOME VIS
off
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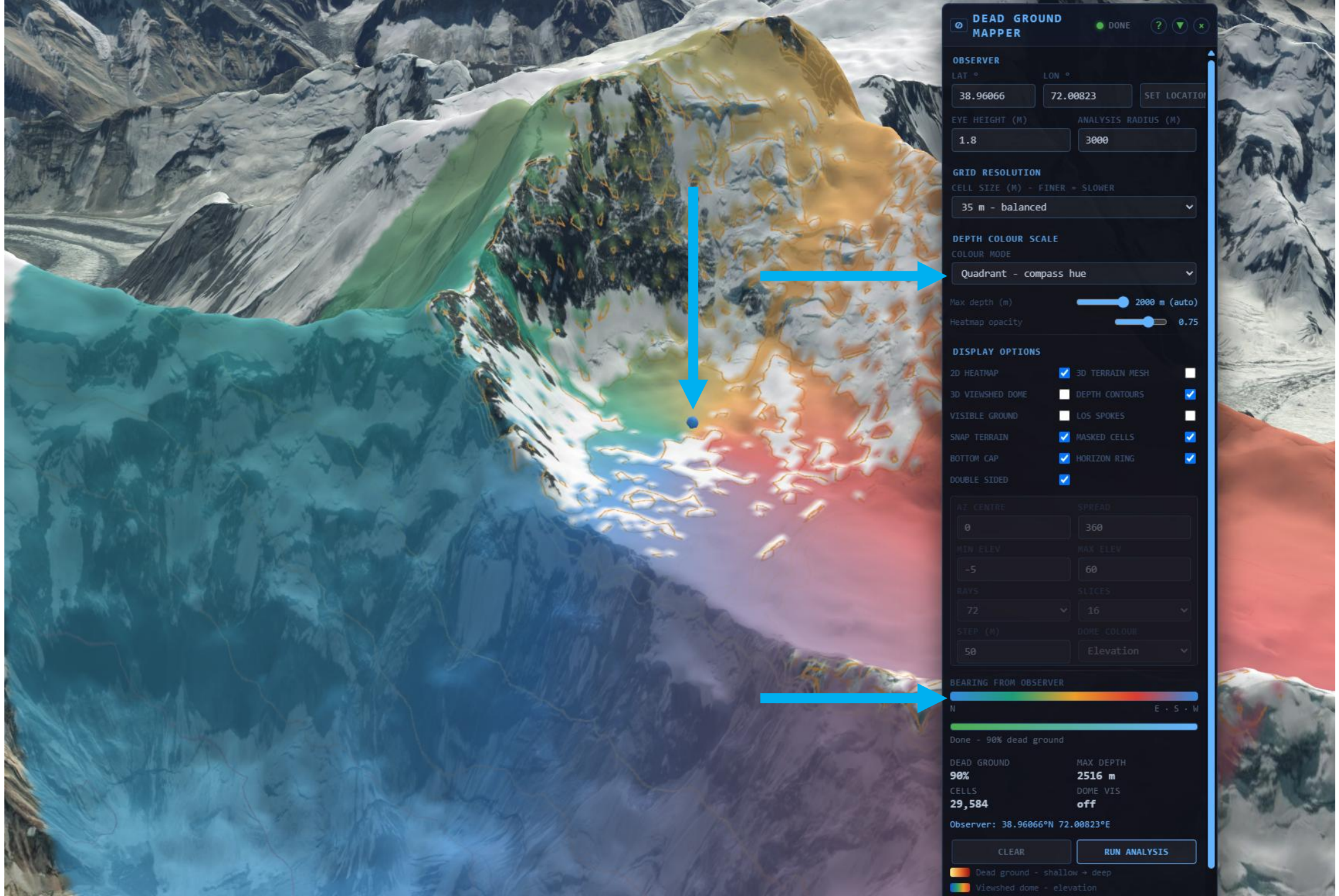
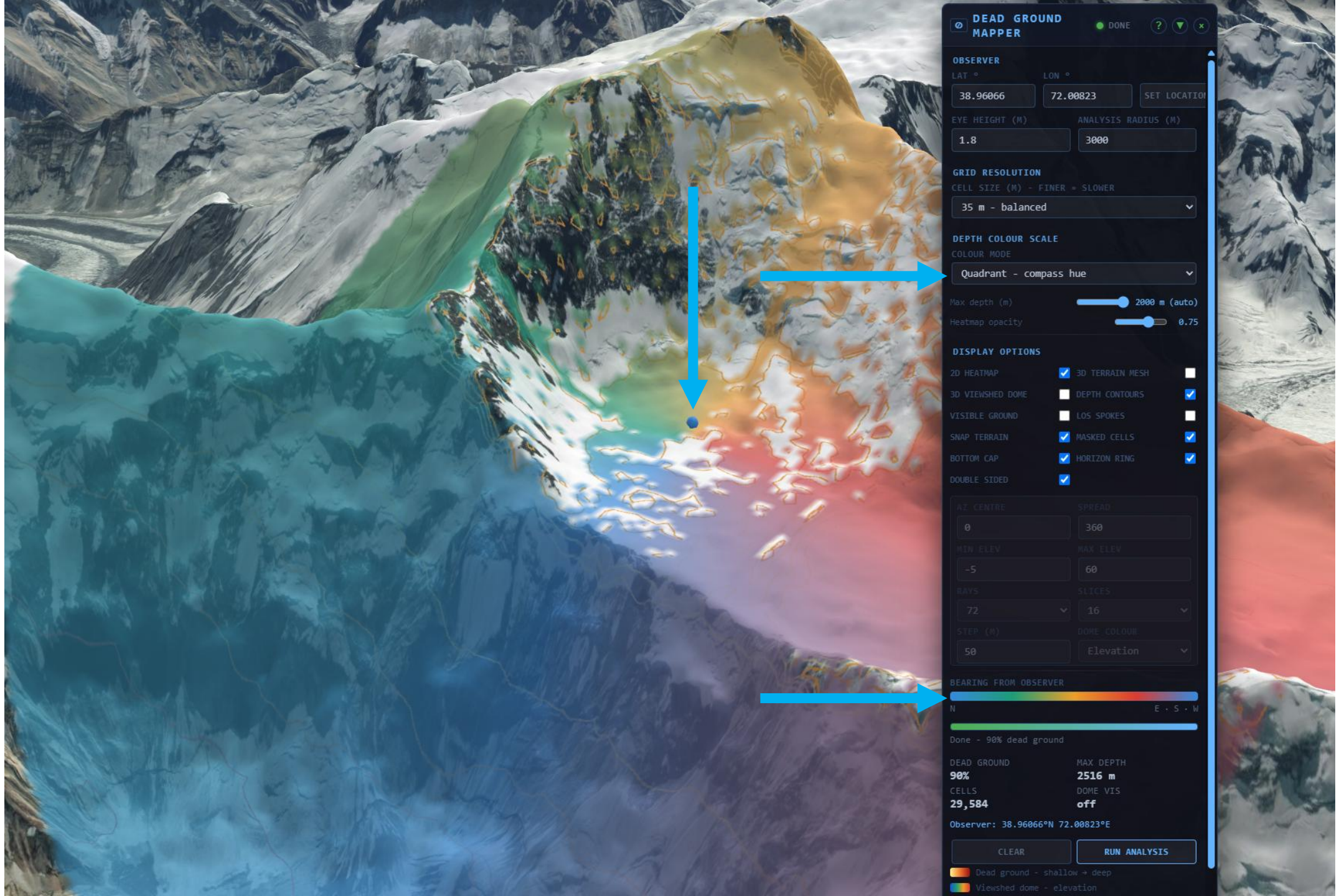
RUN ANALYSIS
Dead ground - shallow → deep


DEAD GROUND MAPPER
DONE
OBSERVER
LAT °
LON °
38.96066
72.00823
SET LOCATIO
EYE HEIGHT (M)
ANALYSIS RADIUS (M)
1.8
3000
GRID RESOLUTION
CELL SIZE (M) - FINER = SLOWER
35 m - balanced
DEPTH COLOUR SCALE
COLOUR MODE
Quadrant - compass hue
Max depth (m)
2000 m (auto)
Heatmap opacity
0.75
DISPLAY OPTIONS
2D HEATMAP
3D TERRAIN MESH
3D VIEWSHED DOME
DEPTH CONTOURS
VISIBLE GROUND
LOS SPOKES
SNAP TERRAIN
MASKED CELLS
BOTTOM CAP
HORIZON RING
DOUBLE SIDED
AZ CENTRE
SPREAD
0
360
MIN ELEV
MAX ELEV
-5
60
RAYS
SLICES
72
16
STEP (M)
DOME COLOUR
50
Elevation
BEARING FROM OBSERVER
N
E · S · W
Done - 90% dead ground
DEAD GROUND
90%
MAX DEPTH
2516 m
CELLS
29,584
DOME VIS
off
Observer: 38.96066°N 72.00823°E
CLEAR
RUN ANALYSIS
Dead ground - shallow → deep
Viewshed dome - elevation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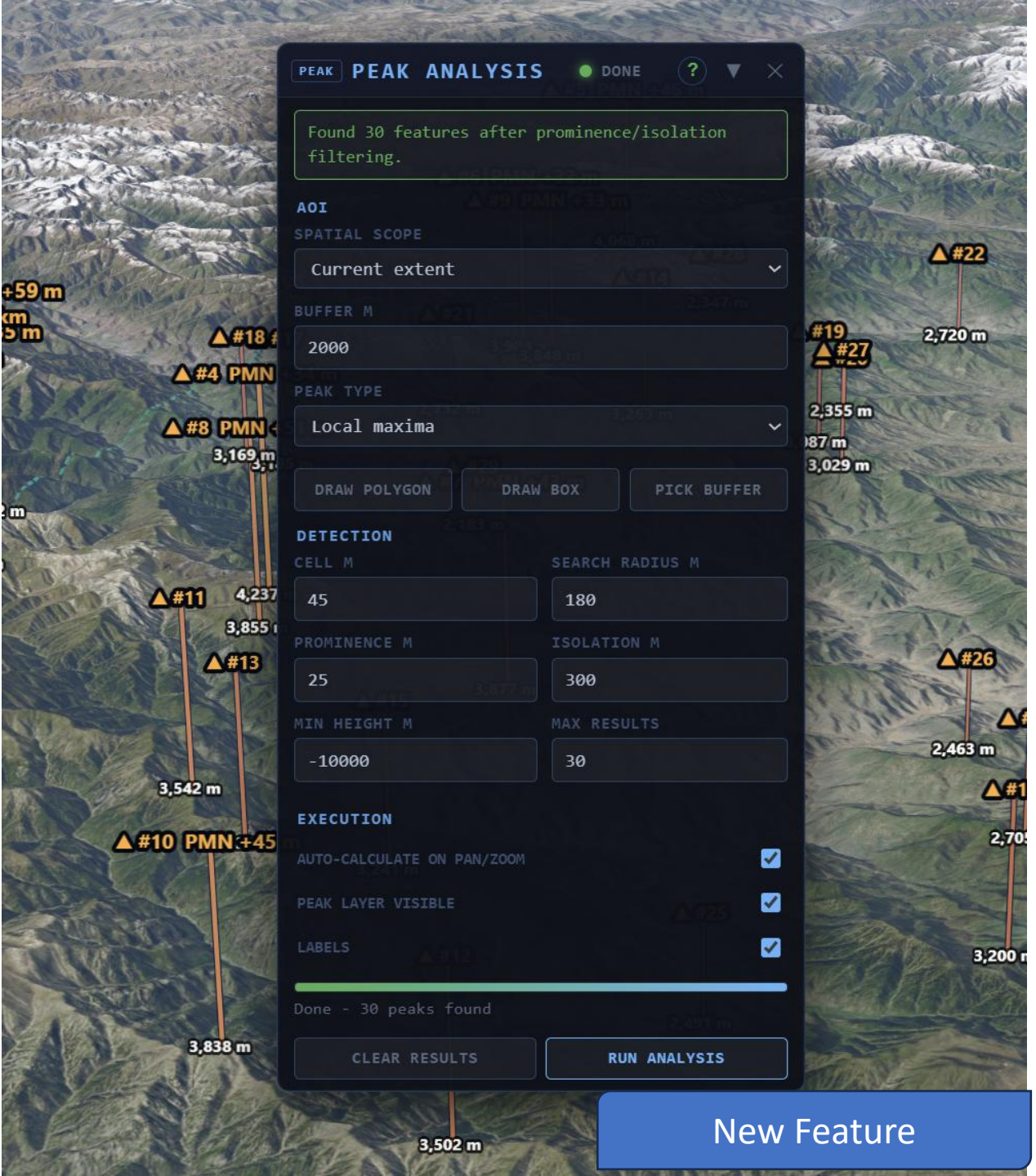

New Feature

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Mission Planner Dashboard

- Combines terrain, threat, and force data into a single integrated planning view.

Analysis Tools

RANK **PEAK RESULTS** 30

COUNT	AVG ELEV	MAX PROM
30	2913 m	60 m

WHAT THE METRICS MEAN

- **▲ PK / ▼ VLY** Feature type — Peak (local high point) or Valley/pit (local low point), with its rank.
- **Prom Prominence** — how far the summit rises above the highest saddle linking it to higher ground (m).
- **Iso Isolation** — straight-line distance to the nearest higher feature ("top" = highest in the area).
- **ΔMean Delta vs mean** — height of this feature above the average elevation of its surrounding neighborhood (m).

SORT

Rank

▲ PK 1	4,622 m	SELECT	FLY ↗

PROM 60 m ISO top ΔMEAN +250 m
35.4603° N, 71.0453° E

▲ PK 2	4,602 m	SELECT	FLY ↗

PROM 31 m ISO 43 km ΔMEAN +220 m
35.3185° N, 70.6089° E

▲ PK 3	4,362 m	SELECT	FLY ↗

PROM 36 m ISO 12 km ΔMEAN +216 m
35.3789° N, 71.1368° E

▲ PK 4	3,678 m	SELECT	FLY ↗

PROM 36 m ISO 85 km ΔMEAN +328 m
35.4422° N, 72.1082° E

PEAK **PEAK ANALYSIS** DONE

Found 30 features after prominence/isolation filtering.

AOI

SPATIAL SCOPE: Current extent

BUFFER M: 2000

PEAK TYPE: Local maxima

DRAW POLYGON | DRAW BOX | PICK BUFFER

DETECTION

CELL M	SEARCH RADIUS M
45	180

PROMINENCE M	ISOLATION M
25	300

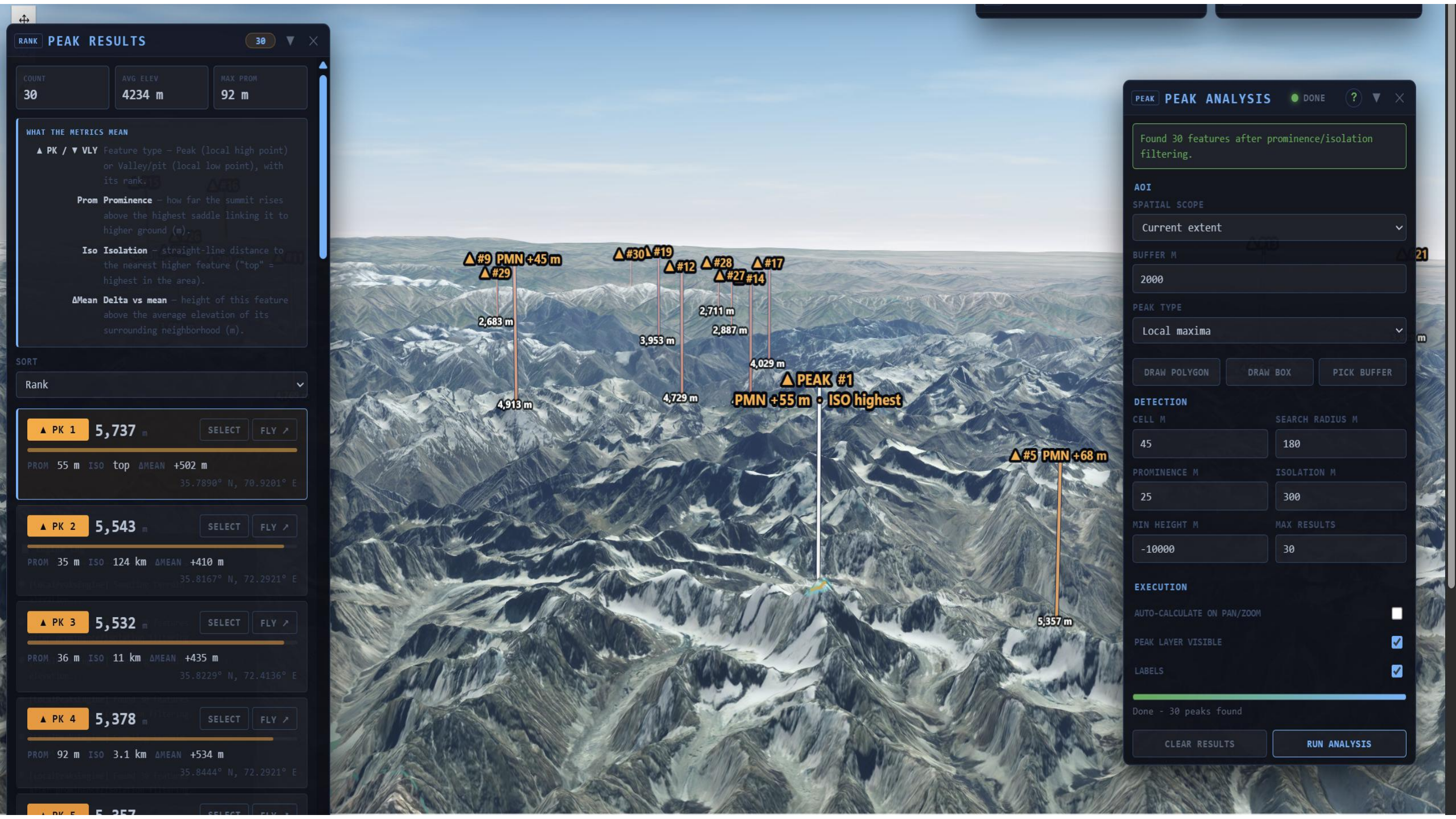
MIN HEIGHT M	MAX RESULTS
-10000	30

EXECUTION

- AUTO-CALCULATE ON PAN/ZOOM ☑
- PEAK LAYER VISIBLE ☑
- LABELS ☑

Done - 30 peaks found

CLEAR RESULTS | RUN ANALYSIS



RANK **PEAK RESULTS** 30

COUNT	AVG ELEV	MAX PROM
30	4234 m	92 m

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- **Iso Isolation** – straight-line distance to the nearest higher feature ("top" = highest in the area).
- **ΔMean Delta vs mean** – height of this feature above the average elevation of its surrounding neighborhood (m).

SORT

Rank

▲ PK 1	5,737 m	SELECT	FLY ↗
PROM 55 m ISO top ΔMEAN +502 m			35.7890° N, 70.9201° E
▲ PK 2	5,543 m	SELECT	FLY ↗
PROM 35 m ISO 124 km ΔMEAN +410 m			35.8167° N, 72.2921° E
▲ PK 3	5,532 m	SELECT	FLY ↗
PROM 36 m ISO 11 km ΔMEAN +435 m			35.8229° N, 72.4136° E
▲ PK 4	5,378 m	SELECT	FLY ↗
PROM 92 m ISO 3.1 km ΔMEAN +534 m			35.8444° N, 72.2921° E

PEAK **PEAK ANALYSIS** DONE ?

Found 30 features after prominence/isolation filtering.

AOI

SPATIAL SCOPE: Current extent

BUFFER M: 2000

PEAK TYPE: Local maxima

DRAW POLYGON | DRAW BOX | PICK BUFFER

DETECTION

CELL M	SEARCH RADIUS M
45	180
PROMINENCE M	**ISOLATION M**
25	300
MIN HEIGHT M	**MAX RESULTS**
-10000	30

EXECUTION

- AUTO-CALCULATE ON PAN/ZOOM ☐
- PEAK LAYER VISIBLE ☑
- LABELS ☑

Done - 30 peaks found

CLEAR RESULTS | RUN ANALYSIS

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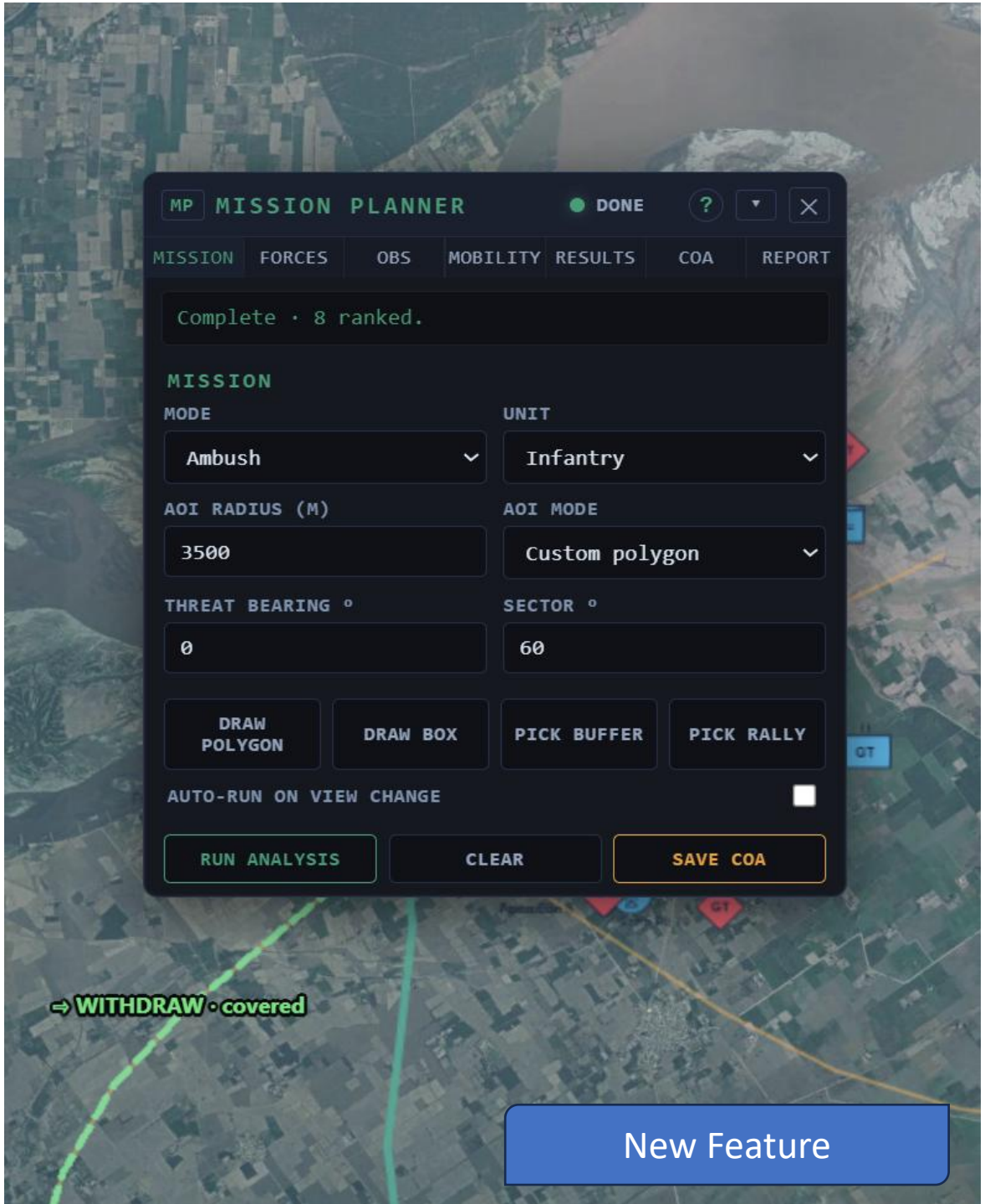
Analysis Tools

New Feature

OCOKA — AVENUES
9 AVENUES RANKED BY COMPOSITE SCORE
CORRIDORS
9
BEST AVENUE
82
WORST SCORE
55
1
Avenue 1
82
PRIMARY AVENUE - ~621M WIDE - 6320M
Width 73 Masking 85
Traff. 91 Observ. 100
Cover 88 Obstacle 44
No terrain chokepoints detected along this avenue.
Road approach — 6.3 km, 5 min · SLOW-GO (limiting: Secondary).
HIGH THREAT AVENUE - MONITOR, DENY, OR BLOCK
2
Avenue 2
81
PRIMARY AVENUE - ~495M WIDE - 6310M
Width 58 Masking 63
Traff. 100 Observ. 97
Cover 76 Obstacle 90
No terrain chokepoints detected along this avenue.
Road approach — 6.3 km, 5 min · GO (limiting: Primary).
HIGH THREAT AVENUE - MONITOR, DENY, OR BLOCK
3
Avenue 3
79
SECONDARY AVENUE - ~478M WIDE - 4210M
Width 56 Masking 62
Traff. 95 Observ. 100
Cover 79 Obstacle 77
No terrain chokepoints detected along this avenue.
Road approach — 4.2 km, 5 min · SLOW-GO (limiting: Secondary).
HIGH THREAT AVENUE - MONITOR, DENY, OR BLOCK
4
Avenue 4
78
SECONDARY AVENUE - ~417M WIDE - 6660M
A4
A2
A7
A6
A9
A8
A3
A5
Centre set - 34.0797N 71.5980E - click Run OCOKA
NLS, OS, NMA, Geodatastyrelsen and the GIS User Community
OCOKA CONFIG
DONE
34.0797
71.5980
PICK
ANALYSIS RADIUS
5 km - company level
CORRIDOR EXTRACTION
GRID CELL SIZE (M)
50 m - balanced
Max corridors 9
Slope threshold (deg) 12deg
FORCE TYPE
TRAFFICABILITY STANDARD
Wheeled vehicles
SCORING WEIGHTS (DRAG TO ADJUST)
WIDTH 8
MASKING 4
TRAFFICABILITY 10
OBSERVATION 4
COVER & CONCEALMENT 3
OBSTACLE 3
DISPLAY OPTIONS
SLOPE HEATMAP OVERLAY
CORRIDOR CENTRELINES
WIDTH POLYGONS
CHOKEPOINT MARKERS
SCORE COLOUR KEY
80-100 - Primary avenue of approach
60-79 - Secondary avenue
40-59 - Restricted / limited use
0-39 - Unlikely / obstacle-rich
Done - 9 avenues, best score 82
CLEAR
RUN OCOKA ANALYSIS

OCOKA — AVENUES
9 AVENUES RANKED BY COMPOSITE SCORE
CORRIDORS
9
BEST AVENUE
85
WORST SCORE
69
1
Avenue 1
85
PRIMARY AVENUE - ~622M WIDE - 7520M
Width 73 Masking 86
Traff. 100 Observ. 100
Cover 88 Obstacle 43
No terrain chokepoints detected along this avenue.
Road approach — 7.5 km, 6 min · GO (limiting: Primary).
HIGH THREAT AVENUE - MONITOR, DENY, OR BLOCK
2
Avenue 2
81
PRIMARY AVENUE - ~487M WIDE - 4600M
Width 57 Masking 62
Traff. 100 Observ. 97
Cover 75 Obstacle 92
No terrain chokepoints detected along this avenue.
Road approach — 4.6 km, 3 min · GO (limiting: Primary).
HIGH THREAT AVENUE - MONITOR, DENY, OR BLOCK
3
Avenue 3
80
PRIMARY AVENUE - ~485M WIDE - 9950M
Width 57 Masking 64
Traff. 100 Observ. 100
Cover 79 Obstacle 74
No terrain chokepoints detected along this avenue.
Road approach — 9.9 km, 7 min · GO (limiting: Primary).
HIGH THREAT AVENUE - MONITOR, DENY, OR BLOCK
4
Avenue 4
78
SECONDARY AVENUE - ~425M WIDE - 5020M
Width 50 Masking 58
Traff. 100 Observ. 92
A9
A2
A8
A1
A3
Centre set - 33.5890N 71.5069E - click Run OCOKA
Geodatastyrelsen and the GIS User Community
OCOKA CONFIG
DONE
33.5890
71.5069
PICK
ANALYSIS RADIUS
5 km - company level
CORRIDOR EXTRACTION
GRID CELL SIZE (M)
50 m - balanced
Max corridors 9
Slope threshold (deg) 12deg
FORCE TYPE
TRAFFICABILITY STANDARD
Tracked / armour
SCORING WEIGHTS (DRAG TO ADJUST)
WIDTH 8
MASKING 4
TRAFFICABILITY 10
OBSERVATION 4
COVER & CONCEALMENT 3
OBSTACLE 3
DISPLAY OPTIONS
SLOPE HEATMAP OVERLAY
CORRIDOR CENTRELINES
WIDTH POLYGONS
CHOKEPOINT MARKERS
SCORE COLOUR KEY
80-100 - Primary avenue of approach
60-79 - Secondary avenue
40-59 - Restricted / limited use
0-39 - Unlikely / obstacle-rich
Done - 9 avenues, best score 85
CLEAR
RUN OCOKA ANALYSIS



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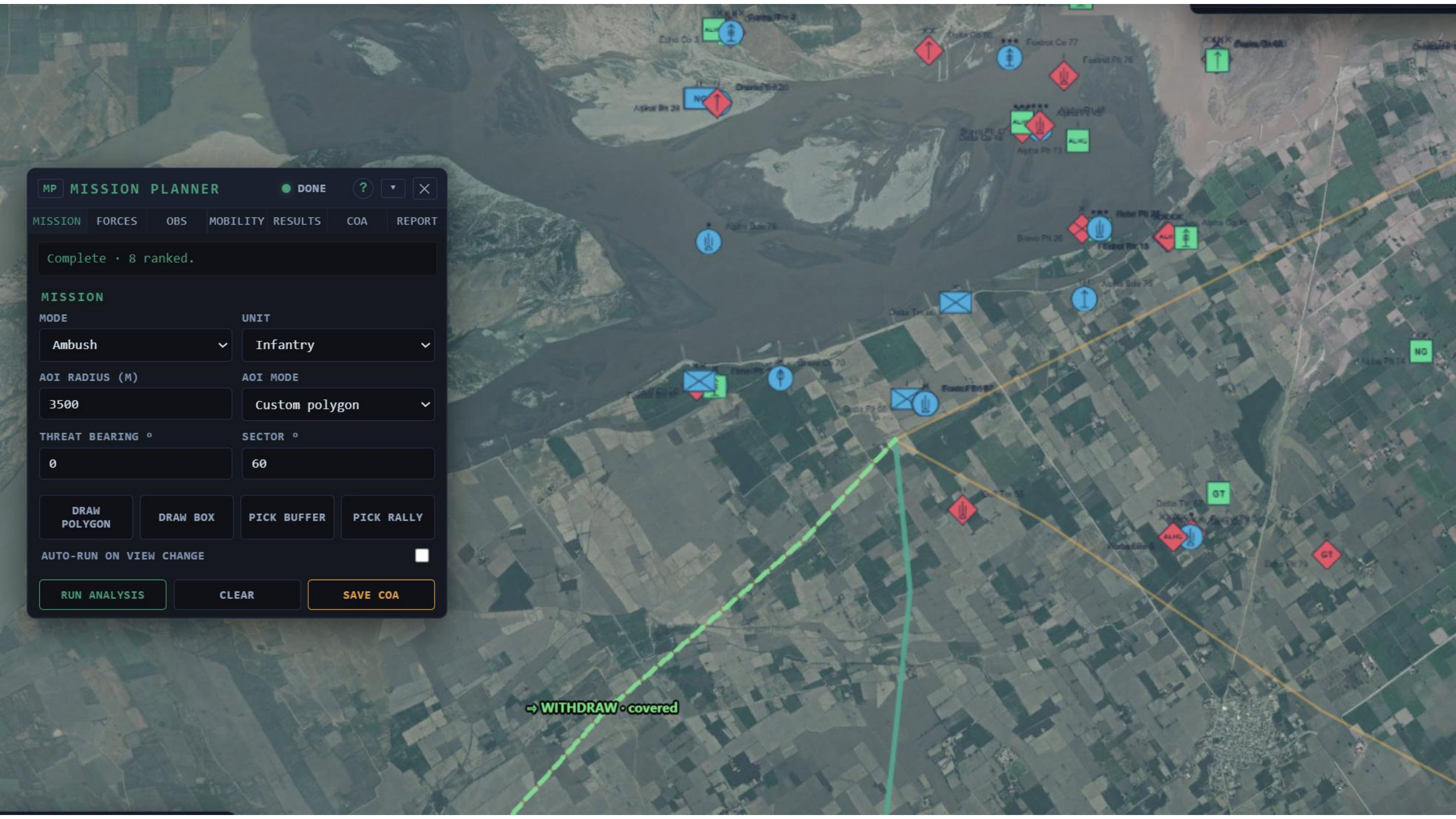
Analysis Tools


MP
MISSION PLANNER
DONE
MISSION
FORCES
OBS
MOBILITY
RESULTS
COA
REPORT
Complete · 8 ranked.
MISSION
MODE
Ambush
UNIT
Infantry
AOI RADIUS (M)
3500
AOI MODE
Custom polygon
THREAT BEARING °
0
SECTOR °
60
DRAW POLYGON
DRAW BOX
PICK BUFFER
PICK RALLY
AUTO-RUN ON VIEW CHANGE
RUN ANALYSIS
CLEAR
SAVE COA
⇨ WITHDRAW · covered

MP MISSION PLANNER DONE ? ▾ ×

MISSION | FORCES | OBS | MOBILITY | RESULTS | COA | REPORT

Complete · 8 ranked.

MISSION

MODE: Ambush

UNIT: Infantry

AOI RADIUS (M): 3500

AOI MODE: Custom polygon

THREAT BEARING °: 0

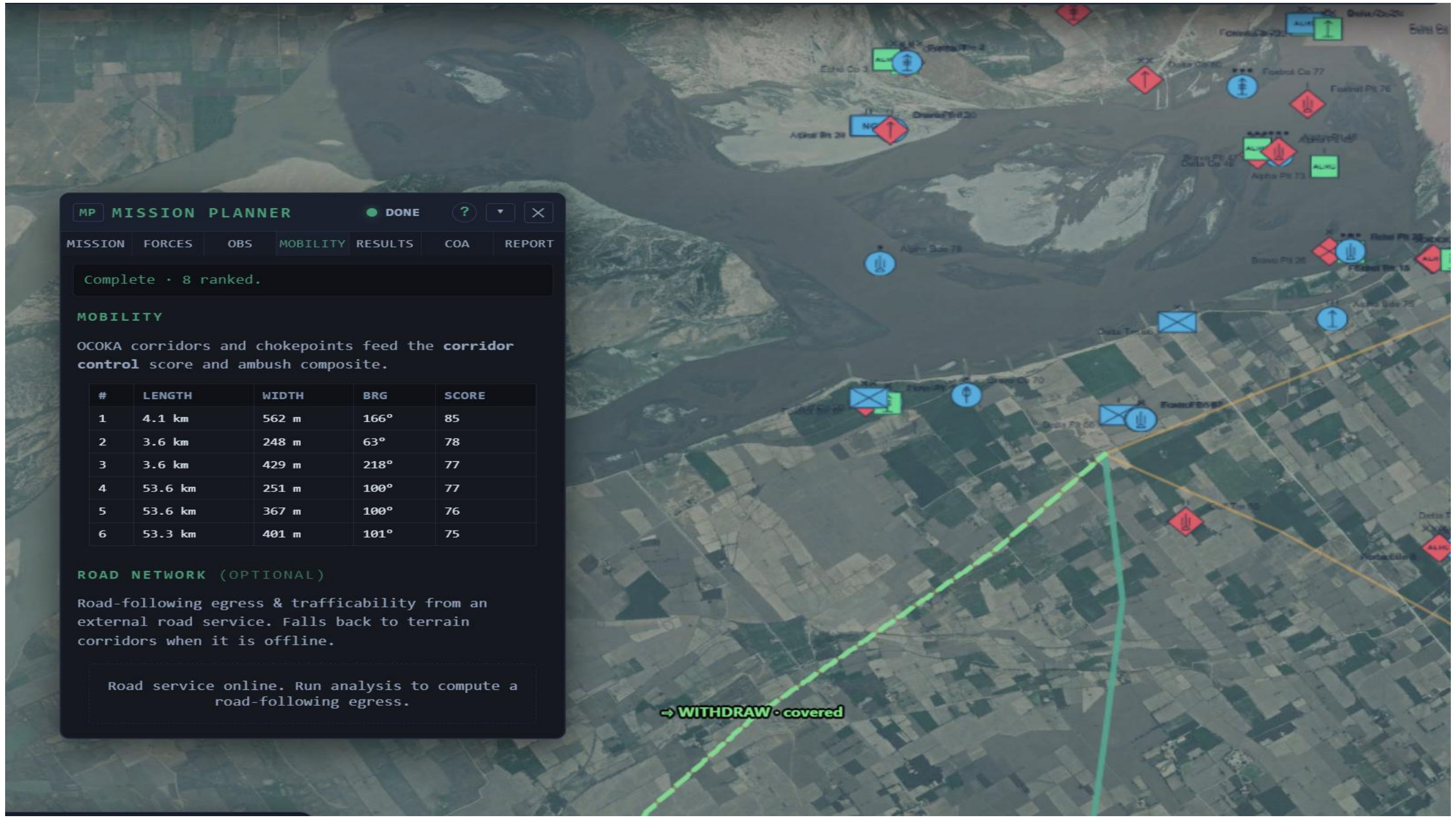
SECTOR °: 60

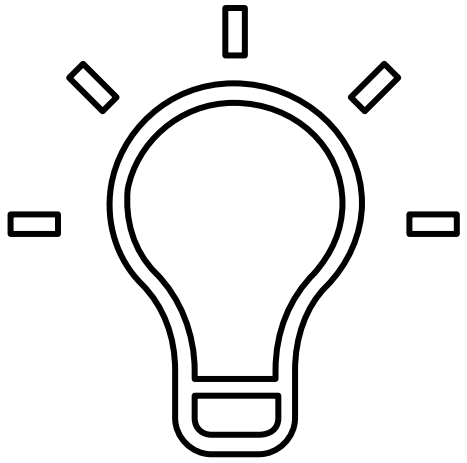
OBJECTIVE (OFFENSIVE): Enemy / threat

DRAW POLYGON | DRAW BOX | PICK BUFFER | PICK DEST

AUTO-RUN ON VIEW CHANGE ☐

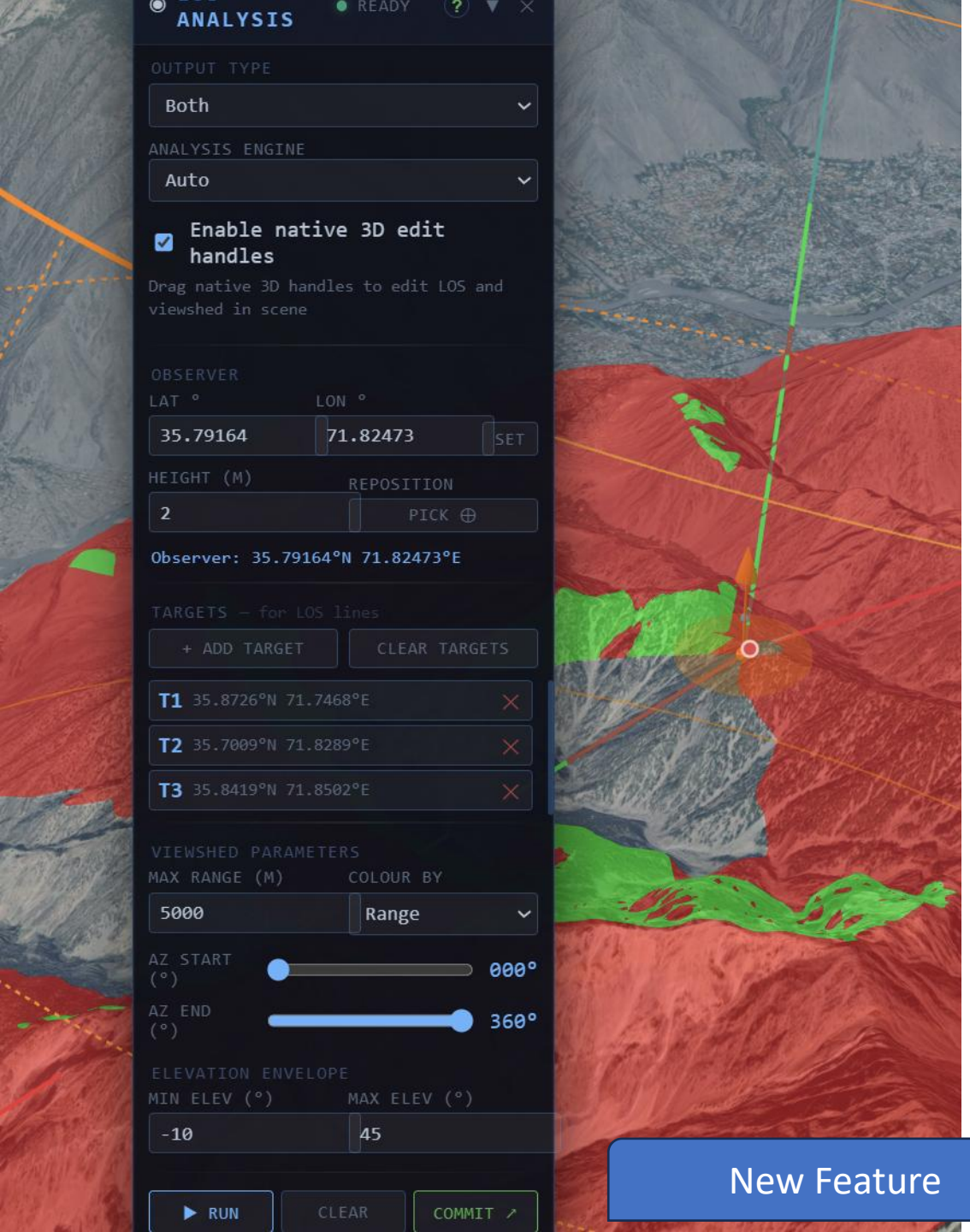
RUN ANALYSIS | CLEAR | SAVE COA


MP MISSION PLANNER
DONE
MISSION FORCES OBS MOBILITY RESULTS COA REPORT
Complete · 8 ranked.
MOBILITY
OCOKA corridors and chokepoints feed the corridor control score and ambush composite.
LENGTH WIDTH BRG SCORE
1 4.1 km 562 m 166° 85
2 3.6 km 248 m 63° 78
3 3.6 km 429 m 218° 77
4 53.6 km 251 m 100° 77
5 53.6 km 367 m 100° 76
6 53.3 km 401 m 101° 75
ROAD NETWORK (OPTIONAL)
Road-following egress & trafficability from an external road service. Falls back to terrain corridors when it is offline.
Road service online. Run analysis to compute a road-following egress.
⇨ WITHDRAW · covered



Analysis Tools

OP & Position Defensibility



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OP & Position Defensibility

Line of Sight / Viewshed

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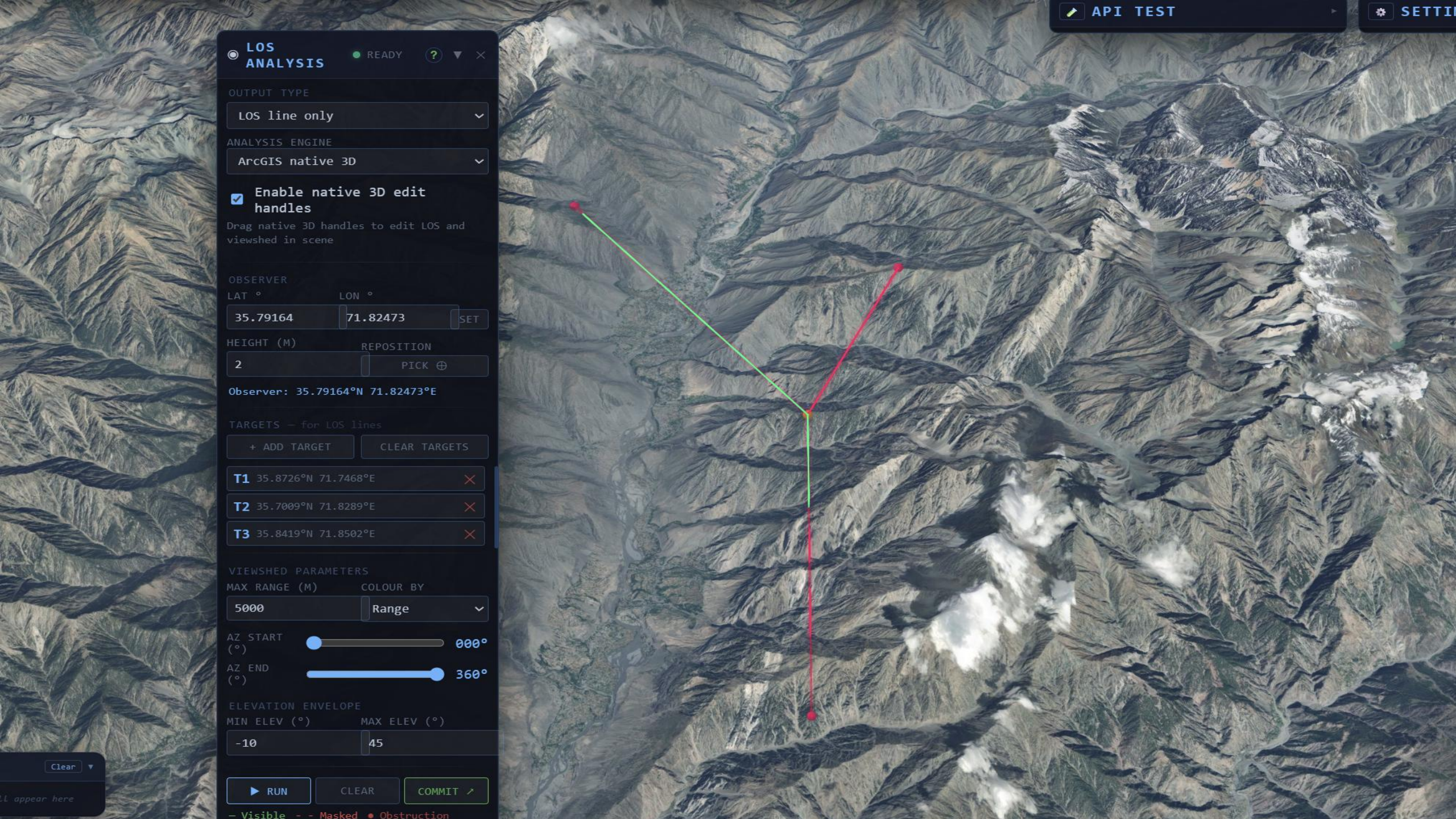
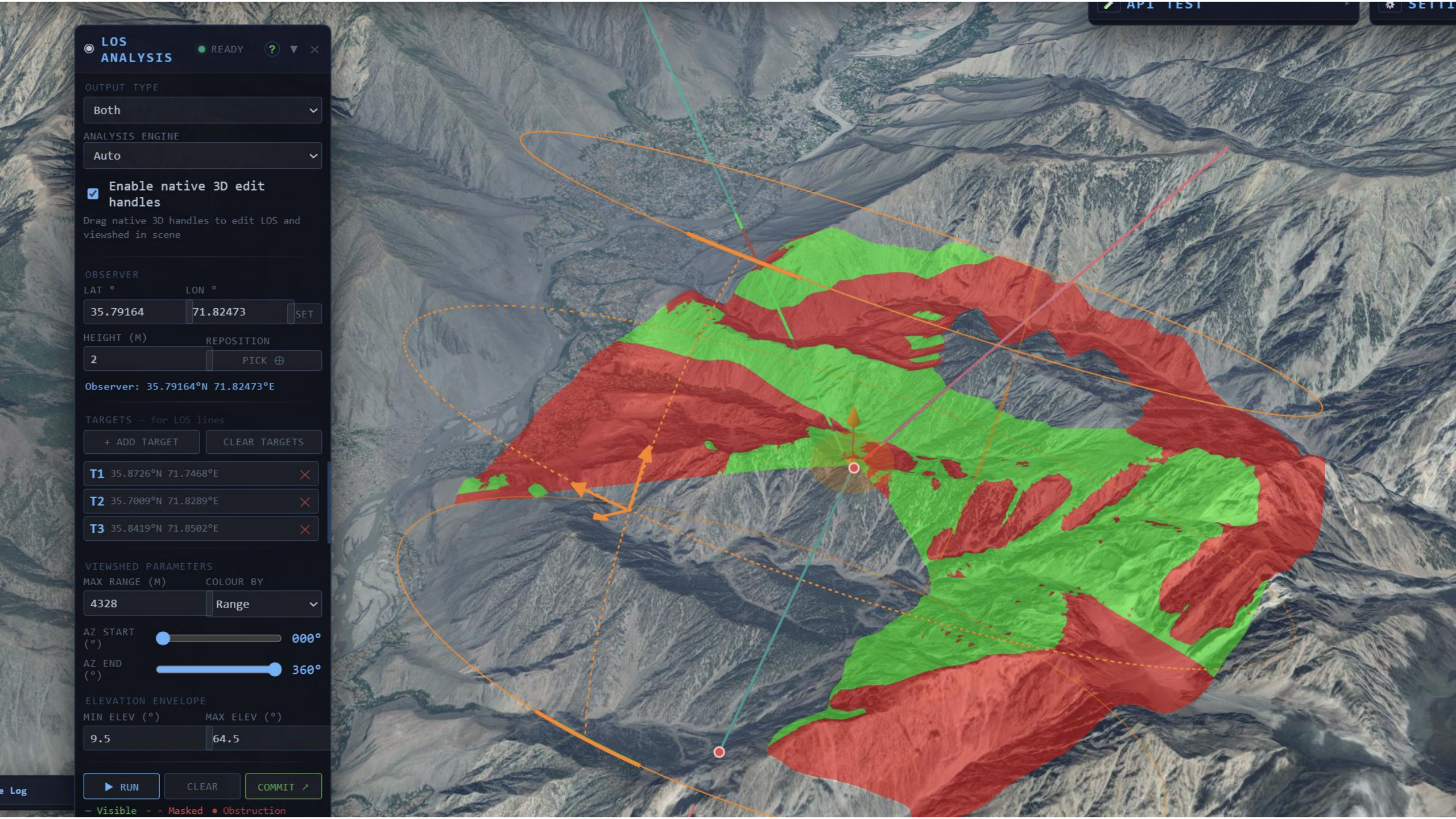
Observation Post Ranker

- Scores and ranks candidate OP positions by visibility coverage and defensibility.

Position Defensibility Scorer

- Rates a position across six military factors to assess how well it can be held.


API TEST
LOS ANALYSIS
READY
OUTPUT TYPE
LOS line only
ANALYSIS ENGINE
ArcGIS native 3D
Enable native 3D edit handles
Drag native 3D handles to edit LOS and viewshed in scene
OBSERVER
LAT °
LON °
35.79164
71.82473
SET
HEIGHT (M)
REPOSITION
2
PICK
Observer: 35.79164°N 71.82473°E
TARGETS — for LOS lines
+ ADD TARGET
CLEAR TARGETS
T1 35.8726°N 71.7468°E
T2 35.7009°N 71.8289°E
T3 35.8419°N 71.8502°E
VIEWSHED PARAMETERS
MAX RANGE (M)
COLOUR BY
5000
Range
AZ START (°)
000°
AZ END (°)
360°
ELEVATION ENVELOPE
MIN ELEV (°)
MAX ELEV (°)
-10
45
RUN
CLEAR
COMMIT
Visible
Masked
Obstruction
Clear


LOS ANALYSIS
READY
OUTPUT TYPE
Both
ANALYSIS ENGINE
Auto
Enable native 3D edit handles
Drag native 3D handles to edit LOS and viewshed in scene
OBSERVER
LAT °
LON °
35.79164
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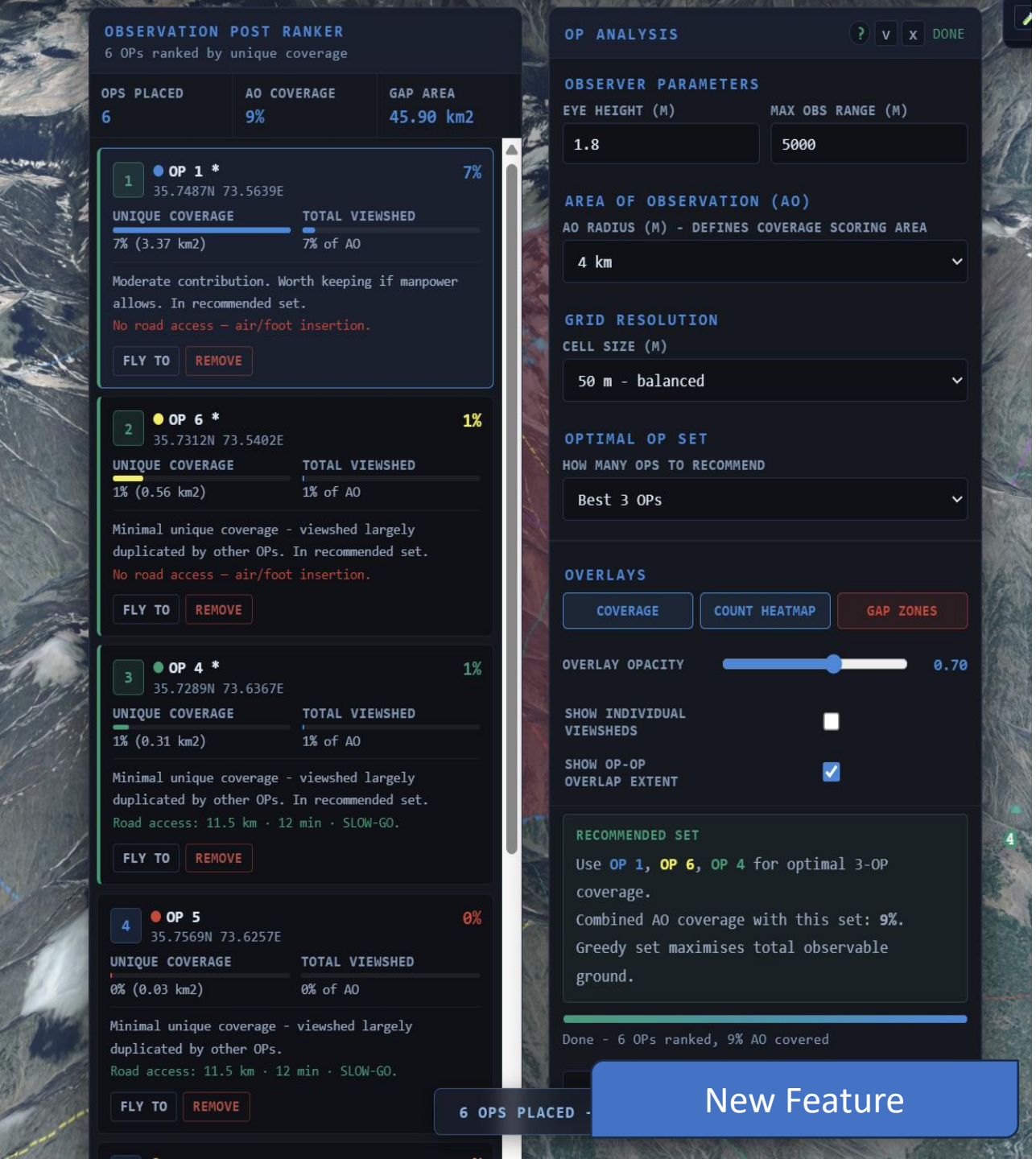
CLEAR TARGETS
T1 35.8726°N 71.7468°E
T2 35.7009°N 71.8289°E
T3 35.8419°N 71.8502°E
VIEWSHED PARAMETERS
MAX RANGE (M)
COLOUR BY
4328
Range
AZ START (°)
000°
AZ END (°)
360°
ELEVATION ENVELOPE
MIN ELEV (°)
MAX ELEV (°)
9.5
64.5
RUN
CLEAR
COMMIT
Visible
Masked
Obstruction



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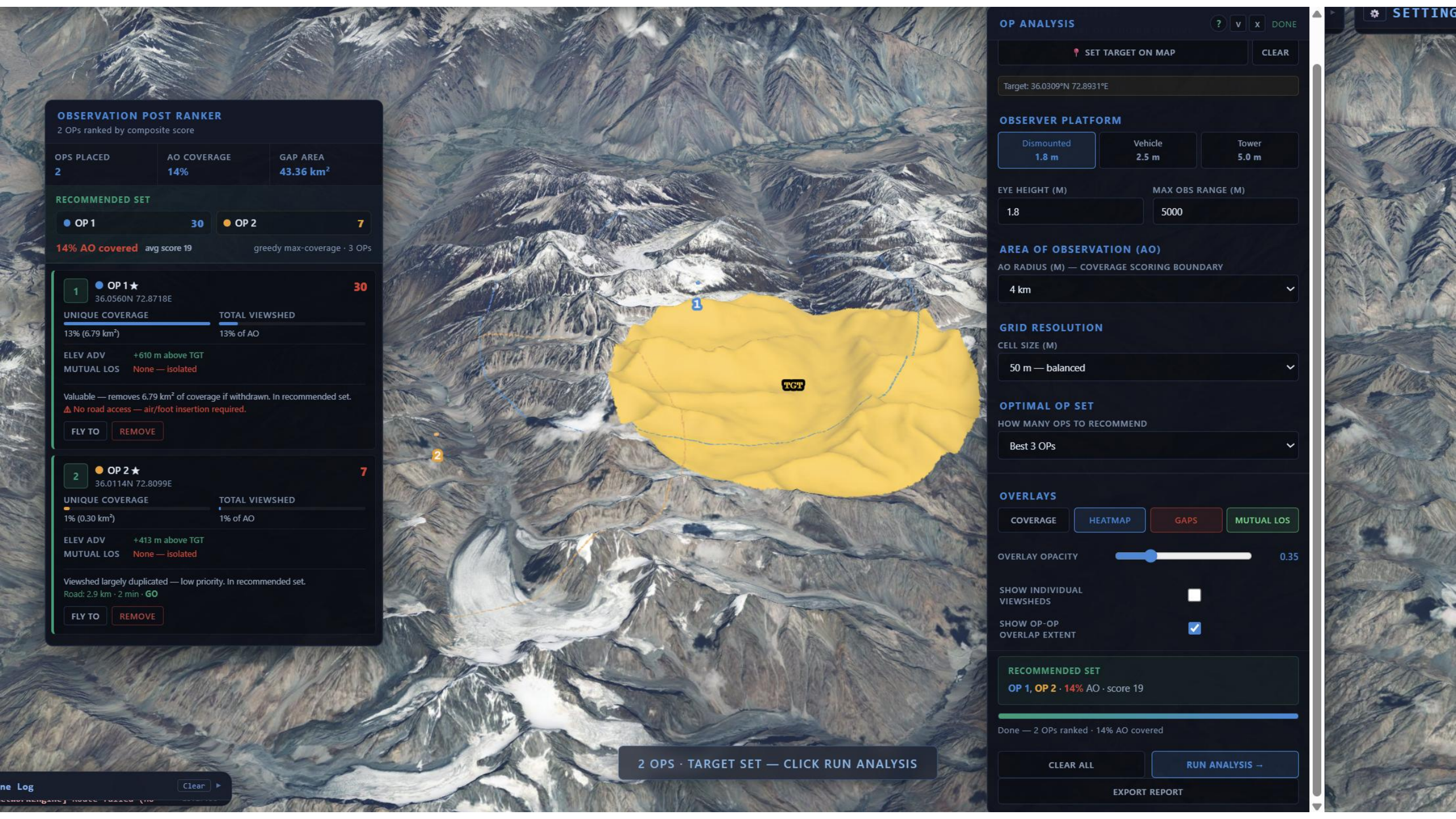


OBSERVATION POST RANKER

2 OPs ranked by composite score

OPS PLACED	AO COVERAGE	GAP AREA
2	14%	43.36 km²

RECOMMENDED SET

OP 1 30 · OP 2 7

14% AO covered avg score 19 — greedy max-coverage · 3 OPs

1 OP 1★ 30
36.0560N 72.8718E

UNIQUE COVERAGE	TOTAL VIEWSHED
13% (6.79 km²)	13% of AO

ELEV ADV +610 m above TGT
MUTUAL LOS None — isolated

Valuable — removes 6.79 km² of coverage if withdrawn. In recommended set.
⚠ No road access — air/foot insertion required.

FLY TO | REMOVE

2 OP 2★ 7
36.0114N 72.8099E

UNIQUE COVERAGE	TOTAL VIEWSHED
1% (0.30 km²)	1% of AO

ELEV ADV +413 m above TGT
MUTUAL LOS None — isolated

Viewshed largely duplicated — low priority. In recommended set.
Road: 2.9 km · 2 min · GO

FLY TO | REMOVE

ne Log Clear

OP ANALYSIS

? v x DONE

SET TARGET ON MAP | CLEAR

Target: 36.0309°N 72.8931°E

OBSERVER PLATFORM

Dismounted	Vehicle	Tower
1.8 m	2.5 m	5.0 m

EYE HEIGHT (M): 1.8
MAX OBS RANGE (M): 5000

AREA OF OBSERVATION (AO)

AO RADIUS (M) — COVERAGE SCORING BOUNDARY: 4 km

GRID RESOLUTION

CELL SIZE (M): 50 m — balanced

OPTIMAL OP SET

HOW MANY OPS TO RECOMMEND: Best 3 OPs

OVERLAYS

COVERAGE | HEATMAP | GAPS | MUTUAL LOS

OVERLAY OPACITY 0.35

SHOW INDIVIDUAL VIEWSHEDS ☐

SHOW OP-OP OVERLAP EXTENT ☑

RECOMMENDED SET

OP 1, OP 2 · 14% AO · score 19

Done — 2 OPs ranked · 14% AO covered

CLEAR ALL | RUN ANALYSIS →

EXPORT REPORT

SETTING

OBSERVATION POST RANKER

3 OPs ranked by composite score

OPS PLACED	AO COVERAGE	GAP AREA
3	25%	37.74 km²

RECOMMENDED SET

OP 1 36 · OP 3 1 · OP 2 29

25% AO covered avg score 22 — greedy max-coverage · 3 OPs

1 — OP 1 ★ 36
36.2135N 73.0188E

UNIQUE COVERAGE: 25% (12.58 km²) | TOTAL VIEWSHED: 25% of AO

ELEV ADV: +2203 m above TGT
MUTUAL LOS: None — isolated

Valuable — removes 12.58 km² of coverage if withdrawn. In recommended set.
Road: 0.0 km · 0 min · GO

FLY TO | REMOVE

2 — OP 2 ★ 29
36.2068N 72.9290E

UNIQUE COVERAGE: 0% (0.00 km²) | TOTAL VIEWSHED: 0% of AO

ELEV ADV: +2303 m above TGT
MUTUAL LOS: None — isolated

Viewshed largely duplicated — low priority. In recommended set.
Road: 4.0 km · 2 min · GO

FLY TO | REMOVE

3 — OP 3 ★ 1
36.1510N 73.0224E

UNIQUE COVERAGE: 1% (0.32 km²) | TOTAL VIEWSHED: 1% of AO

ELEV ADV: +1997 m above TGT
MUTUAL LOS: None — isolated

Viewshed largely duplicated — low priority. In recommended set.
⚠ No road access — air/foot insertion required.

FLY TO | REMOVE

Log Clear
[engine] Place OPs on map 15:21:02

3 OPS · TARGET SET — CLICK RUN ANALYSIS

OP ANALYSIS

? v x DONE

TARGET / AO CENTRE

CLICK TO SET WHERE OPS SHOULD OBSERVE

SET TARGET ON MAP | CLEAR

Target: 36.1849°N 73.0022°E

OBSERVER PLATFORM

Dismounted 1.8 m | Vehicle 2.5 m | Tower 5.0 m

EYE HEIGHT (M): 1.8

MAX OBS RANGE (M): 5000

AREA OF OBSERVATION (AO)

AO RADIUS (M) — COVERAGE SCORING BOUNDARY: 4 km

GRID RESOLUTION

CELL SIZE (M): 50 m — balanced

OPTIMAL OP SET

HOW MANY OPS TO RECOMMEND: Best 3 OPs

OVERLAYS

COVERAGE | HEATMAP | GAPS | MUTUAL LOS

OVERLAY OPACITY: 0.35

SHOW INDIVIDUAL VIEWSHEDS: ☐

SHOW OP-OP OVERLAP EXTENT: ☑

RECOMMENDED SET

OP 1, OP 3, OP 2 · 25% AO · score 22

Done — 3 OPs ranked · 25% AO covered

CLEAR ALL | RUN ANALYSIS →

OBSERVATION POST RANKER

4 OPs ranked by composite score

OPS PLACED	AO COVERAGE	GAP AREA
4	14%	43.38 km²

RECOMMENDED SET

OP 1 12 · OP 3 33 · OP 2 24

14% AO covered avg score 23 — greedy max-coverage · 3 OPs

1 OP 3 ★ 33
35.3832N 72.7837E

UNIQUE COVERAGE 6% (3.19 km²) — TOTAL VIEWSHED 6% of AO

ELEV ADV +147 m above TGT
MUTUAL LOS None — isolated

Moderate contribution. Worth keeping if manpower allows. In recommended set.
Road: 0.8 km · 1 min · GO

FLY TO | REMOVE

2 OP 2 ★ 24
35.4045N 72.6602E

UNIQUE COVERAGE 1% (0.41 km²) — TOTAL VIEWSHED 1% of AO

ELEV ADV -348 m below TGT
MUTUAL LOS None — isolated

Viewshed largely duplicated — low priority. In recommended set.
Road: 0.8 km · 1 min · GO

FLY TO | REMOVE

3 OP 1 ★ 12
35.4512N 72.6889E

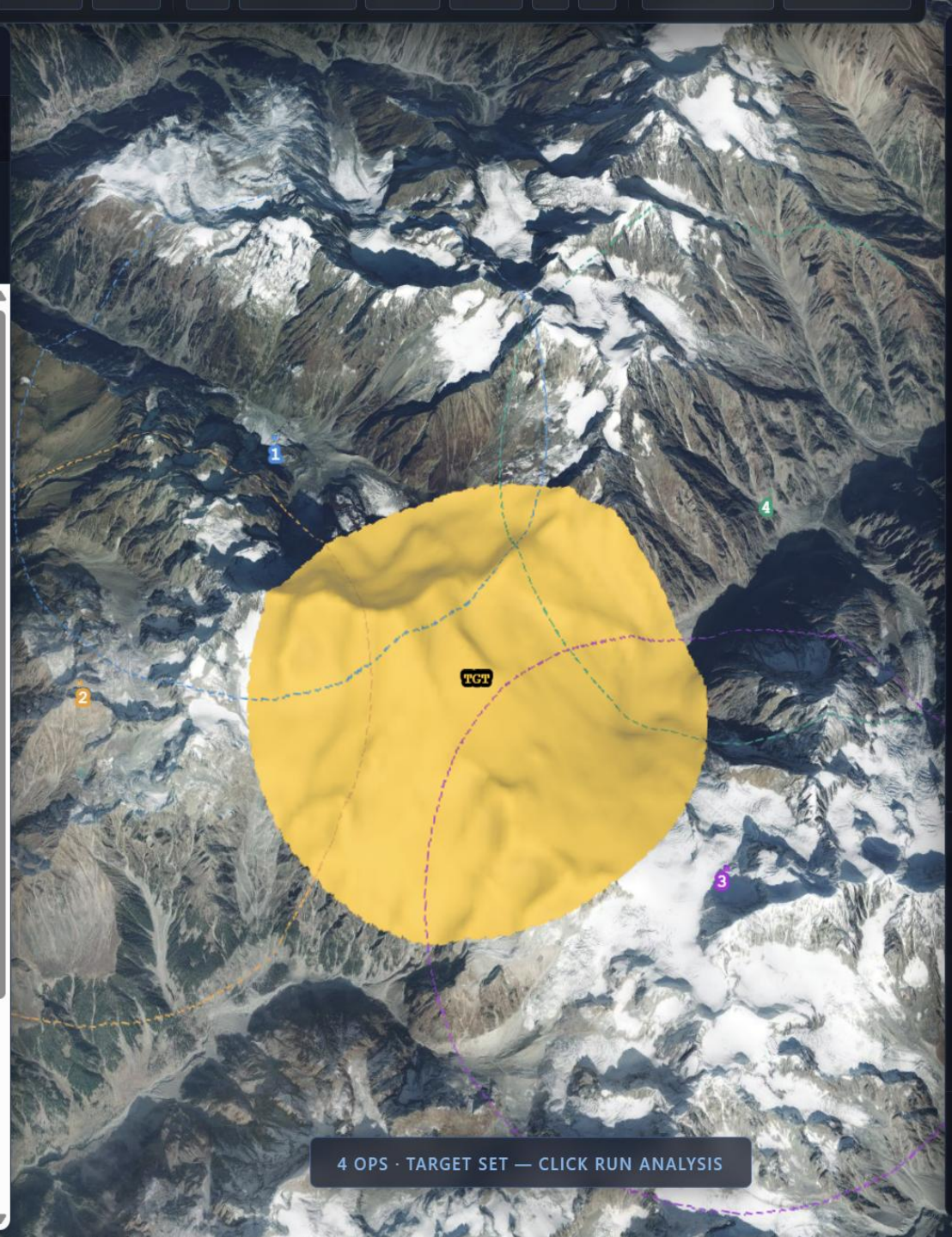
UNIQUE COVERAGE 7% (3.51 km²) — TOTAL VIEWSHED 7% of AO

ELEV ADV -1306 m below TGT
MUTUAL LOS None — isolated

Moderate contribution. Worth keeping if manpower allows. In recommended set.
⚠ No road access — air/foot insertion required.

FLY TO | REMOVE

OP 4 0



OP ANALYSIS

? v x DONE

TARGET / AO CENTRE

CLICK TO SET WHERE OPS SHOULD OBSERVE

SET TARGET ON MAP | CLEAR

Target: 35.4098°N 72.7354°E

OBSERVER PLATFORM

Dismounted	Vehicle	Tower
1.8 m	2.5 m	5.0 m

EYE HEIGHT (M): 1.8

MAX OBS RANGE (M): 5000

AREA OF OBSERVATION (AO)

AO RADIUS (M) — COVERAGE SCORING BOUNDARY: 4 km

GRID RESOLUTION

CELL SIZE (M): 50 m — balanced

OPTIMAL OP SET

HOW MANY OPS TO RECOMMEND: Best 3 OPs

OVERLAYS

COVERAGE | HEATMAP | GAPS | MUTUAL LOS

OVERLAY OPACITY 0.70

SHOW INDIVIDUAL VIEWSHEDS ☐

SHOW OP-OP OVERLAP EXTENT ☑

RECOMMENDED SET

OP 1, OP 3, OP 2 · 14% AO · score 23

Done — 4 OPs ranked · 14% AO covered

OP & Position Defensibility

Line of Sight / Viewshed

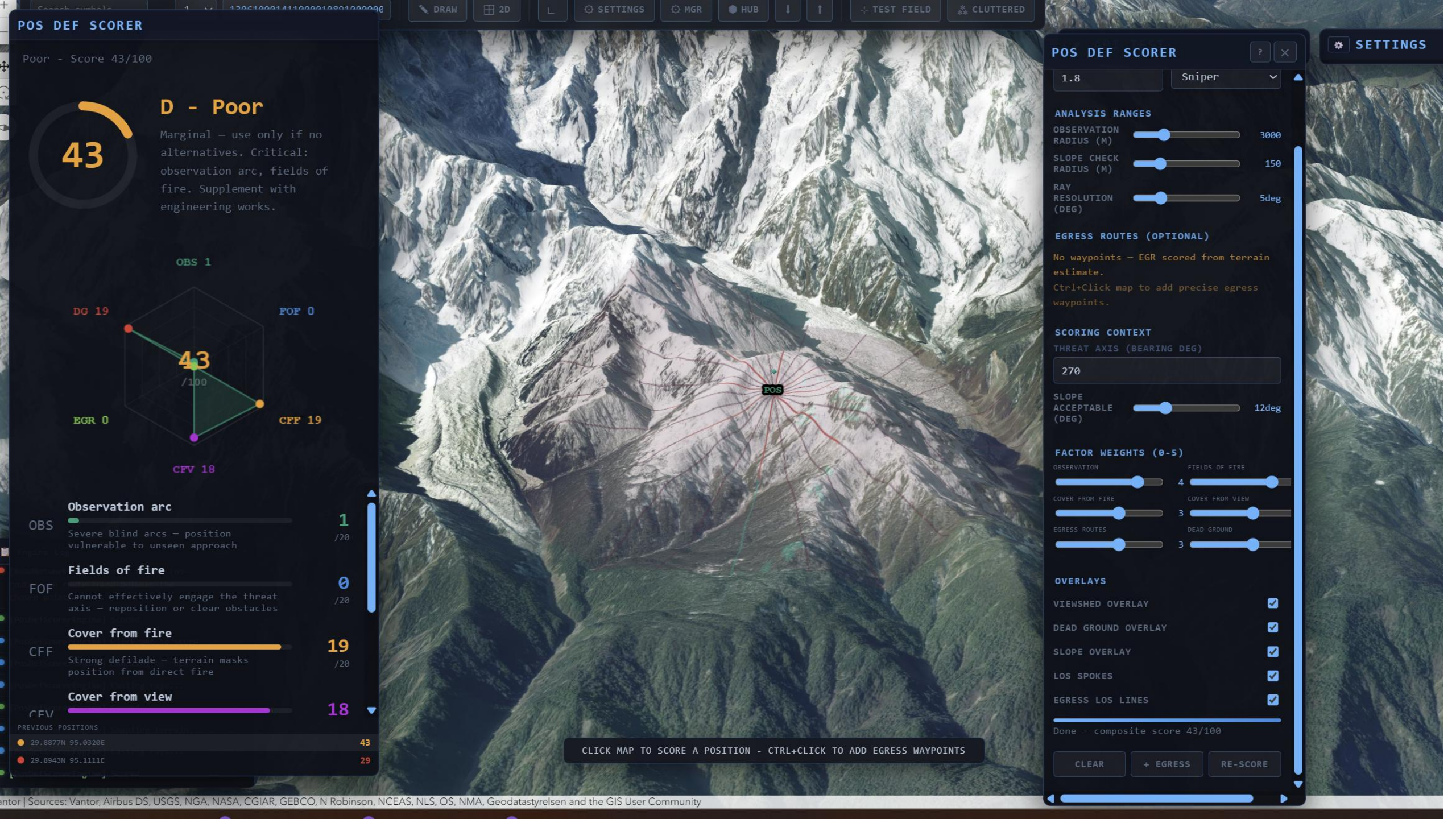
- Computes what an observer can and cannot see from any position on the map.

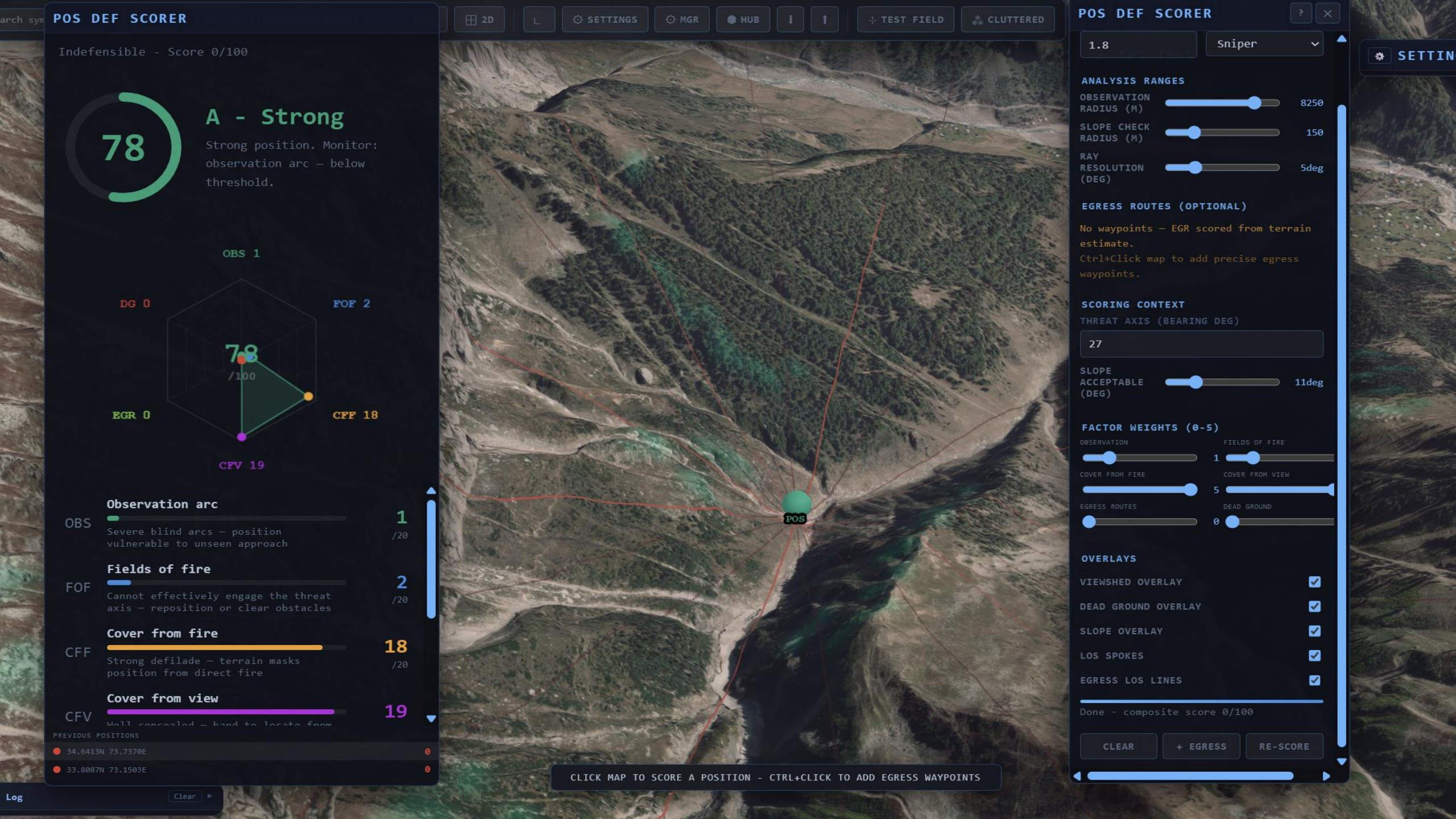
Observation Post Ranker

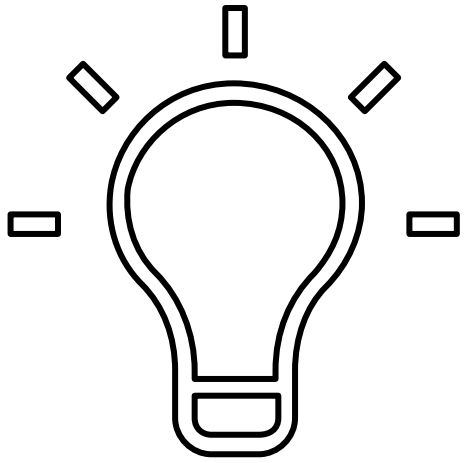
- Scores and ranks candidate OP positions by visibility coverage and defensibility.

Position Defensibility Scorer

- Rates a position across six military factors to assess how well it can be held.

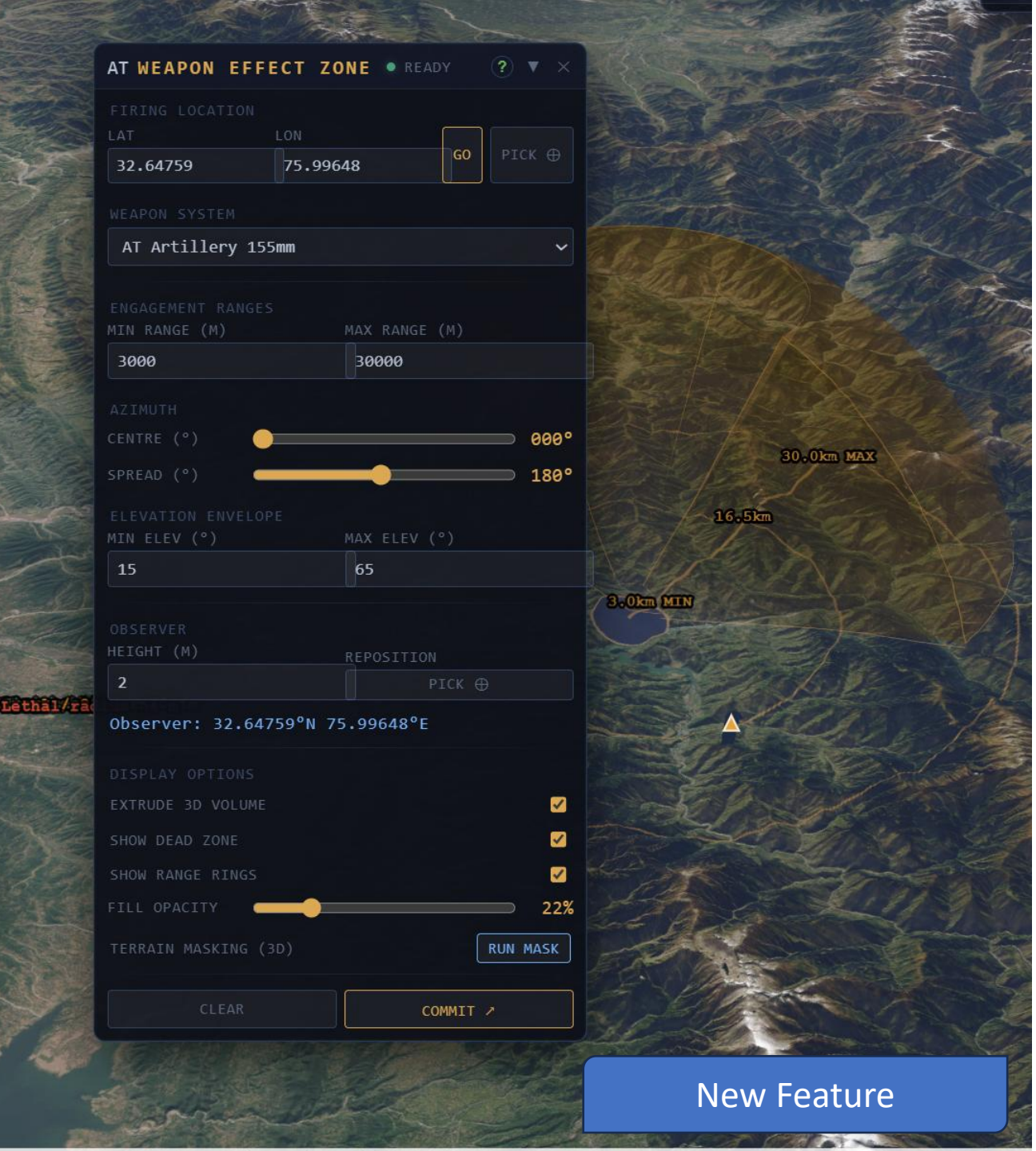

POS DEF SCORER
Poor - Score 43/100
43
D - Poor
Marginal — use only if no alternatives. Critical: observation arc, fields of fire. Supplement with engineering works.
OBS 1
DG 19
FOF 0
43
/100
EGR 0
CFF 19
CFV 18
OBS
Observation arc
Severe blind arcs — position vulnerable to unseen approach
1
/20
FOF
Fields of fire
Cannot effectively engage the threat axis — reposition or clear obstacles
0
/20
CFF
Cover from fire
Strong defilade — terrain masks position from direct fire
19
/20
Cover from view
18
PREVIOUS POSITIONS
29.8877N 95.0320E
43
29.8943N 95.1111E
29
DRAW
2D
SETTINGS
MGR
HUB
TEST FIELD
CLUTTERED
POS
CLICK MAP TO SCORE A POSITION - CTRL+CLICK TO ADD EGRESS WAYPOINTS
POS DEF SCORER
1.8
Sniper
ANALYSIS RANGES
OBSERVATION RADIUS (M)
3000
SLOPE CHECK RADIUS (M)
150
RAY RESOLUTION (DEG)
5deg
EGRESS ROUTES (OPTIONAL)
No waypoints — EGR scored from terrain estimate.
Ctrl+Click map to add precise egress waypoints.
SCORING CONTEXT
THREAT AXIS (BEARING DEG)
270
SLOPE ACCEPTABLE (DEG)
12deg
FACTOR WEIGHTS (0-5)
OBSERVATION
FIELDS OF FIRE
4
COVER FROM FIRE
COVER FROM VIEW
3
EGRESS ROUTES
DEAD GROUND
3
OVERLAYS
VIEWSHED OVERLAY
DEAD GROUND OVERLAY
SLOPE OVERLAY
LOS SPOKES
EGRESS LOS LINES
Done - composite score 43/100
CLEAR
+ EGRESS
RE-SCORE
SETTINGS
antor | Sources: Vantor, Airbus DS, USGS, NGA, NASA, CGIAR, GEBCO, N Robinson, NCEAS, NLS, OS, NMA, Geodatastyrelsen and the GIS User Community


POS DEF SCORER
Indefensible - Score 0/100
78
A - Strong
Strong position. Monitor: observation arc — below threshold.
OBS 1
DG 0
FOF 2
78
/100
EGR 0
CFF 18
CFV 19
OBS
Observation arc
Severe blind arcs — position vulnerable to unseen approach
1
/20
FOF
Fields of fire
Cannot effectively engage the threat axis — reposition or clear obstacles
2
/20
CFF
Cover from fire
Strong defilade — terrain masks position from direct fire
18
/20
CFV
Cover from view
19
PREVIOUS POSITIONS
34.6413N 73.7370E
33.8087N 73.1503E
Log
Clear
2D
SETTINGS
MGR
HUB
TEST FIELD
CLUTTERED
POS
POS DEF SCORER
1.8
Sniper
ANALYSIS RANGES
OBSERVATION RADIUS (M)
8250
SLOPE CHECK RADIUS (M)
150
RAY RESOLUTION (DEG)
5deg
EGRESS ROUTES (OPTIONAL)
No waypoints — EGR scored from terrain estimate.
Ctrl+Click map to add precise egress waypoints.
SCORING CONTEXT
THREAT AXIS (BEARING DEG)
27
SLOPE ACCEPTABLE (DEG)
11deg
FACTOR WEIGHTS (0-5)
OBSERVATION
FIELDS OF FIRE
1
COVER FROM FIRE
COVER FROM VIEW
5
EGRESS ROUTES
DEAD GROUND
0
OVERLAYS
VIEWSHED OVERLAY
DEAD GROUND OVERLAY
SLOPE OVERLAY
LOS SPOKES
EGRESS LOS LINES
Done - composite score 0/100
CLEAR
+ EGRESS
RE-SCORE
CLICK MAP TO SCORE A POSITION - CTRL+CLICK TO ADD EGRESS WAYPOINTS
SETTIN



Analysis Tools

UAV & Weapon Effects



UAV & Weapon Effects

Weapon Effect Zone (WEZ)

- Plots the engagement envelope of a weapon system around a selected position.

Ballistic Trajectory

- Models the flight path and impact point of a projectile from a fire position.

Munitions Effect Radius

- Computes the blast and effect zone of a detonation at a placed point.

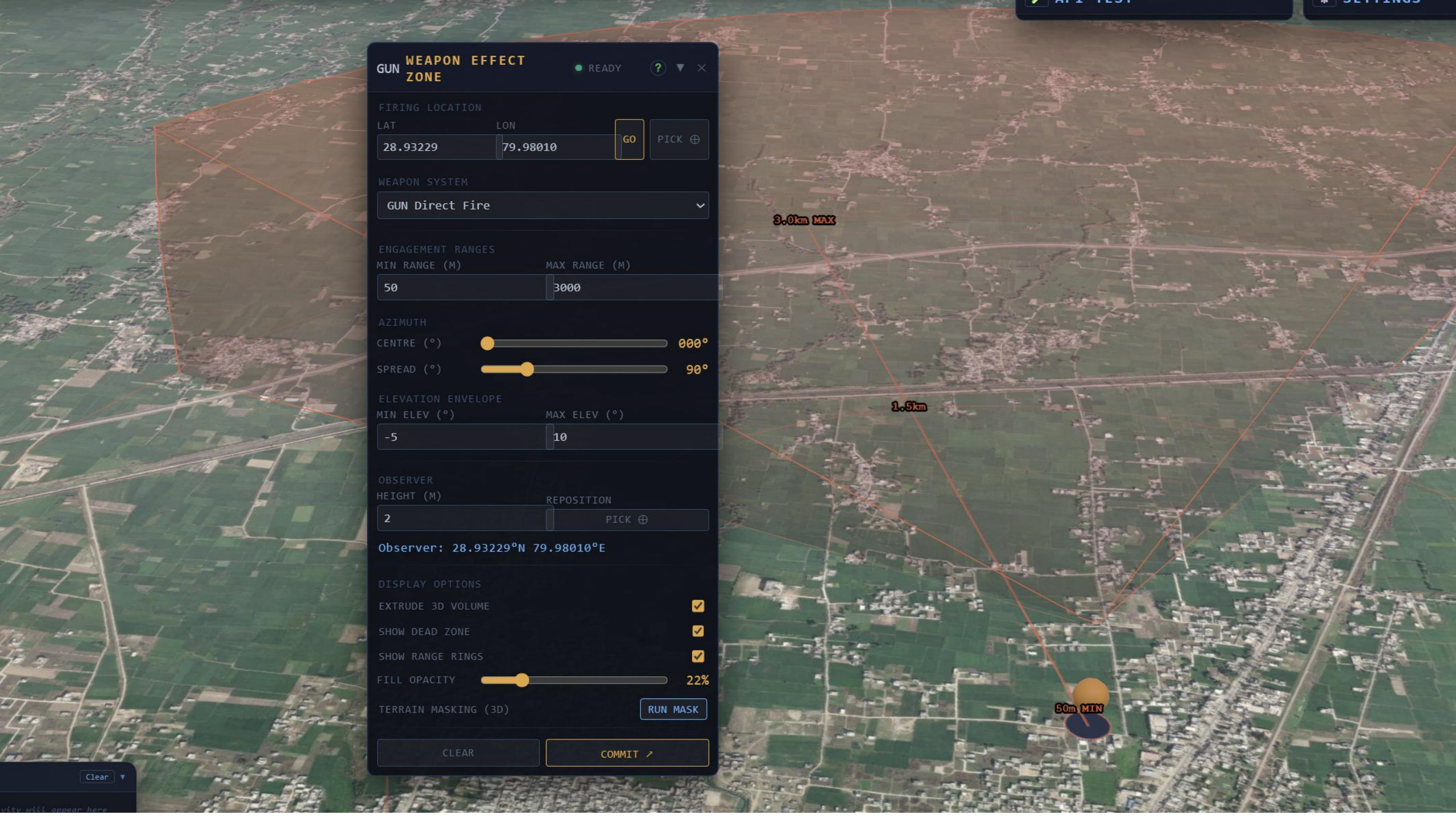
Buffer & Threat Rings

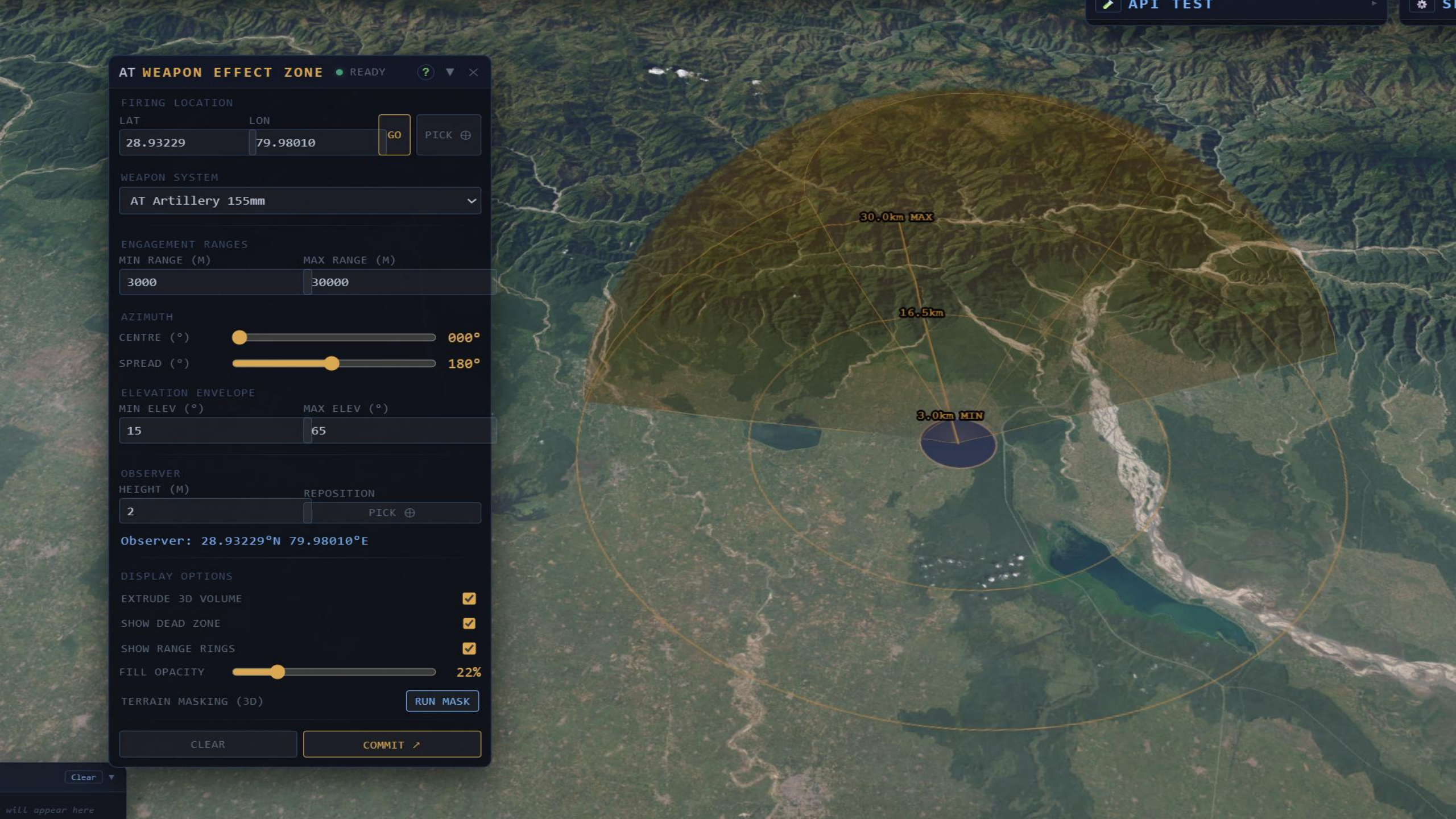
- Draws concentric threat rings around a position at configurable standoff distances.

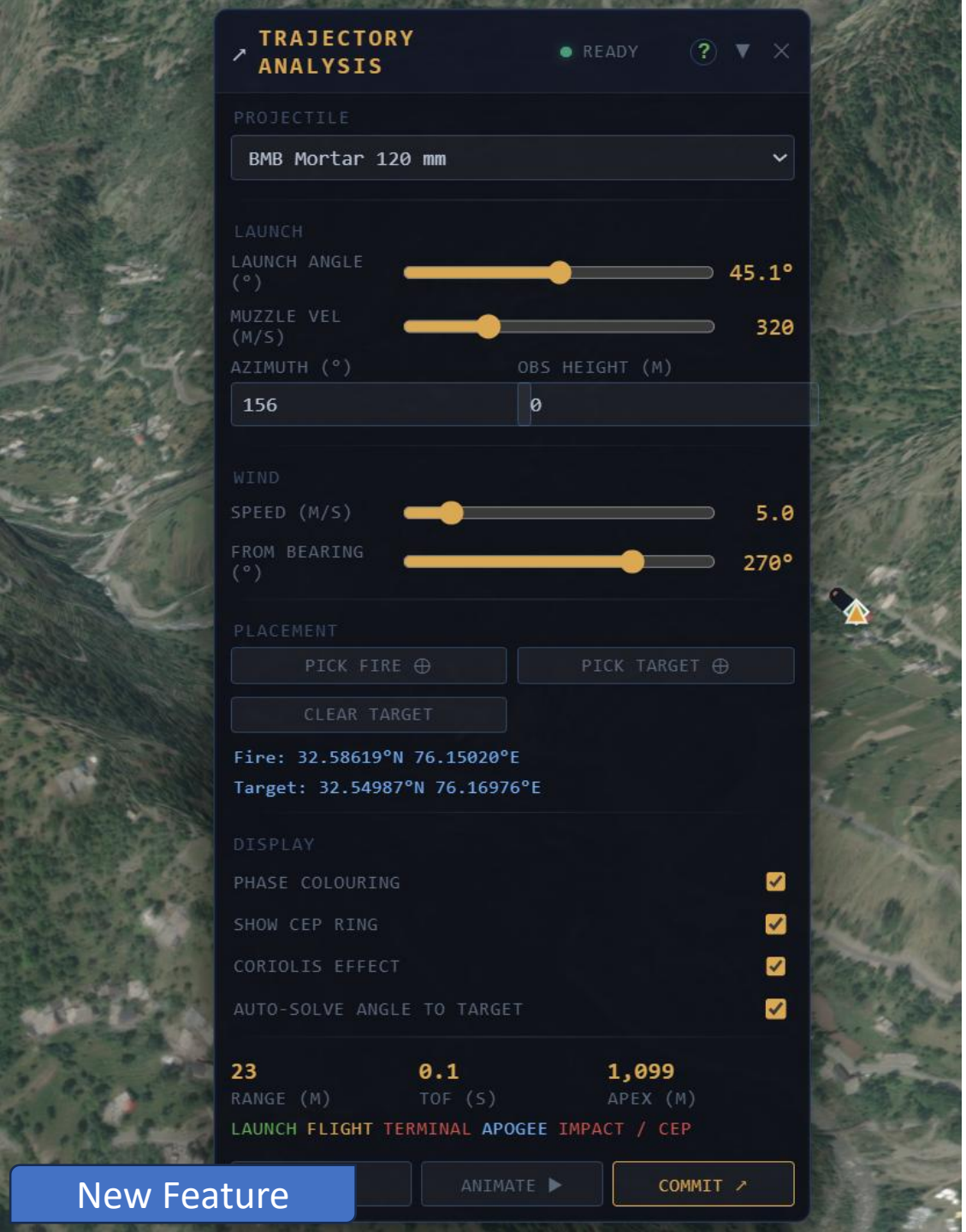
UAV Flight Analysis

- Plans UAV routes and calculates coverage, range, and terrain clearance along the path.

Analysis Tools


GUN WEAPON EFFECT ZONE
READY
FIRING LOCATION
LAT
LON
28.93229
79.98010
GO
PICK
WEAPON SYSTEM
GUN Direct Fire
ENGAGEMENT RANGES
MIN RANGE (M)
MAX RANGE (M)
50
3000
AZIMUTH
CENTRE (°)
000°
SPREAD (°)
90°
ELEVATION ENVELOPE
MIN ELEV (°)
MAX ELEV (°)
-5
10
OBSERVER
HEIGHT (M)
REPOSITION
2
PICK
Observer: 28.93229°N 79.98010°E
DISPLAY OPTIONS
EXTRUDE 3D VOLUME
SHOW DEAD ZONE
SHOW RANGE RINGS
FILL OPACITY
22%
TERRAIN MASKING (3D)
RUN MASK
CLEAR
COMMIT ↗
3.0km MAX
1.5km
50m MIN
Clear


API TEST
AT WEAPON EFFECT ZONE
READY
FIRING LOCATION
LAT
LON
28.93229
79.98010
GO
PICK
WEAPON SYSTEM
AT Artillery 155mm
ENGAGEMENT RANGES
MIN RANGE (M)
MAX RANGE (M)
3000
30000
AZIMUTH
CENTRE (°)
000°
SPREAD (°)
180°
ELEVATION ENVELOPE
MIN ELEV (°)
MAX ELEV (°)
15
65
OBSERVER
HEIGHT (M)
REPOSITION
2
PICK
Observer: 28.93229°N 79.98010°E
DISPLAY OPTIONS
EXTRUDE 3D VOLUME
SHOW DEAD ZONE
SHOW RANGE RINGS
FILL OPACITY
22%
TERRAIN MASKING (3D)
RUN MASK
CLEAR
COMMIT ↗
30.0km MAX
16.5km
3.0km MIN
Clear
will appear here



New Feature

UAV & Weapon Effects

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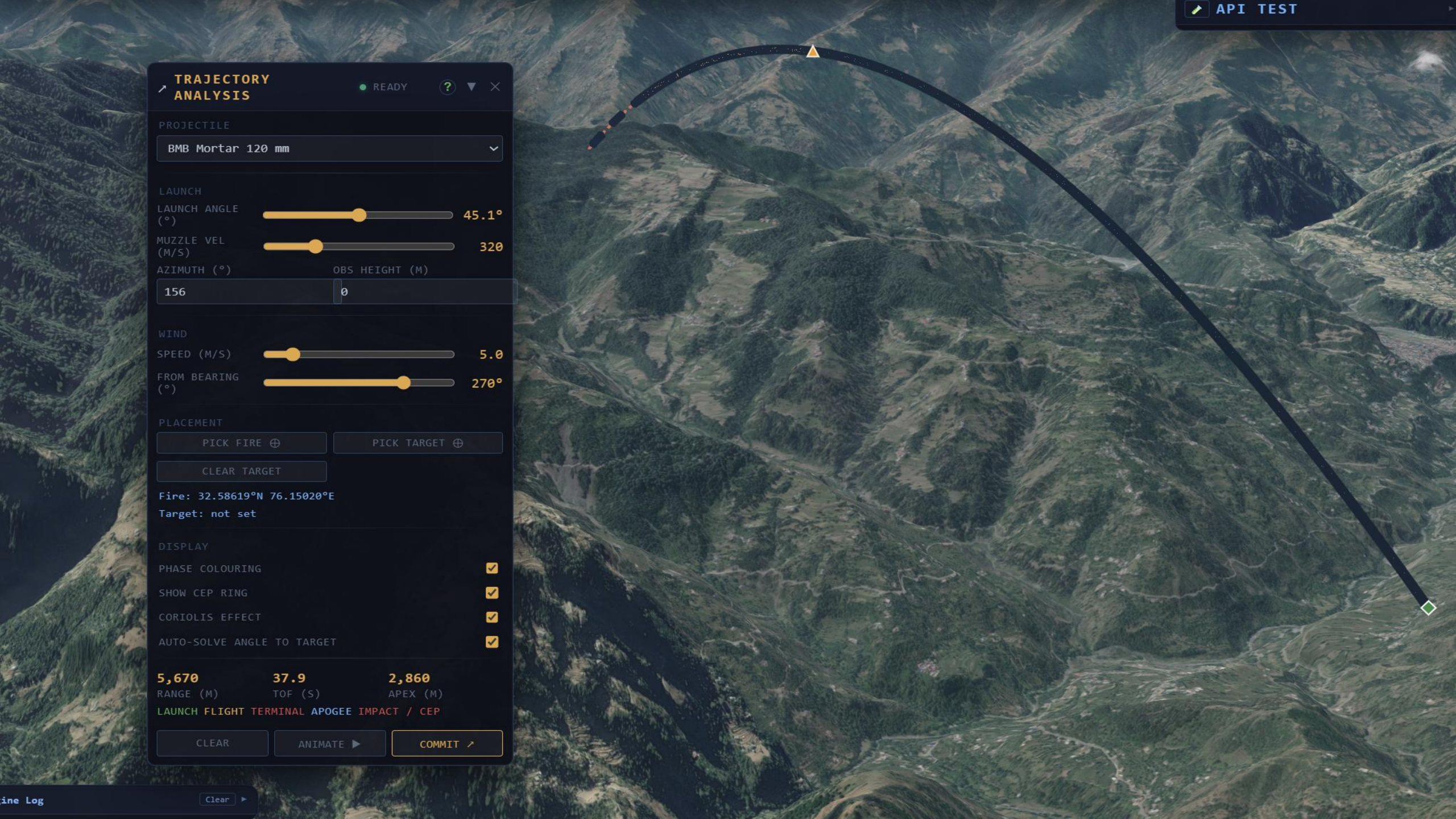
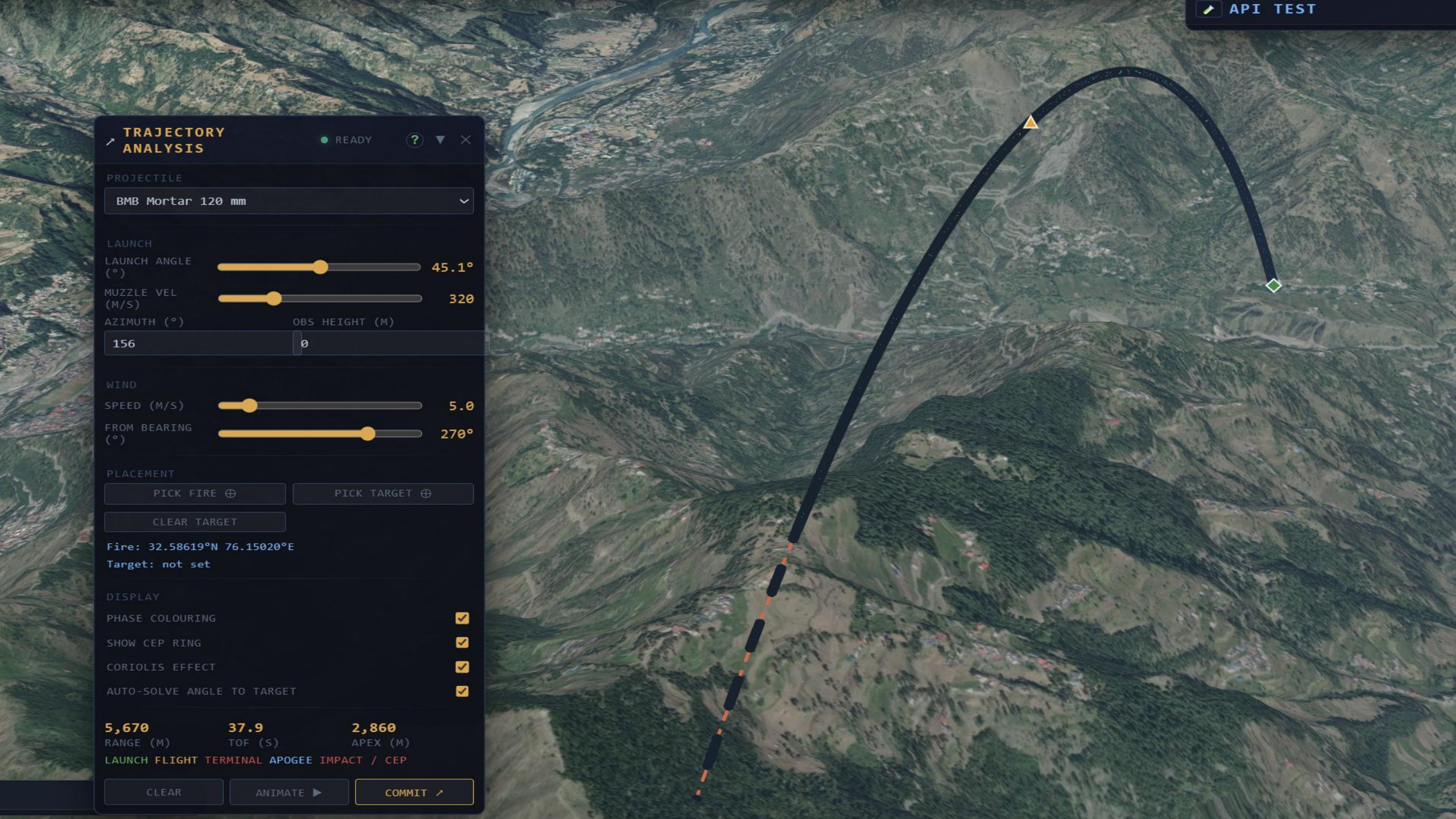
UAV Flight Analysis

- Plans UAV routes and calculates coverage, range, and terrain clearance along the path.

Analysis Tools


API TEST
TRAJECTORY ANALYSIS
READY
PROJECTILE
BMB Mortar 120 mm
LAUNCH
LAUNCH ANGLE (°)
45.1°
MUZZLE VEL (M/S)
320
AZIMUTH (°)
OBS HEIGHT (M)
156
0
WIND
SPEED (M/S)
5.0
FROM BEARING (°)
270°
PLACEMENT
PICK FIRE
PICK TARGET
CLEAR TARGET
Fire: 32.58619°N 76.15020°E
Target: not set
DISPLAY
PHASE COLOURING
SHOW CEP RING
CORIOLIS EFFECT
AUTO-SOLVE ANGLE TO TARGET
5,670
RANGE (M)
37.9
TOF (S)
2,860
APEX (M)
LAUNCH FLIGHT TERMINAL APOGEE IMPACT / CEP
CLEAR
ANIMATE
COMMIT
ine Log
Clear


API TEST
TRAJECTORY ANALYSIS
READY
PROJECTILE
BMB Mortar 120 mm
LAUNCH
LAUNCH ANGLE (°)
45.1°
MUZZLE VEL (M/S)
320
AZIMUTH (°)
OBS HEIGHT (M)
156
0
WIND
SPEED (M/S)
5.0
FROM BEARING (°)
270°
PLACEMENT
PICK FIRE
PICK TARGET
CLEAR TARGET
Fire: 32.58619°N 76.15020°E
Target: not set
DISPLAY
PHASE COLOURING
SHOW CEP RING
CORIOLIS EFFECT
AUTO-SOLVE ANGLE TO TARGET
5,670
RANGE (M)
37.9
TOF (S)
2,860
APEX (M)
LAUNCH FLIGHT TERMINAL APOGEE IMPACT / CEP
CLEAR
ANIMATE
COMMIT

UAV & Weapon Effects

Weapon Effect Zone (WEZ)

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Buffer & Threat Rings

- Draws concentric threat rings around a position at configurable standoff distances.

UAV Flight Analysis

- Plans UAV routes and calculates coverage, range, and terrain clearance along the path.

Analysis Tools

API TEST

WEAPON EFFECT READY ?

MUNITION / DEVICE

TYPE

GBU-12 500 lb

TNT EQUIV. (KG) 89

DET. HEIGHT (M) 0

ENVIRONMENT / STRUCTURE

STRUCTURAL FACTOR

Reinforced concrete

DISPLAY OPTIONS

DONUT RINGS (PUNCH INNER)

RING LABELS

SHOW BLAST WAVE

SHOW IMPACT DOME

MULTI-STRIKE UNION

ANIM SPEED 1×

DOME OPACITY 60%

BLAST OPACITY 35%

COMPUTED RADII — HOPKINSON-CRANZ MODEL

LETHAL (BLAST+FRAG) 4 m	INJURY — BLAST 6 m
FRAG CASUALTY 6 m	THERMAL / 3° BURN 5 m
SAFE — BLAST 14 m	QD INHABITED 99 m

STRIKES PLACED (CLICK MAP TO ADD)

1 32.3409°N 75.7293°E

Strike 1: 32.3409°N 75.7293°E

CLEAR ALL | UNDO LAST | ▶ BLAST WAVE

COMMIT TO MAP ↗

RAG)
UALTY
N
BLAST
LDINGS
ON
Clear

API TEST
SETTINGS
WEAPON EFFECT
READY
MUNITION / DEVICE
TYPE
IED 10 kg TNT
TNT EQUIV. (KG)
DET. HEIGHT (M)
10
0
ENVIRONMENT / STRUCTURE
STRUCTURAL FACTOR
Open area
DISPLAY OPTIONS
DONUT RINGS (PUNCH INNER)
RING LABELS
SHOW BLAST WAVE
SHOW IMPACT DOME
MULTI-STRIKE UNION
ANIM SPEED
1×
DOME OPACITY
60%
BLAST OPACITY
35%
COMPUTED RADII — HOPKINSON-CRANZ MODEL
LETHAL (BLAST+FRAG)
4 m
INJURY — BLAST
10 m
FRAG CASUALTY
7 m
THERMAL / 3° BURN
6 m
SAFE — BLAST
22 m
QD INHABITED
48 m
STRIKES PLACED (CLICK MAP TO ADD)
1 32.3409°N 75.7293°E
Strike 1: 32.3409°N 75.7293°E
CLEAR ALL
UNDO LAST
▶ BLAST WAVE
COMMIT TO MAP ↗
QD inhabited 48 m
Safe — blast 22 m
Injury — blast 10 m
Frag casualty 7 m
Thermal / 3° burn 6 m
Lethal radius 4 m

UAV & Weapon Effects

Weapon Effect Zone (WEZ)

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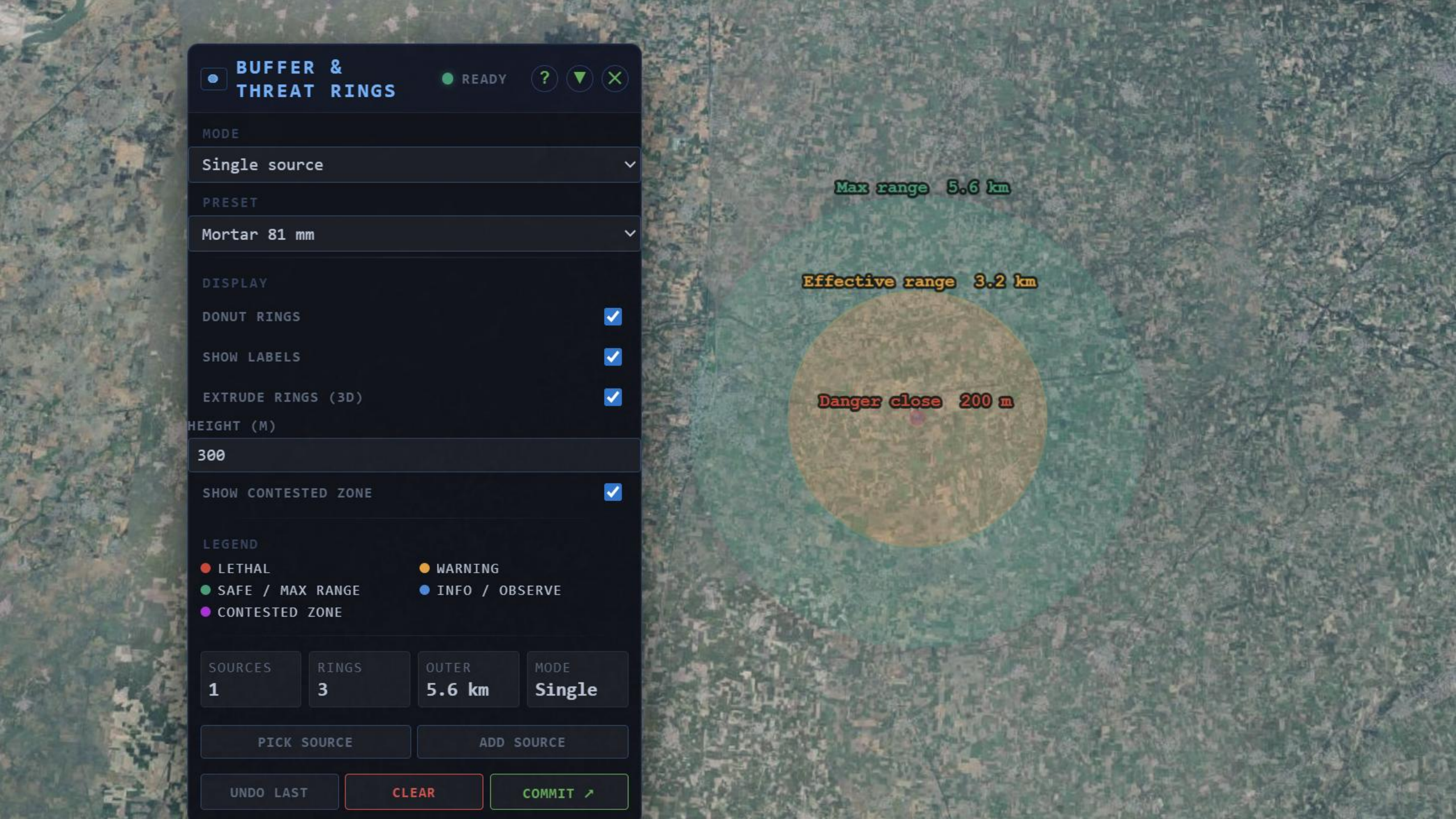
- Plans UAV routes and calculates coverage, range, and terrain clearance along the path.


BUFFER & THREAT RINGS
READY
MODE
Single source
PRESET
Mortar 81 mm
DISPLAY
DONUT RINGS
SHOW LABELS
EXTRUDE RINGS (3D)
HEIGHT (M)
300
SHOW CONTESTED ZONE
LEGEND
LETHAL
WARNING
SAFE / MAX RANGE
INFO / OBSERVE
CONTESTED ZONE
SOURCES
1
RINGS
3
OUTER
5.6 km
MODE
Single
PICK SOURCE
ADD SOURCE
UNDO LAST
CLEAR
COMMIT ↗
Max range 5.6 km
Effective range 3.2 km
Danger close 200 m

BUFFER &
THREAT RINGS
READY
MODE
Multi-source union
PRESET
Artillery 155 mm
DISPLAY
DONUT RINGS
SHOW LABELS
EXTRUDE RINGS (3D)
HEIGHT (M)
300
SHOW CONTESTED ZONE
LEGEND
LETHAL
WARNING
SAFE / MAX RANGE
INFO / OBSERVE
CONTESTED ZONE
SOURCES
4
RINGS
3
OUTER
30.0 km
MODE
Union
PICK SOURCE
ADD SOURCE
UNDO LAST
CLEAR
COMMIT ↗
Ef:Danger:close:0.600:m km

UAV & Weapon Effects

Weapon Effect Zone (WEZ)

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Ballistic Trajectory

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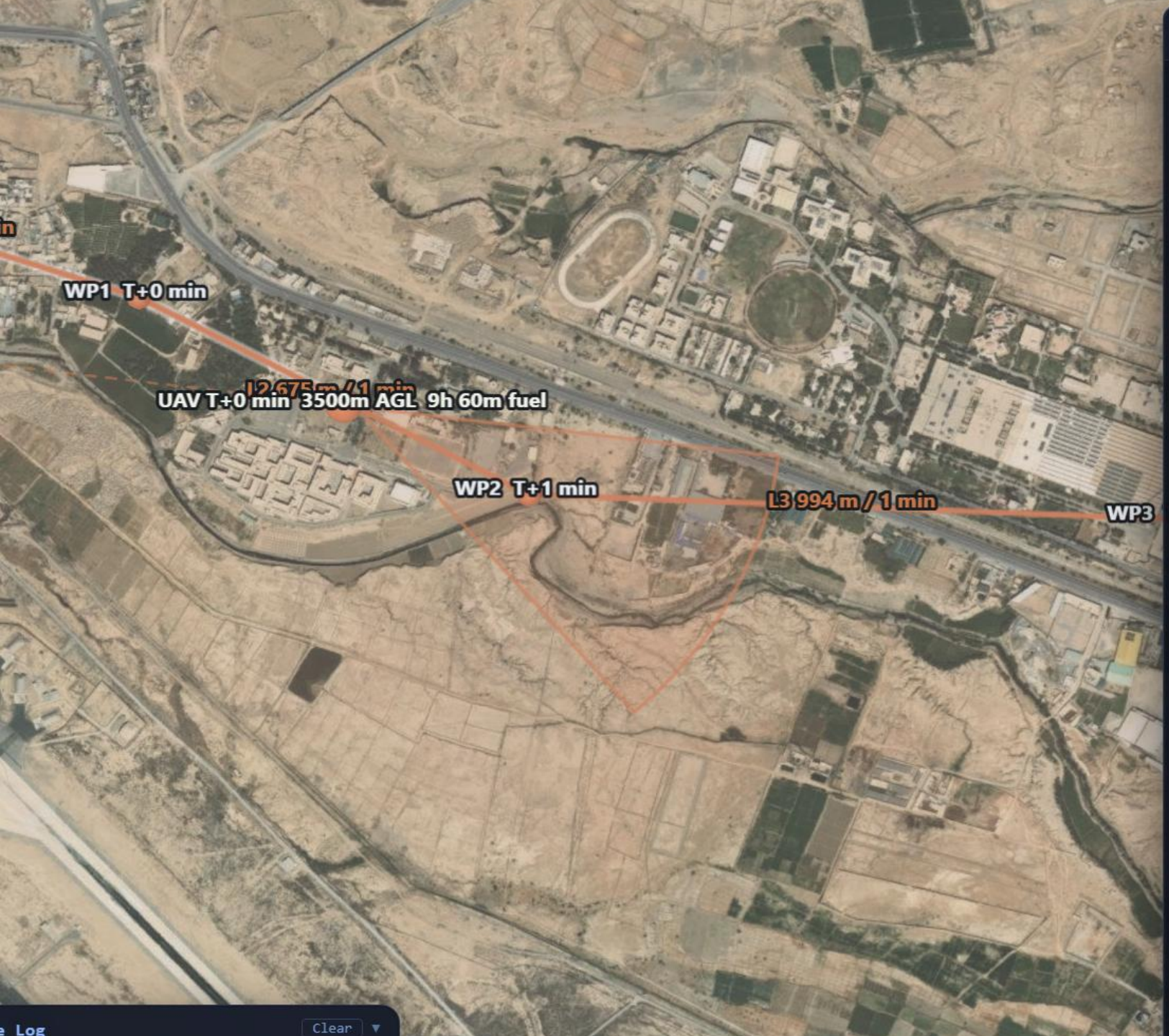
Buffer & Threat Rings

- Draws concentric threat rings around a position at configurable standoff distances.

UAV Flight Analysis

- Plans UAV routes and calculates coverage, range, and terrain clearance along the path.

Analysis Tools



UAV **FLIGHT ANALYSIS** READY ? v x

Feasible with 9h 58m reserve after return-to-base.

PLATFORM

Armed UAV

Reconnaissance plus standoff engagement

MISSION PERFORMANCE

SPEED KM/H	ENDURANCE MIN
155	600
ALTITUDE M	SENSOR RANGE M
3500	650
SENSOR FOV DEG	WEAPON RANGE M
42	8000

DISPLAY OPTIONS

- [x] ARMED
- [x] SENSOR FOOTPRINT
- [x] WEAPON ENVELOPE
- [x] ENDURANCE BOUNDARY
- [x] COVERAGE TRAIL

TIMELINE

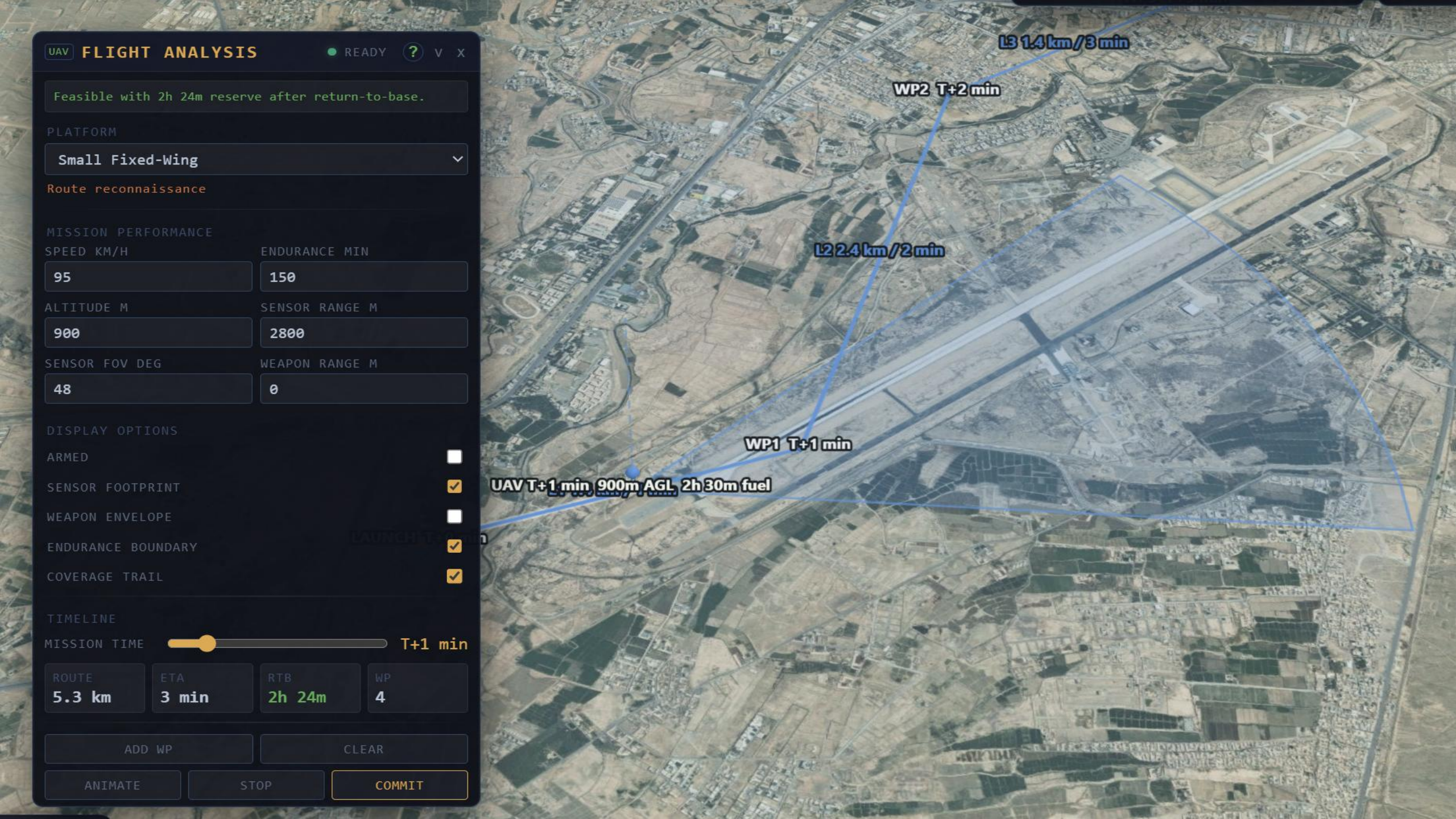
MISSION TIME T+0 min

ROUTE	ETA	RTB	WP
3.3 km	1 min	9h 58m	5

ADD WP | CLEAR

ANIMATE | STOP | COMMIT

Log Clear



UAV **FLIGHT ANALYSIS** ● READY ? V X

Feasible with 2h 24m reserve after return-to-base.

PLATFORM

Small Fixed-Wing

Route reconnaissance

MISSION PERFORMANCE

SPEED KM/H	ENDURANCE MIN
95	150
ALTITUDE M	SENSOR RANGE M
900	2800
SENSOR FOV DEG	WEAPON RANGE M
48	0

DISPLAY OPTIONS

- ARMED ☐
- SENSOR FOOTPRINT ☑
- WEAPON ENVELOPE ☐
- ENDURANCE BOUNDARY ☑
- COVERAGE TRAIL ☑

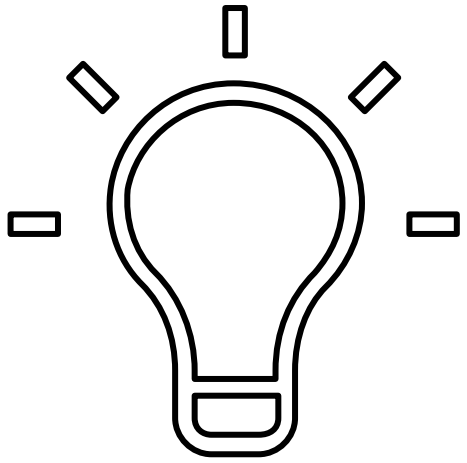
TIMELINE

MISSION TIME T+1 min

ROUTE	ETA	RTB	WP
5.3 km	3 min	2h 24m	4

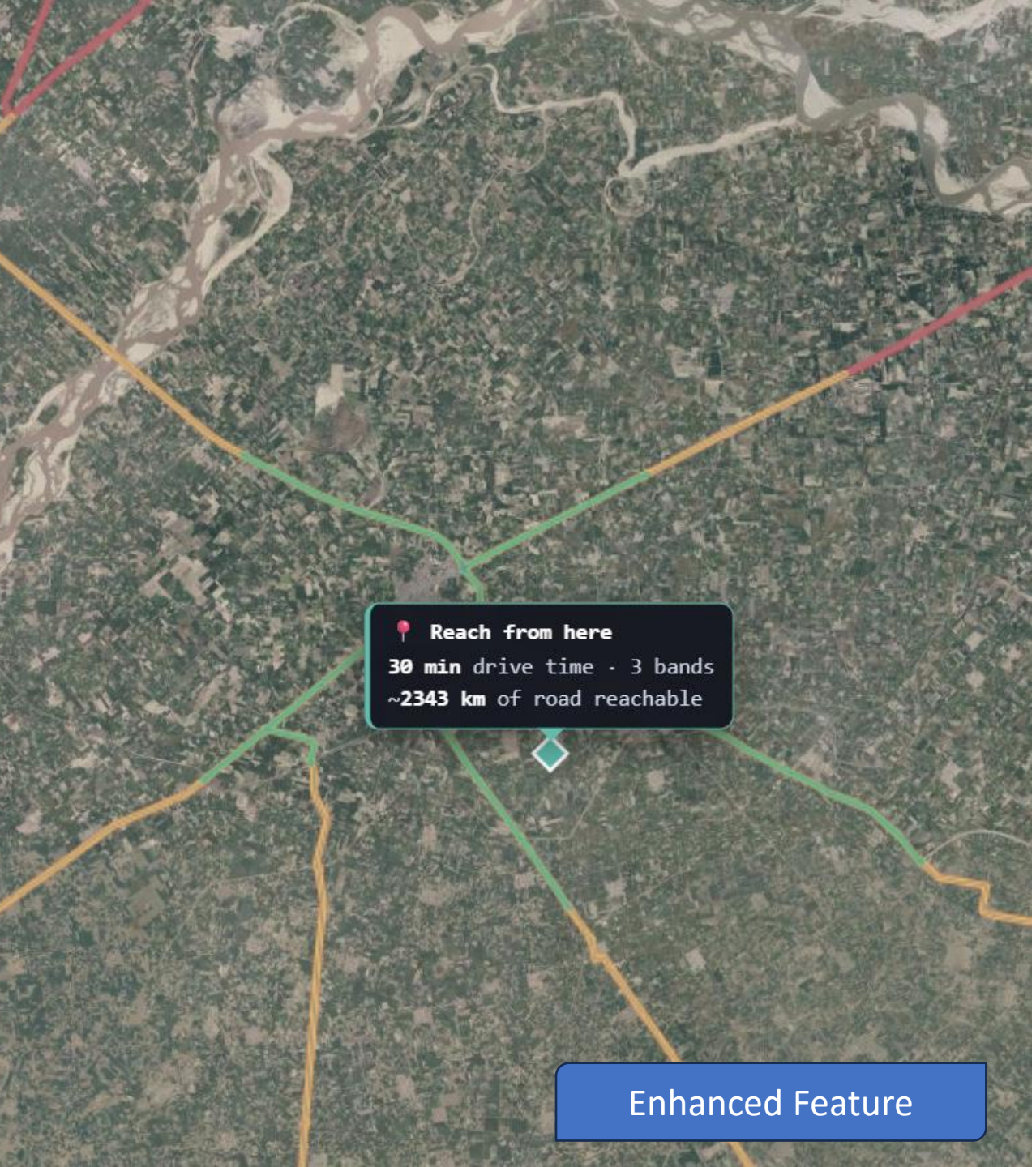
ADD WP | CLEAR

ANIMATE | STOP | COMMIT



Analysis Tools

MSRs, Service Areas & Convoy Fuel Estimates



Enhanced Feature

MSRs, Service Areas & Convoy Fuel Estimates

Road Network Engine

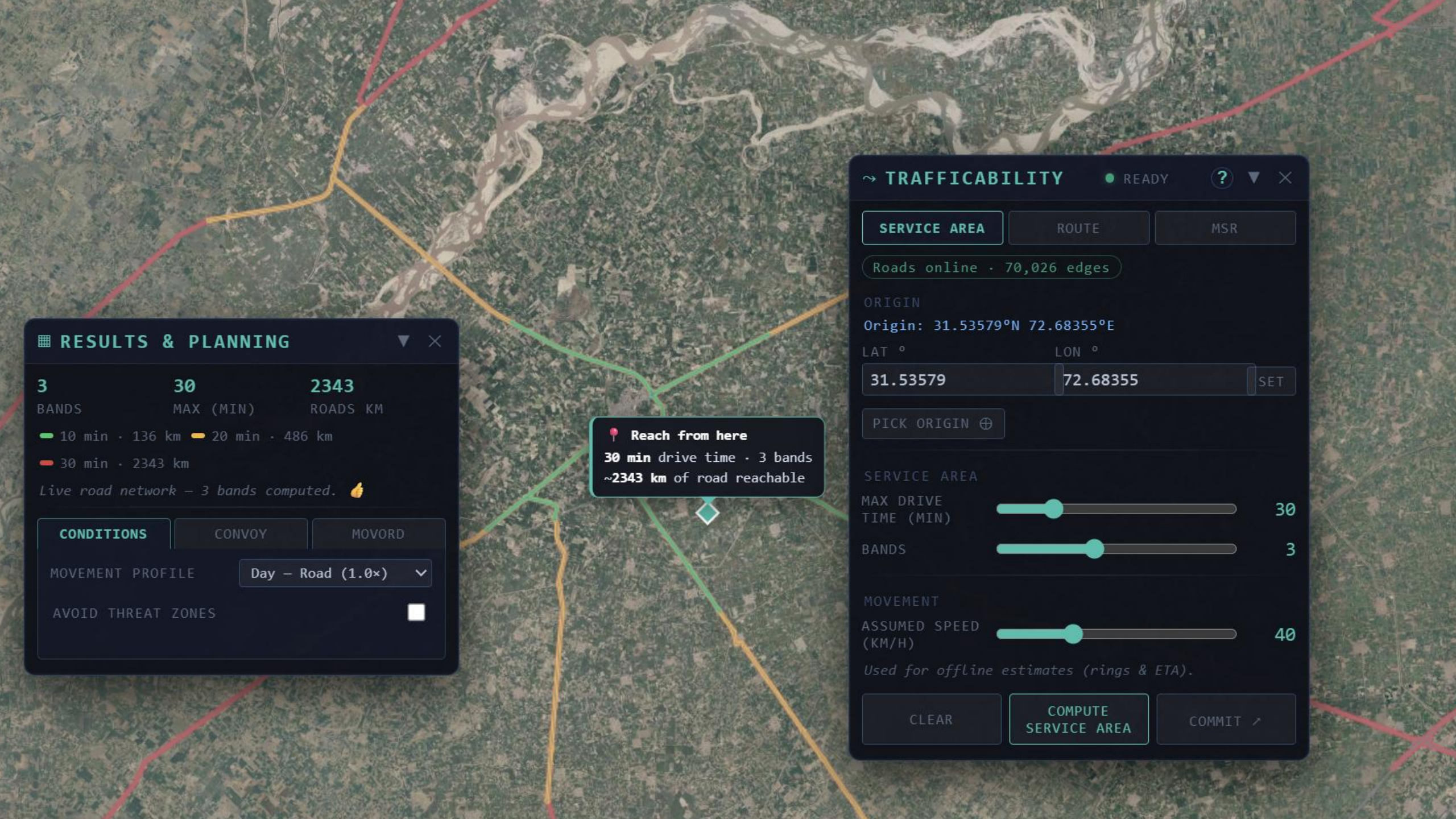
- Calculates road-following routes, travel times, and fuel estimates between two points.

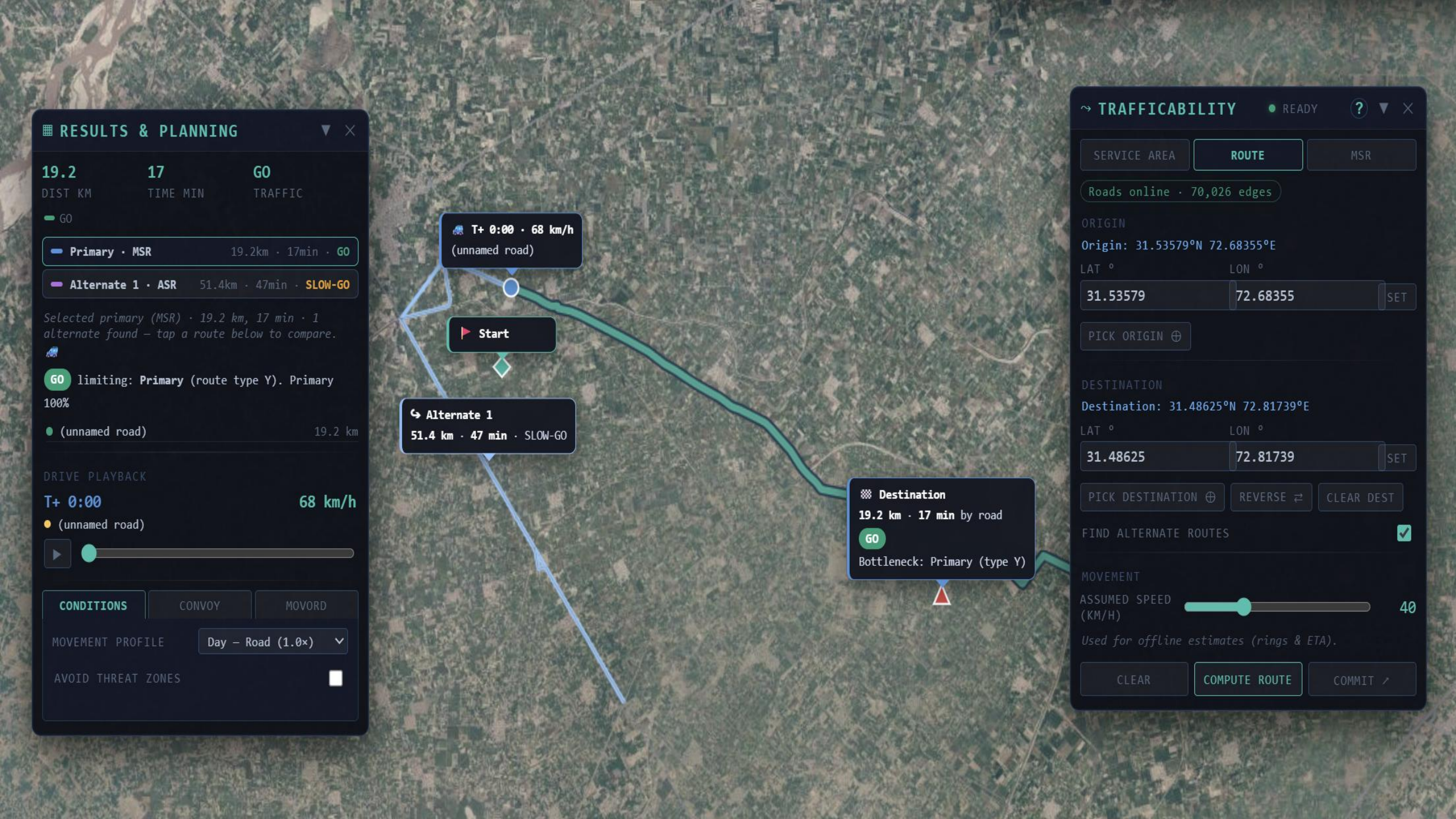
Trafficability

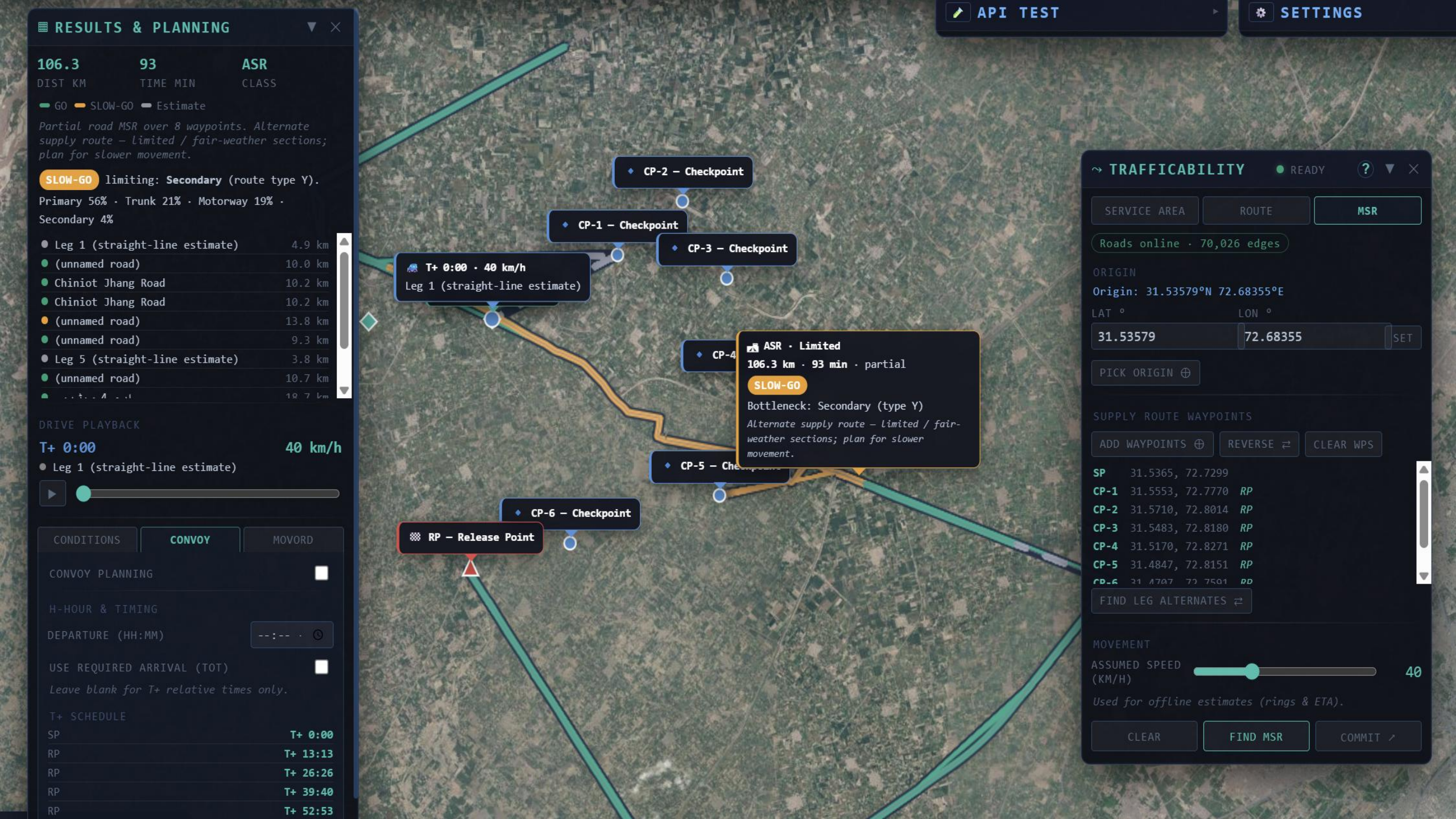
- Classifies terrain and roads as GO, SLOW-GO, or NO-GO for vehicle movement.

Corridor Analysis

- Evaluates route width, chokepoints, and threat exposure along a selected MSR.


RESULTS & PLANNING
3
BANDS
30
MAX (MIN)
2343
ROADS KM
10 min · 136 km
20 min · 486 km
30 min · 2343 km
Live road network — 3 bands computed.
CONDITIONS
CONVOY
MOVORD
MOVEMENT PROFILE
Day — Road (1.0×)
AVOID THREAT ZONES
Reach from here
30 min drive time · 3 bands
~2343 km of road reachable
TRAFFICABILITY
READY
SERVICE AREA
ROUTE
MSR
Roads online · 70,026 edges
ORIGIN
Origin: 31.53579°N 72.68355°E
LAT °
LON °
31.53579
72.68355
SET
PICK ORIGIN
SERVICE AREA
MAX DRIVE TIME (MIN)
30
BANDS
3
MOVEMENT
ASSUMED SPEED (KM/H)
40
Used for offline estimates (rings & ETA).
CLEAR
COMPUTE SERVICE AREA
COMMIT


RESULTS & PLANNING
19.2
DIST KM
17
TIME MIN
GO
TRAFFIC
GO
Primary · MSR 19.2km · 17min · GO
Alternate 1 · ASR 51.4km · 47min · SLOW-GO
Selected primary (MSR) · 19.2 km, 17 min · 1 alternate found — tap a route below to compare.
GO limiting: Primary (route type Y). Primary 100%
(unnamed road) 19.2 km
DRIVE PLAYBACK
T+ 0:00
68 km/h
(unnamed road)
CONDITIONS
CONVOY
MOVORD
MOVEMENT PROFILE
Day — Road (1.0×)
AVOID THREAT ZONES
T+ 0:00 · 68 km/h
(unnamed road)
Start
Alternate 1
51.4 km · 47 min · SLOW-GO
Destination
19.2 km · 17 min by road
GO
Bottleneck: Primary (type Y)
TRAFFICABILITY
READY
SERVICE AREA
ROUTE
MSR
Roads online · 70,026 edges
ORIGIN
Origin: 31.53579°N 72.68355°E
LAT °
LON °
31.53579
72.68355
SET
PICK ORIGIN
DESTINATION
Destination: 31.48625°N 72.81739°E
LAT °
LON °
31.48625
72.81739
SET
PICK DESTINATION
REVERSE
CLEAR DEST
FIND ALTERNATE ROUTES
MOVEMENT
ASSUMED SPEED (KM/H)
40
Used for offline estimates (rings & ETA).
CLEAR
COMPUTE ROUTE
COMMIT



RESULTS & PLANNING

106.3	93	ASR
DIST KM	TIME MIN	CLASS

GO · SLOW-GO · Estimate

Partial road MSR over 8 waypoints. Alternate supply route — limited / fair-weather sections; plan for slower movement.

SLOW-GO limiting: **Secondary** (route type Y).

Primary 56% · Trunk 21% · Motorway 19% · Secondary 4%

Leg 1 (straight-line estimate)	4.9 km
(unnamed road)	10.0 km
Chiniot Jhang Road	10.2 km
Chiniot Jhang Road	10.2 km
(unnamed road)	13.8 km
(unnamed road)	9.3 km
Leg 5 (straight-line estimate)	3.8 km
(unnamed road)	10.7 km

DRIVE PLAYBACK

T+ 0:00 — 40 km/h

Leg 1 (straight-line estimate)

CONDITIONS | CONVOY | MOVORD

CONVOY PLANNING ☐

H-HOUR & TIMING

DEPARTURE (HH:MM) --:--

USE REQUIRED ARRIVAL (TOT) ☐

Leave blank for T+ relative times only.

T+ SCHEDULE

SP	T+ 0:00
RP	T+ 13:13
RP	T+ 26:26
RP	T+ 39:40
RP	T+ 52:53

TRAFFICABILITY

READY

SERVICE AREA | ROUTE | MSR

Roads online · 70,026 edges

ORIGIN

Origin: 31.53579°N 72.68355°E

LAT °	LON °	
31.53579	72.68355	SET

PICK ORIGIN ⊕

SUPPLY ROUTE WAYPOINTS

ADD WAYPOINTS ⊕ | REVERSE ⇄ | CLEAR WPS

SP	31.5365, 72.7299	
CP-1	31.5553, 72.7770	RP
CP-2	31.5710, 72.8014	RP
CP-3	31.5483, 72.8180	RP
CP-4	31.5170, 72.8271	RP
CP-5	31.4847, 72.8151	RP

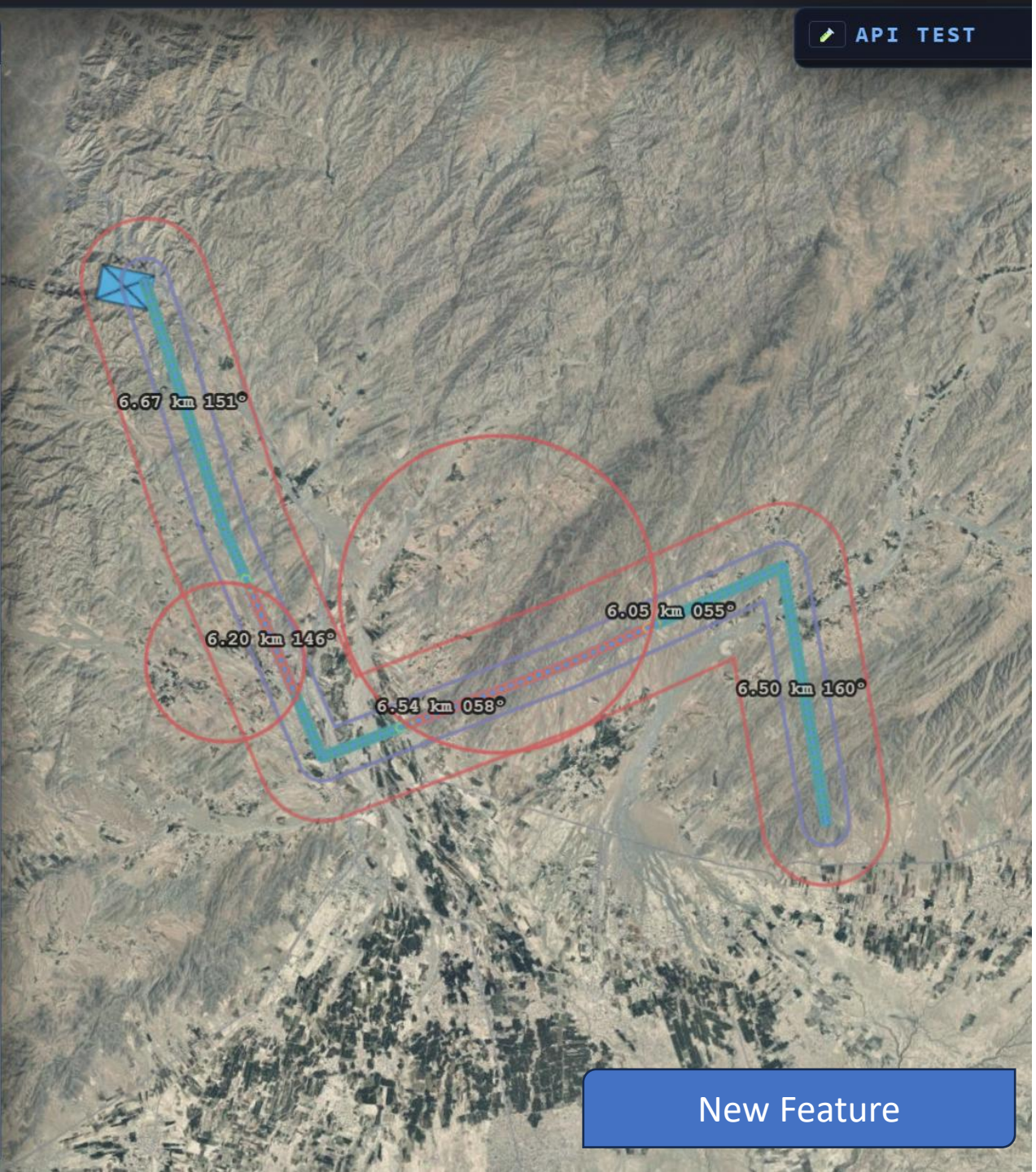
FIND LEG ALTERNATES ⇄

MOVEMENT

ASSUMED SPEED (KM/H) — 40

Used for offline estimates (rings & ETA).

CLEAR | FIND MSR | COMMIT ↗



MSRs, Service Areas & Convoy Fuel Estimates

Road Network Engine

- Calculates road-following routes, travel times, and fuel estimates between two points.

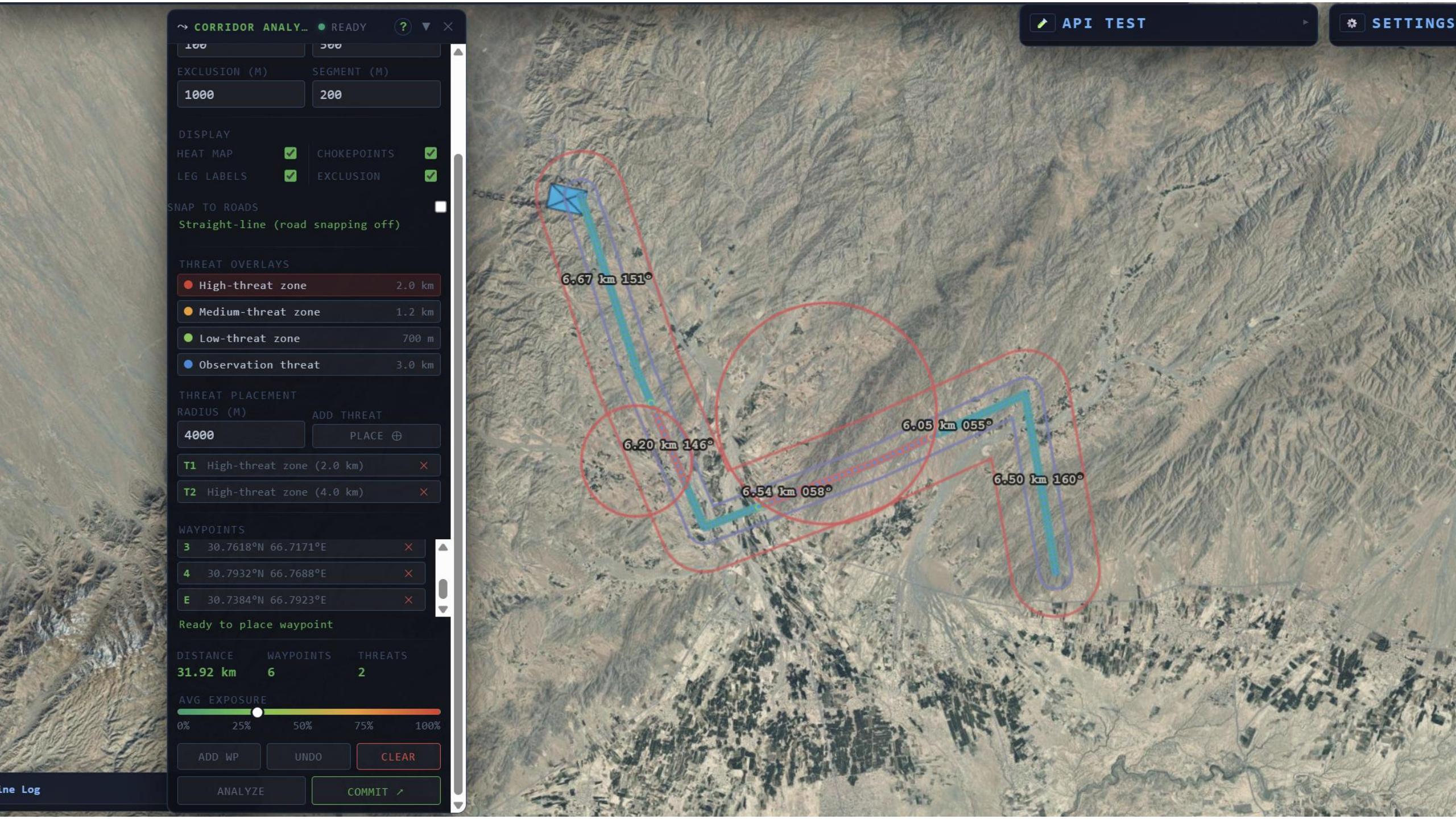
Trafficability

- Classifies terrain and roads as GO, SLOW-GO, or NO-GO for vehicle movement.

Corridor Analysis

- Evaluates route width, chokepoints, and threat exposure along a selected MSR.

Analysis Tools


CORRIDOR ANALY…
READY
EXCLUSION (M)
SEGMENT (M)
1000
200
DISPLAY
HEAT MAP
CHOKEPOINTS
LEG LABELS
EXCLUSION
SNAP TO ROADS
Straight-line (road snapping off)
THREAT OVERLAYS
High-threat zone 2.0 km
Medium-threat zone 1.2 km
Low-threat zone 700 m
Observation threat 3.0 km
THREAT PLACEMENT
RADIUS (M)
ADD THREAT
4000
PLACE
T1 High-threat zone (2.0 km)
T2 High-threat zone (4.0 km)
WAYPOINTS
3 30.7618°N 66.7171°E
4 30.7932°N 66.7688°E
E 30.7384°N 66.7923°E
Ready to place waypoint
DISTANCE
WAYPOINTS
THREATS
31.92 km
6
2
AVG EXPOSURE
0%
25%
50%
75%
100%
ADD WP
UNDO
CLEAR
ANALYZE
COMMIT
API TEST
SETTINGS
6.67 km 151°
6.20 km 146°
6.54 km 058°
6.05 km 055°
6.50 km 160°

Thank You